GARRISON ENGINEER (EAST), DELHI CANTT-10

NAME OF WORK: PROVN OF THREE SINGLE OFFRS ACCN AT ARMY AIR DEFENCE OFFICERS MESS 8 PARADE ROAD AND 01 MINOR WORK UNDER GE (EAST) DELHI CANTT-10

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Contractor

AGE (Contracts)
for Accepting Officer

REGISTERED /ACK DUE
Military Engineer Services
Garrison Engineer (East)
Delhi Cantt-10

8100/DE-100/2024-25/ /E8 05 Aug 2025

E-TENDER

CA NO : GE/DE-100/2024-25: PROVN OF THREE SINGLE OFFRS ACCN AT ARMY AIR DEFENCE OFFICERS MESS 8 PARADE ROAD AND 01 MINOR WORK UNDER GE (EAST) DELHI CANTT-10

Dear Sir,

1. Tender documents in respect of above work are uploaded on the site **www.defproc.gov.in**. The tender is base on single stage two cover/two stage three cover e-tendering system. The contents of Cover I & Cover II are specified in NOTICE OF TENDER.

2. Bids will be received online by ACCEPTING OFFICER upto the date and time mentioned in the **NOTICE INVITING TENDER (NIT)**. No tender/bid will be received in physical form and any tender/bid received in such manner will be treated as non bona fide.

3. Bid will be opened on line on a due date fixed for opening as per critical dates given in the portal. Cover 1 will be opened first. Tenderers may see the result of the opening of Cover 1 on the tender portal. Date of opening of Cover 2 shall be decided after technical evaluation of Cover 1 keeping in view the various eligibility criteria given in the NIT and the same will be intimated to the tenderers while uploading the technical evaluation on the tender portal.

4. Your attention is also drawn to instruction on filling and submission of tender attached herewith. You may forward your points on tender documents through e-mail and/or depute your technical representative for discussion on tender/drawing and to clarify doubts, if any, before the bid submission starts. You are requested not to write piece meal points and forward your points duly consolidated in one go. You may also attend the pre bid meeting on the date given in the tender.

5. Un-enlisted contractors are required to submit the scanned copies (in pdf file) of documents required as per eligibility criteria mentioned in instructions for filling the tender documents and Appendix ‘A’ to NIT along with EARNEST MONEY DEPOSIT (EMD) and tender fee on procurement portal mentioned above and submit the physical documents in the office of **GE (East) Delhi Cantt-10** within time limit specified in NIT. Inadequacy /deficiency of documents shall make the bid liable for rejection resulting in disqualification for opening of finance bid.

6. Enlisted contractors of MES shall submit the scanned copies (pdf file) of enlistment letter, tender fee and such other documents as mentioned in Appx ‘A’ to NIT on e-procurement portal and submit physical documents in the office of **GE (East) Delhi Cantt-10** before date & time fixed for this purpose.

7. The contractor must ensure that the tender/bid is uploaded on the tender portal using the authorised DSC in time as the Accepting Officer will take no cognizance of any quotations/offer received in any other electronic or physical form like email/fax/by hand/through post from tenderer/bidder.

8. Keeping in view delays due to system failure or other communication related failures, it is suggested that tender/bid may be uploaded sufficiently in advance of the last due date and time fixed on the portal.

9. General Conditions of Contracts (IAFW-2249) (1989 Print) and errata and amendments thereto, Schedule of Minimum Fair Wages and MES SSR (Part-I 2009 Print and Part-II 2020 Print) are not enclosed with these documents. These are available for perusal in the Office of **GE (East) Delhi Cantt-10**. Tenderer is also requested to keep the copy of these documents with him as same are available in the market.

10. **PERFORMANCE SECURITY DEPOSIT**

After acceptance of the Tender, the contractor will be required to lodge with the Accepting Officer PERFORMANCE SECURITY DEPOSIT @ 5% of contract amount. The amount is required to be lodged within 28 (Twenty eight) days of the receipt by the contractor of notification of acceptance of tender/bid, failing which action as stipulated in Condition 19 of IAFW-2249 General Conditions of Contracts shall be taken.

11. ANY TENDERER, WHICH PROPOSES ALTERATIONS TO ANY OF THE CONDITIONS, SPECIFICATIONS LAID DOWN IN THE TENDER DOCUMENTS OR ANY NEW CONDITION, WHATSOEVER, IS LIABLE TO BE REJECTED.

Yours faithfully

Sig of Contractor

AGE (Contracts)
for Accepting Officer

INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER

1. EARNEST MONEY DEPOSIT (EMD)

Contractor(s) who are not enlisted with MES/who are enlisted but have not executed the Standing Security Bond shall submit Earnest Money Deposit as detailed in Notice of Tender in one of the following forms, along with their tender/bid:-

(a) Deposit at Call Receipt from a Scheduled Bank in favour of **GE (East) Delhi Cantt-10**.

(b) Receipted Treasury Challan, the amount being credited to the Revenue Deposit of **GE (East) Delhi Cantt-10.**

It is advisable that Earnest Money is deposited in the form of deposit call receipt from an approved Schedule Bank for easy refund. In case the tenderer/ bidder wants to lodge 'Earnest Money Deposit' in any other form allowed by MES, a confirmation about its acceptability will be obtained from the Accepting Officer well in advance of the bid submission end date and time. Earnest Money Deposit shall be submitted in the name of **GE (East) Delhi Cantt-10.**

NOTES:- Earnest Money Deposit (EMD) in the form of cheque/ Bank Guarantee etc will not be accepted. **Non-Submission of Earnest Money Deposit (EMD) (scanned copy along with Technical Bid & hard copy before the date & time fixed for opening of BOQ) will render the bid disqualified for opening of Cover -II (finance bid)**.

2. GENERAL INSTRUCTIONS FOR COMPLIANCE

2.1 The bids received only in the electronic form will be considered. All bids shall be submitted on defproc.gov.in portal. Documents should be scanned and forwarded in 'pdf' format and 'xls' form as indicated. Bids shall be uploaded on 'defproc.gov.in' portal on or before the bid closing date mentioned in the tender. No tender/bid in any other electronic or physical form like date email/fax/by hand/through post will be considered.

2.2 The bid shall be DIGITALLY signed using authorised DSC. All pages of tender documents, drawings, corrections/alterations shall be signed/initialed by the lowest bidder after acceptance of tender for making original and CTC of the contract.

2.3 Digital signatory of the bid/tender shall ensure that he is competent to bind the contractor (through partnership deed, general power of attorney or Memorandum and Article of Association of the Company) in all the matters pertaining to the contract with Union of India including arbitration clause. A scanned copy of the documents in confirmation of such authority shall be attached with the tender/bid in 'pdf' form in Cover 1. It shall be ensured that power of attorney shall be executed in accordance with the constitution of the company as laid down in its Memorandum & Article of Association.

INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER

2.4 Drawings, if issued in physical form, must be returned duly initialed by the tenderer/bidder in separate envelope indicating his name and address.

2.5 The tender shall be signed, dated and witnessed at all places provided for in the documents after acceptance. All corrections shall be initialed. The Contractor shall initial every page of tender and shall sign all drawings forming part of the tender. Any tender/bid, which proposes alterations to any of the conditions whatsoever, is liable to be rejected.

3. BOQ:-

3.1 The tenderer shall quote his rates on the BOQ EXCEL FILE only as per guide line of e-procure web portal. No alteration to the format will be accepted and such bid will be disqualified.

3.2 In case any tenderer wishes to revise/modify the rates quoted in the BOQ file, he can do so only in the BOQ files before uploading the tender through https://defproc.gov.in site only before closing date & time.

3.3 After the uploading of tender, Department may upload the errata/ amendment through corrigendum. The tenderer/bidder should submit their offer considering the errata/amendment carried out through corrigendum issued from time to time.

3.4 While uploading the bid, the tenderers/bidders should specifically check whether any **revised BOQ** has been uploaded by department through corrigendum prior to Bid **submission start date**. Tenderers/bidders attention is specifically drawn to the fact that they should submit their offer on revised BOQ only. In case any tenderer/bidder submits offer on pre-revised BOQ in lieu of Revised BOQ, **it will be considered as a willful negligence by the tenderer/ bidder and quotation shall be considered non- bonafide**.

4. Tenders/bidders who uploaded their priced tenders/bids and are desirous of being present at the time of opening of the tenders/bids, may do so at the appointed time.

5. Hard copies of all above documents shall be sent by the contractor to the Tender issuing authority well in advance to be received before the date & time fixed for the same.

6. The contractor shall employ Indian Nationals after verifying their antecedents and loyalty. Attention is also drawn to relevant provisions in Special Condition referred herein-after and also Conditions 24 & 25 of IAFW-2249 (General Conditions of Contract).

Sig of Contractor

AGE (Contracts)
for Accepting Officer

MILITARY ENGINEER SERVICES
NOTICE INVITING TENDER (NIT)

1. A tender is invited for the work as mentioned in Appendix 'A' to this NOTICE INVITING TENDER (NIT).

2. The work is estimated to cost as indicated in aforesaid Appendix 'A'. This estimate, however, is not a guarantee and is merely given as a rough guide and if the work costs more or less, a tenderer/bidder will have no claim on that account.

3. The work is to be completed within the period as indicated in aforesaid Appendix 'A' in accordance with the phasing, if any, indicated in the tender from the date of handing over site, which will be on or about two weeks after the date of acceptance of tender.

4. Contractors whose names are on the MES approved list and within whose financial category the estimated amount would fall and un-enlisted contractors may submit tender/bid subject to other criteria mentioned in Appendix A. However in case of term contracts, enlisted contractors of Class SS to E may submit tender. Not more than one tender shall be submitted/uploaded by one contractor/ firm. Under no circumstances will a father and his son(s) or other close relations who have business dealing with one another be allowed to tender/bid for the same contract as separate competitors. Two firms shall be deemed to have business dealing if any of the partners/proprietor/director is common among both of them. A breach of this condition will render the tenders/bids of both the parties liable for rejection.

5. The Office of **GE (East) Delhi Cantt-10** will be the Accepting Officer here-in-after referred to as such for the purpose of this contract.

6. Not more than one tender/bid shall be submitted/uploaded by one bidder firm. Under no circumstances will a father and his son(s) or other close relations who have business dealing with one another be allowed to tender/bid for the same tender as separate competitors. A breach of this condition will tender the tender/bids of both the parties liable for rejection.

7. The Technical Bid and Financial Bid (Cover-1 and Cover-2) shall be uploaded by the contractor on or before the date & time mentioned in NIT. A scanned copy of DD with enlistment details and other documents as specified in Appendix A shall be uploaded as Cover-1 (Technical bid) of the tender on e-tendering portal. DD is refundable in case the contractor is not considered eligible in technical evaluation of Cover 1 resulting in non opening of Cover1. The applicant contractor shall bear the cost of bank charges for procuring and encashing the DD including revalidation of DDs and shall not have any claim from Government whatsoever on this account.

8. Tender form and conditions of contract and other necessary documents shall be available on website defproc.gov.in for download and shall form part of contract agreement in case the tender/bid is accepted.

9. In case of MES enlisted contractor who has not executed the Standing Security Bond and un-enlisted contractor, the Cover-I shall be accompanied by Earnest Money for the amount mentioned in Appendix 'A' in the form of deposit at call receipt in favour of **GE (East) Delhi Cantt-10** by a Scheduled Bank or in received treasury Challan the amount being credited to the revenue deposit of the **GE (East) Delhi Cantt-10**. The **GE (East) Delhi Cantt-10** will return the Earnest Money, wherever applicable, to all unsuccessful tenderers/ bidders by endorsing an authority on the deposit at call receipt for its refund, on receipt of intimation from the Accepting Officer to do that.

MILITARY ENGINEER SERVICES
NOTICE INVITING TENDER (NIT)

10. In case of successful contractor i e the lowest contractor having submitted EMD, he shall have the option of converting the EMD instrument into part of the Performance Security to be deposited by him within 28 days from the receipt of intimation of acceptance of tender from Accepting Officer.

11. Sample of materials and stores to be supplied by the contractor will also be available for inspection by the bidder at the office of **GE (East) Delhi Cantt-10** during working hours. The bidder is advised to visit the site of work by making prior appointment with **GE (East) Delhi Cantt-10**, who is the Executing Agency of the work. The bidder shall be deemed to have full knowledge of all relevant documents, samples, site etc whether he has inspected them or not.

12. Any bid which proposes any alteration to any of the conditions laid down or proposes any other new condition whatsoever, is liable to be rejected.

13. The uploading of bid by a bidder implies that bidder has read this notice and the conditions of contract and has made himself aware of the scope and specification of work to be done and of the conditions and rates at which stores (as applicable) etc will be issued to him and local conditions and other factors having bearing on the execution of the work.

14. The bidder must be in possession of a copy of the MES Schedule (SSR) (Part-I 2009 Print & Part-II 2020 Print) including amendments and errata thereto.

15. Accepting Officer does not bind himself to accept the lowest or any tender/bid or to give any reason for not doing so.

16. The Accepting Officer reserves the right to accept a tender submitted by a Public Undertaking/Small & Medium Enterprises (SMEs), giving a price preference/purchase preference over other tender(s)/bids which may be lower, as are admissible under the Government Policy. No claim for any compensation or otherwise shall be admissible for such tenderer/bidder whose tender/bid is rejected.

17. The **Notice Inviting Tender (NIT)** including Appendix 'A' and Annexures thereto, if any, shall form part of the contract agreement.

Sig of Contractor

AGE (Contracts)
for Accepting Officer

APPENDIX 'A' TO NOTICE INVITING TENDER (NIT)

1.	Name of work	**PROVN OF THREE SINGLE OFFRS ACCN AT ARMY AIR DEFENCE OFFICERS MESS 8 PARADE ROAD AND 01 MINOR WORK UNDER GE (EAST) DELHI CANTT-10**
2.	Estimated Cost of Work	Rs 43,36,000.00
3.	Cost of tender documents	Rs 500/- in the form of DD / Bankers cheque from any Scheduled/ Nationalised bank in favour of **GE (East) Delhi Cantt-10** and payable at **Delhi Cantt-10**
4.	Period of completion	**04 (Four)** months
5.	Website/portal address	https://defproc.gov.in
6.	Type of Contract	The tender shall be Item rate on IAFW-2249 with list of works to be priced by tenderer. The tenderers are required to quote their rates against items of list of works.
7.	Timeline Details:-	
(a)	Bid submission start date	Refer critical dates on website.
(b)	Bid submission end date	
(c)	Date of bid opening	
8.	Eligibility Criteria:-	
	(a) For MES enlisted contractors	Contractors shall be enlisted with MES in Class **'D'** and above and category **a (i)** subject to satisfactory remarks wrt performance in respect of works in hand as reflected in WLR or any other report circulated by any competent engineer authority.
	(b) For other Contractors	(i) The firm not enlisted with MES shall meet the enlistment criteria of class **'D'** and category **a (i)** i.e. with regard to having satisfactory completion of requisite value works with Central/ State Government, / Central/ State PSUs/ AWHO/ AFNHB/ CGEWHO/ DGMAP, Annual turnover, Bank Solvency, working capital, and other requirements given in Para 1.4 & 1.5 of Section 1 of MES Manual on Contracts - 2020 as available in all MES formations as well as MES website (www.mes.gov.in). (ii) Not carrying adverse remarks in Work Load Report (WLR) or any other similar report circulated by any competent authority, if already working in MES. (iii) Not suspended/debarred/blacklisted (either permanently or temporarily) from participating in any bid or for business dealings by any Central/State Government department or any Central/State Government PSU or any Autonomous Body under Central/State Government or any Local Body as on bid submission end date. (iv) Details of works completed and under progress in MES be submitted in the following format:-

Srl No	CA No & Name of Work	Value of CA	Date of Commencement	Date of Completion	Extended Date of Completion

(v) Un-enlisted Contractor who have secured two works in MES should get themselves registered in the appropriate designated Class with any Registering Authority, else the firm will not be eligible for participation in the tender unless until the firm is enlisted with MES.

9.	Tender issuing authority and Accepting Officer and contact details	**GE (East) Delhi Cantt.-10, email id- edcntdz4-mes.nic.in, Contact No.011-20892445.**
10.	Executive Agency	**GE (East) Delhi Cantt-10**
11.	Earnest Money	**Rs. 86720.00** in form of Deposit Call Receipt or Receipted Treasury Challan from any Scheduled/ **Nationalized** bank in favour of **GE (East) Delhi Cantt-10**

APPENDIX 'A' TO NOTICE INVITING TENDER (NIT)

NOTES:-

1. In case after opening of Cover 1, if the number of MES enlisted contractors of eligible class as well as un-enlisted contractors, if any, fulfilling the other eligibility criteria given in NIT are less than 7 (Seven), applications in respect of contractors one class below the eligible class shall also be considered subject to fulfillment of other eligibility criteria given in the NIT. Therefore MES contractor's one class below may also bid for this tender. Such contractors shall not be considered in case their present residual work in hand is more than five times their present tendering limit. However in case such contractors fulfill the criteria of upgradation to the stipulated eligible class based on past experience of completed works (individual work experience and/ or average annual turnover, as applicable) and financial soundness (solvency/financial soundness) and working capital), the ceiling of present residual work will not apply and they will be considered for issue of tender. Such contractors shall upload in their Cover-1 bid details of works in hand showing names of work names of Accepting Officers, Contract amounts, dates of commencement and completion (stipulated) and progress as on bid submission date. Such contractors, if claim to fulfill the criteria of upgradation shall also upload the requisite information/documents in support of upgradation. These details shall be verified by the Tender Issuing Authority from concerned formations in case bids of such contractors are considered for evaluation.

2. In case after opening of Cover 1, the number of MES enlisted contractors of eligible class as well as un-enlisted contractors, if any, fulfilling the other eligibility criteria given in NIT, are 7 (Seven) or more, applications of only those one class below the eligible class bidders shall be considered, who have previously completed similar works satisfactorily and are meeting the criteria of upgrading in respect of past experience of completed works (individual work experience and/or average annual turnover as applicable) and financial soundness (solvency/financial soundness and working capital) as per details given in Manual on Contracts. Therefore such contractors shall upload the requisite information/documents in the Cover-1.

3. Unenlisted contractors shall be considered provided he meets the criteria. Foreign firms shall not be eligible for this tender. However Indian Firms having foreign national/ Indian nationals staying abroad / Indian national having taken foreign citizenship, as directors (s) shall be considered subject to security clearance from the concerned authorities.

4. Contractors enlisted with MES will upload following documents in Cover I for checking eligibility.

(a) Application for tender on Firm's letterhead.
(b) Enlistment letter issued by the Registering Authority duly renewed for the cycle period in vogue.
(c) Scanned copy of DD/Bankers Cheque towards cost of tender and EMD instrument in case SSD bond is not signed at the time of registration.
(d) Scanned copy of GST registration number.
(e) Scanned copy of EPF registration number.
(f) Documents required as per srl No 1 & 2 of Notes:-

(i) Experience of work completed.
(ii) Financial soundness (Solvency).
(iii) Working Capital.
(iv) Details of work in hand showing names of work names of Accepting Officers, Contract amounts, dates of commencement and completion (stipulated) and progress as on bid submission date.

5. Contractors not enlisted with MES will be required to upload following documents in Cover I for checking eligibility.

(a) Application for tender on Firm's letterhead.
(b) Scanned copy of DD/Bankers cheque toward cost of tender and Earnest Money Deposit (EMD) instrument.
(c) Copy of Police Verification Report/Police Clearance Certificate/Character Certificate from the Police Authority of the area where the registered office of the firm is located/notarized copy of valid passport of Proprietor/each Partner/each Director.

(d) Scanned copy of GST registration number.
(e) Scanned copy of EPF registration number.

APPENDIX 'A' TO NOTICE INVITING TENDER (NIT)

(f) The following documents required for enlistment in MES for the class mentioned in Para 8(b) above as per Para 1.4 & 1.5 of section I of MES Manual on Contracts 2020:-

(i) Past experience of completed works in last 5 years.

(ii) Average turnover for last two consecutive years (in Rs.)

(iii) Solvency certificate from scheduled bank.

(iv) Working Capital from schedule bank.

(v) Application for enrolment as contractor on IAFW-2190 (Revised 2020). Coloured Photographs affixed on application shall be self-attested (In case of becoming Lowest Bidder).

(vi) Affidavit for constitution of firm.

(vii) Audited balance sheet of last five years in case of Limited companies.

(viii) Affidavit from contractor that there is no Govt. dues/ recovery outstanding against him. If there is any Govt. dues/ recovery outstanding from the contractor enlistment /upgradation shall not be carried out.

(ix) Affidavit from contractor giving brief details of arbitration/ litigation cases he was involved in. Registering authority will critically examine such details and if contractor is considered habitual litigant, his enlistment / upgradation shall not be carried out. Contractor shall be deemed to be a habitual litigant if he moves the court more than once ignoring arbitration clause and court does not uphold his view.

(x) Copy of partnership deed in case of partnership firm.

(xi) Memorandum and Articles of Association in case of Limited Companies.

(xii) Copy of registration certificate of firm with Registrar of Companies/ Registrar of Firms.

(xiii) Affidavit that no near relative(s) of the contractor or his / their employees / agents is /are working as Gazetted / Commissioned Officer(s) in MES/ Corps of Engineers/ Ministry of Defence. If the near relative(s) is /are working in such capacity in any formation upto Zone CE/ CCE, he/they shall furnish details. The contractor shall not be entitled to tender for the works in entire area of CE Zone/ CCE.

(xiv) Affidavit that no near relative(s) of the contractor or his/ their employees / agents is/ are working as Junior Engineer(s) in MES / Corps of Engineers. If his near relatives(s) is / are working in such capacity in any formation upto CWE/ GE (I) office, the contractor shall furnish details. The contractor shall not be entitled to tender for the works in entire area of that CWE/GE(I).

(xv) Copy of power of attorney/ resolution of the board in favour of any Partner/ Director of the firm.

(xvi) Copy of immediate last Income Tax Return.

(xvii) A separate sheet containing specimen signatures (signed in black ink) and affixed with photographs of Proprietor/ Partners/ Directors (in case of becoming Lowest Bidder).

(xviii) Self attested copy of PAN Card of Proprietor, Partners, Directors and the firm / Company.

(g) Details of works being executed in MES, if any.

APPENDIX 'A' TO NOTICE INVITING TENDER (NIT)

6. Tenders not accompanied by scanned copies of requisite DD/Bankers Cheque towards cost of tender and earnest money (as applicable) in Cover I shall not be considered for validation of 'T' bid and their Financial Bids will not be opened.

7. Contractors should ensure that their original physical DDs and Earnest Money Deposit (EMD) instruments (as applicable) reach the office of Accepting Officer within **07 (Seven)** days of bid submission end date failing which following action shall be taken.

(a) In case of tenders from and enlisted contractors of MES, where scanned copies of requisite DD/Bankers Cheque towards cost of tender have been uploaded in Cover I but physical copies are not received within the stipulated period, their financial bids (Cover 2) will be opened. However non-submission of physical copies of cost of tender shall be considered as willful negligence of the tenderer with ulterior motives and such tenderer shall be banned from bidding for a period of six months commencing from the date of opening of Financial Bid (Cover 2).

(b) In case of tenders from un-enlisted contractor, where scanned copies of requisite DD/Bankers Cheque towards cost of tender have been uploaded in Cover I but physical copies are not received within the stipulated period, their financial bids (Cover 2) will not be opened. Named of such contractors along with complete address shall be circulated for not opening of their bids for a period of six months commencing from the date of opening of Financial bid (Cover 2).

(c) In case of tenders from enlisted and un-enlisted contractors, where scanned copies of instruments for Earnest Money Deposit (as applicable) have been uploaded in Cover-1 but the same are not received in physical form within stipulated period, such tenders shall not qualify for opening of financial bid (Cover-2).

8. Contractor will not be allowed to execute the work by subletting or through power of attorney to a third party/another firm on his behalf. However a contractor can execute the work through power of attorney to sons/daughters/spouse of Proprietor/Partner/Director and firm's own employees, director, project manager provided they are not having a separate enlisted firm in MES in their name as Proprietor/Partner/Director.

9. After opening of Cover 1 and during its technical evaluation, in case any deficiency is noticed in the documents required to be uploaded by the tenderers as per NIT, as communication in the form of e-mail/SMS/Speed Post etc shall be sent to the contractor to rectify the deficiency within a period of seven days from date of communication failing which their financial bid (Cover 2) shall be opened and contractor shall have any claim on the same.

10. Invitation for e-tender does not constitute any guarantee for validation of Technical bid and subsequent opening of financial bid of any applicant/bidder merely by virtue of enclosing DD. Accepting Officer reserves the right to reject the Technical bid and not to open the financial bid of any applicant/bidder. Technical bid validation shall be decided by the Accepting Officers based on eligibility of the firm as per criteria given in this Appendix. Tenderer/bidder will be informed regarding no-validation of his Technical bid assigning reasons therefore through tender evaluation report which shall be uploaded on the website. Such tenderer, if desires, may appeal to the next higher Engineer Authority (NHEA) viz **CWE Delhi Cantt. on email id dcntdz3-mes@nic.in** with copy to the Accepting Officer on email before the scheduled date of opening of Cover 2. NHEA shall decide the matter within a period of seven working days from the date of receipt of appeal. The decision of the NHEA shall be final and binding. The tenderer/bidder shall not be entitled for any compensation whatsoever for rejection of his bid.

11. In case an un-enlisted contractor is already executing works in MES, he shall not be considered eligible for the subject tender if the total value of such works is more than twice the tendering limit of the MES class of contractor for which it is eligible. For this purpose details of the works being executed by such a contractor shall be uploaded in the Cover-1 of the bid and shall be checked/verified by the Accepting Officer.

APPENDIX 'A' TO NOTICE INVITING TENDER (NIT)

12. In case the BOQ is revised through the corrigendum and the bidder has failed to quote on revised BOQ (i.e he has quoted on pre revised BOQ), such bid shall be treated as willful negligence by the bidder and his quotation shall be considered non-bonafide. In such cases the lowest tender shall be determined from amongst the valid/bonafide bids only. Accepting Officer may decide whether to re-tender or consider the lowest bonafide tender for acceptance.

13. Revoking the offer or revising the rates upward or offering voluntary reduction by the lowest tenderer after opening of Cover 2 shall be considered as a willful default. For this default a penalty of an amount equal to Earnest Money shall be levied. In case of an un-enlisted tenderer, Earnest Money deposited by him shall be forfeited. In case of MES enlisted tenderer having deposited the Standing Security Bond, an amount equal to the earnest money stipulated in the NIT, shall be notified to the tenderer for depositing through MRO and consideration of such tenderer in tender evaluation for future works shall remain suspended till the aforementioned amount is deposited in the Government Treasury. No other disciplinary/administrative action shall be taken against such tenderers. In such a situation, the next lowest offer shall not be considered for acceptance, instead, retendering shall be resorted to in a transparent and fair manner and the defaulting tenderer and his related firm if any, shall not be eligible for this tender in second call or subsequent calls.

13A. "Irrespective of whatever is mentioned in condition 19.3 of IAFW-2249 with regard to suspension of tenders on account of non-submission of Performance Security, issue of tenders to such tenderers shall remain suspended for a period of six month from the date of cancellation of contract under the condition of 19.3 of IAFW-2249 in case of Unenlisted contractors. In case of MES enlisted contractor, issue of tenders shall remain suspended till deposit of EMD or six months from date of cancellation whichever is later"

14. Tender to related firms shall not be issued simultaneously. Firms shall be termed as related if Proprietor/ one or more Partners/ Directors are common, Decision of Accepting Officer on issue/deny the tender to any one of the related firms shall be final and binding.

Sig of Contractor

AGE (Contracts)
for Accepting Officer

File No 8100/DE-100/2024-25/ /E8
Garrison Engineer (East)
Delhi Cantt-10
05 Aug 2025

ITEM RATE CONTRACT IAFW-1779-A TO BE READ IN CONJUNCTION WITH GENERAL CONDITIONS OF CONTRACT (IAFW-2249) 1989 PRINT

Tele:- 25692671

Garrison Engineer (East)
Delhi Cantt-10

8100/DE-100/2024-25/ /E8

05 Aug 2025

CA NO : GE/DE-100/2024-25: PROVN OF THREE SINGLE OFFRS ACCN AT ARMY AIR DEFENCE OFFICERS MESS 8 PARADE ROAD AND 01 MINOR WORK UNDER GE (EAST) DELHI CANTT-10

1. E-tender in the www.defproc.gov.in portal shall be submitted in electronic form duly digitally signed by the authorised signatory of the tenderer and the requisite documents as mentioned in the tender shall be physically delivered at the Office of the GE (East) Delhi Cantt 110010.

(i)	Closing date and time of e-bid submission (Cover No 1 & 2)	:	**Refer critical dates on website.**
(ii)	Date and time of e-bid opening (Cover No 1)	:	**Refer critical dates on website.**
(iii)	Date and time of e-bid opening. (Cover No 2)	:	Will be intimated online after completion of evaluation of Fee / Pre-Qualification / Technical**.**

2. Any correspondence concerning to this tender should be addressed as indicated on top of this sheet quoting the reference as given.

3. All documents must be returned whether or not tender has been accepted.

THE PRESIDENT OF INDIA DOES NOT BIND HIMSELF TO ACCEPT THE LOWEST OR ANY TENDER

Contractor

AGE (Contracts)
For Accepting Officer

SCHEDULE 'A'
(IAFW 1779-A REVISED 1955)
(LIST OF WORKS AND PRICES

1. The quantities shown in Column 3 are approximate and are inserted as a guide only. However, these shall not be varied beyond the limits laid down in condition 7 of IAFW-2249 General Condition of contracts.

2. The bidder shall Quote/enter unit rate in Figures at **green space (Column 5) on the BOQ (SCHEDULE 'A').** The tenderer shall calculate his own details and insert unit rates in Column 5 (**green space**) against each item of Schedule 'A' (BOQ). It is an express condition of this contract that the rates quoted by the tenderer in the tender for various items shall be deemed to include for the entire and final completion of the items of work in accordance with provision of the tender. Government shall not entertain any claim whatsoever on account of inaccuracies/misunderstanding, if any in the aforesaid rates.

3. Tenderers shall be deemed to include in his unit rate for each item for the provision of all materials, labour, operations, tools/plants as required and other requirements as detailed in the Particular Specifications irrespective of whether these appear as specific item or mentioned in the description of items in Schedule 'A' (BOQ) or not and the tenderer shall not be paid any extra for the same.

4. The Entire work covered under this contract shall be completed within a period of **04 (Four) Calendar Months** from the date of commencement of work with phase wise or otherwise completion as mentioned in Work Order which shall be issued by GE after acceptance of the tender.

5. The rate quoted by the tenderer shall be inclusive of all types of taxes, levies including GST, labour welfare tax etc. as prevailing on the date of submission of tender.

6. Certain taxes such as central excise duty, service tax, additional custom duty, state level value added tax, octori and other levies which were applicable on interstate transportation of goods are subsumed by GST, thus special condition for **"Reimbursement/refund on variation in prices in taxes directly related to contract value"** as specified here-in-after in tender shall be deemed to be amended incorporating GST in lieu of taxes mentioned in the condition but subsumed by GST.

7. Cost of Testing including Transportation of Materials or Equipments or Items, provision of all facilities for Testing in accordance with Specifications and/or IS shall be borne by the Contractor unless otherwise mentioned.

8. The Work shall be carried out with utmost care to ensure that no damage to Existing/Adjoining Work/Fittings is done failing which the damage, if any done shall be rectified by the Contractor to match with the Existing/ Adjoining Work /Fittings to the entire satisfaction of Engineer-in-Charge under Contractor's own arrangement and at his own expenses.

9. Suitable Tools, Plants, Equipments, Mechanism, etc., as considered necessary shall be adopted during execution of the Work. The Contractor shall take all Precautions, Safety Measures, etc., to avoid any Damage, Miss-happening, Accident, etc., to the Workmen engaged by him to carry out the Work. The UNIT RATES quoted by Contractor shall be deemed to have included the Element of adopting Safety Measures, Precautions and also the Risks, etc. involved in Work and nothing EXTRA shall be admissible on this account.

10. **The Contractor has to follow all the directives issued by MES / Station Authority / Concerned State Authority / GOI to avoid the spread of Corona Virus (COVID-19). For this account nothing extra shall be admissible.**

11. The method of Measurements to be adopted in the Contract shall be as laid down in MES STANDARD SCHEDULE OF RATES PART-I & II, **IS 1200** and other relevant BIS standard. The General Rules, Special Conditions, Preambles and Other Provisions in MES STANDARD SCHEDULE OF RATES PART-I & II and relevant BIS standard shall be deemed to apply to the Work under this Contract unless otherwise mentioned in the Tender documents. However, where the UNITS of Work given in **Column 4** against Items of SCHEDULE 'A' (BOQ) are at variance with that mentioned against respective Items in MES STANDARD SCHEDULE OF RATES, the Measurements of Work against such Items of SCHEDULE 'A' (BOQ) shall be done as per the UNITS given in SCHEDULE 'A' (BOQ).

SCHEDULE 'A' NOTES (Contd..)

14. In the event of deviation order involving fixation of Special (Star) Rate, Draft Rate shall be prepared by GE (within a maximum period of 30 days) while initiating the proposal for deviation seeking approval of Accepting Officer and notified to contractor. While notifying the Draft Rate, it will be clearly stipulated that the same is merely an estimated rate and firm rate shall be fixed based on actuals and receipt of supporting documents from contractor such as vouchers / literature of product / test certificates, etc. (as applicable) on completion of the work involving Star Rate. Any objection to the method of fixing Star Rate will be dealt as per Condition 7 of IAFW-2249.

15. Draft Star rate shall be made based on market enquiry through telephonic enquiry / quotations / email / rate list / internet based sources, material and labour constants available in various Civil Engg books and record available in respect of Star Rates approved in the past for similar items of work etc. Contractor may also assist GEs office in preparation of Draft Star Rate.

16. The Draft Star Rate shall be purely a draft rate and shall not be used for claiming final payment during execution of work. However GE shall allow part payment to the tune of 80% during execution to avoid any financial hardship to contractor.

17. After completion of the items of work involving Star Rate, contractor shall submit the vouchers / literature of product / test certificates (as applicable, decision of GE being final in case of any disagreement) for finalisation of Star Rate. The Star Rate shall be technically checked by DCWE (C) / Director (C) depending upon the financial effect and approved by competent authority within a period of one month from submission of the relevant documents by contractor as mentioned above.

18. The Star Rate as approved by competent authority after technical check by DCWE (C) / Director (C) depending upon the financial effect shall be referred as "the rate decided by GE" under Para 62(G) of IAFW-2249.

19. **INSPECTION OF THE WORKS (DEFECTS LIABILITY PERIOD) :** {Refer Condition 46 of IAFW - 2249 (General Conditions of Contracts) (As amended under amendment No. 49 to IAFW- 2249)}:-

(a) The Defects liability period for the scope of work(s) covered under the contract are (i) **TWELVE CALENDAR MONTHS** after the work have been handed over the the Govt. (hereinafter referred to as the "defects liability period) for works of Periodical Services and Welcome Maintenance, Minor works and Revenue Works and Works for "Maintenance & Operation" including Repairs and Works for Maintenance & Repair/replacement. (ii) **TWENTY FOUR CALENDAR MONTHS** after the works have been handed over to the Govt. (hereinafter referred to as the "defects liability period") for works other than mentioned in sub Para 19 (a) (i) above. Any defects in the work noticed during this period shall be rectified by the contractor forthwith, on demand in writing from the GE specifying the area affected not withstanding that the work executed might have been inadvertently passed, certified and paid. The contractor shall carry out the rectification of the defects notified at his own expense during the defects liability period and in event of his failure to do within a period to be specified by the GE in his demand, the GE shall undertake such rectification work at the risk and expense of the contractor.

(b) If work contains Revenue/Minor works or both alongwith Capital/Special repair/Low Budget work etc the defect liability period shall be **TWENTY FOUR CALENDER MONTHS** after the works have been handed over to the Govt. (hereinafter referred to as the "defects liability period") and Condition 49 of IAFW-2249 shall deemed to has been amended accordingly.

20. **PERFORMANCE SECURITY** :- Condition 19 of IAFW-2249 (amended vide Amendment No 47) amount of performance security deposit shall 5% of Contract Sum/Lump Sum.

21. The unit rate quoted by the contractor in Sch 'A' shall be deemed to include for minor extras, construction details or minor accessories which are not specified in particular specification but are essential for the execution of work in a workmanlike manner. In case of difference of opinion between the contractor and the GE as to whether or not certain items of work constitutes minor extras and construction details or minor accessories included in the contractor's quoted rates, the decision of the Accepting officer shall be final and binding.

SCHEDULE 'A' NOTES (Contd..)

22. Unless otherwise specified, quoted unit rates for all items of schedule "A" includes for all materials and labours. Contractor's special attention is drawn to the fact that all the provisions including use of materials or labours or both, specified in the Particular Specifications but specifically not included in the description of the relevant items of schedule "A" shall be deemed to be included in the unit rate quoted by the tenderer and nothing extra shall be admissible on this account.

23 CPM

23.1 The tender is based on CPM.

23.2 The tenderer is expected to be fully conversant with the technique and employ technical staff who can use the technique in sufficient details. Sufficient books and other literatures on the subject are widely available which the tenderer may make use of.

23.3 The tenderers attention is drawn to special condition of the tender regarding preparation of the detailed net work and time schedules as for the work and his liability for employing sufficient resources to adhere to this schedule. Any inability on the part of the tenderer in using the technique will be taken as his technical insufficiency and will affect his class of enlistment and future prospects for receiving tenders for the works.

24. The quantity of materials such as cement, steel, paints, water proofing compound, chemicals for anti-termite treatment and the like, as directed by Engineer-in-Charge (the quantity of which cannot be checked after incorporation in the works), shall be recorded in Measurement Books and signed by the contractor and the Engineer-in-Charge as a check to ensure that the required quantity has been brought to site for incorporation in the work.

25. All rubbish/mud/mulba/debris obtained from demolition/dismantling/taking down and roof treatment shall be thrown away outside MoD land as per guidelines of Govt of NCT (Delhi)/NGT without any extra cost to the department.

26. Cost of cutting/forming chases/forming chases/holes recesses in masonry/ concrete required for electrical work and making good the same in cement mortar 1:3 for filling upto 20mm and in PCC (1:2:4) type B-O for filling more than 20mm shall be deemed to be included in the unit rates quoted by the contractor in the Schedule 'A'.

27. Total amount quoted by the tenderer is not firm but shall be treated as contract sum as referred to in IAFW-2249.

28. Tenderers are required to get fully acquainted with scope of work as catered for in the tender documents and also the site of work. No claim shall be entrained subsequently for any misunderstanding or any inaccuracy with regard to site conditions and provisions made in tender documents.

29. The hard rock if any, obtained from excavation shall be properly stacked by the contractor at site of work. The method of measurements shall be laid down in clause 3.1 of SSR-2020 (Part-II). The hard rock obtained from excavation shall become the property of the Contractor, for which he will be charged at the rate of Rs.550.00 per Cum of stack measurement. The quantity of hard rock excavated shall be reduced by 50% to allow for voids. The measurement shall be properly recorded in MB and signed by Engr-in-Charge and Contractor. Nothing extra shall be paid for stacking and removal of such material from site work.

30 The department may exercise the right to call/correspond with the manufacturers/suppliers for the verification of the quality and quantity of the materials/items as per his discretion.

31. In case make of any item is not specified in the tender documents, these shall be of the standard make conforming to relevant IS and duly ISI marked. If ISI marked materials/items are not available in the market, these shall be at least manufactured by an ISO company.

32. All material obtained during dismantling/demolition except those catered for in Schedule of Credit shall become the property of the Govt as such the same shall be deposited by the Contractor in MES Store Yard all as directed by Engineer-in-Charge. The rates quoted by the tenderer in Sch 'A' shall be deemed to include for this aspect and nothing extra shall be payable on this account.

33. Contractor shall necessarily use steel scaffolding upto required height of building as approved by GE for guniting, external plaster, repair to structures such as chajjas, beams, columns, slabs, staircase, walls etc. and as directed by GE.

34. Barricading and view cutter to be fixed immediately after demolition work by the contractor at his own cost. Quoted rates shall include all expenditure to be incurred for following the NGT guidelines and no extra payment shall be paid in this regard. Contractor will be solely responsible for following up the guidelines passed by Govt. Agencies which are to be abided by MES for execution of work.

SCHEDULE 'A' NOTES (Contd..)

35. Damage caused to the pipe lines, water lines, cable or any other underground entity during execution if any to NDMC, IGL and other Govt/Pvt Depts to be borne by Contractor.

36. There may be VIP movement in the area causing temporary hindrances but contractor shall have no additional claim on this account.

37. The contractor shall ensure that the labour employed on the site have no criminal background and are in possession of Aadhar card/Govt recoganised identity card. Contractor shall also arrange security passes for his labour with the help of Engr-in-Charge and shall arrange multiple gang of labour to complete the work in time.

38. The contractor shall also facilitate police verification of his labour where required at his own cost.

39. The contractor shall submit updated inventory of various articles such as fittings and fixtures such as wash basin, WC etc. before issuance of completion certificate.

40. The tenderers are requested to note that if any item/items is/are repeating in BOQ and tenderers has quoted different rate for the same item, the minimum rate shall be considered for acceptance & payment purpose.

41. Irrespective of details/procedure given elsewhere the following procedure will be following for submission and approval of samples :-

(a) Contractor will submit sample of materials well in advance (minimum one week before intended/ expected date of use) with a certificate that "samples submitted are as per specifications given in the contract as per terms and conditions of contract". Wherever required, GE may ask for submission of various test certificates as requested to ascertain the quality of material before approval of samples.

(b) Approval of samples by the GE/department does not absolve the responsibility of the contractor to provide material as per specifications given in the contract and hence,if noticed subsequently that material/ materials used in the work is not as per specification and notified to the contract even one year after defect liability period, it will be responsibility of the contractor to remove the item/ material alongwith associated material/items of work which may be required and provide material/item of correct specifications alongwith associated items of work. In case of failure on the part of contractor to rectify the defects, department/accepting officer reserves the right to initiate necessary administrative action (including downgrading of class/removal from approved list/blacklisting) as deemed fit.

42. It shall be ensured that the variation limit of 25% given for freakishly high and low item is strictly adhered to, in case of any unavoidable and inevitable variation due to site or any cognizable requirement, prior approval of Chief Engineer shall be obtained. No departure of any kind shall be acceptable on this account and shall be liable to strict action against the defaulters found responsible.

43. The work requiring use of sand shall be executed by incorporation of fine/coarse aggregate as per particular specification. In case of shortage or non-availability of the same approved manufactured sand/crushed stone sand shall be used with prior approved crushed store sand of Accepting Officer.

44. In case of repair of painting work, existing surface of finishes shall be completely removed and scrapping shall be done with mechanical means and thereafter the treatment all as specified shall be done to the wall i.e. applying putty primer and paint strictly as per manufacturers instruction.

45. It may be noted that the some of the work in BOQ shall be fixed in repair as such the same shall be made good with adjoining surfaces with same existing specification without any extra cost i.e. the cost element is deemed to be included in the respective items quoted rates.

46. Makes of certain item given in Sch 'A" & corresponding item given in approved list of makes i.e. Appendix 'A' both shall supplementary to each other. However in all other cases the order of precedence given in condition 6A of IAFW 2249, shall hold good. In case of non-availability of item with the manufactures/makes given in Sch 'A', then it shall be procured all as approved by Accepting Officer.

SCHEDULE 'A' NOTES (Contd..)

47. Existing floor tiles in qtrs to be covered with foam and tarpaulin sheets tp prevent damage during the execution of other works. Any damages to be made good in the same pattern by the contractor without any additional cost. Entire site of work to be covered with green cloth of adequate size to prevent the surrounding from dust/pollution.

48 The contractor shall take necessary dust control measures as per NGT guidelines. The rate quoted by him shall be deemed to be inclusive of arrangements required for taking control measures.

49. Contractor shall carry out water testing required for work as and when asked by Engineer-in-Charge/GE.

50. The contractor has to keep in mind while quoting the rate in the tender that all the necessary scaffolding for bldg shall be provided by the contractor for the respective items of Sch'A'/BOQ and no extra payment for the height of the bldg shall be paid by the department.

51. **EPF Code & UAN**: All contractors shall mandatorily be in possession of the EPF Code Number and UAN of employees working under their establishment under EPF & MP Act 1952. So far this tender is concern, before release of work order the contractor shall submit EPF Code Number to GE. The contractor shall also ensure that Sub Contractor if engaged by him shall also comply with provisions of EPF & MP Act 1952.

52. **Payment to Labour during execution work**: The payment to labour engaged by the contractor shall be made as per Wages Act. No payments to the contractor /firms will be made unless proof of payment of wages is submitted by the contractor/firm. In case payments are made through current coins or currency notes, certificate will be submitted by the contractor alongwith undertaking by the concern workmen asking for payments other than by cheques or by crediting in the bank account of the employee/workmen. Adherence to these instructions will be checked during various inspection. Being Principal Employer, the GE shall ensure that the contractor fully comply with above provisions. The documents held by the contractor shall be subject to scrutiny by law enforcement agencies. The contractor shall also provide proof of monthly payment details under EPF & ESI with account numbers alongwith details of regular payments which shall be verified by GE.

53. Irrespective of whatever specified elsewhere in tender documents, all the debris/waste materials produced out of above mentioned work shall be removed out of Cantonment land and disposed off at locations notified by the Delhi Cantonment Board for disposal of waste building materials within NCR area, without any extra cost to Department under own arrangement of contractors. The rates quoted by the contractors against the items catered here-in-above in the BOQ are inclusive of said provision.

54. In case makes are given against any of the item/ items of Sch 'A'/BOQ the same shall take precedence over whatever makes specified in list of makes and make shall be provided out of any of the make indicated therein as approved by GE.

55. In case of single make or make not available with list of make in the market for whatsoever reasons, the make shall be approved out of makes given in list of makes of Chief Engineer Delhi Zone any running contract as approved by GE. No claim whatsoever shall be admissible on this account.

56. On acceptance of contract, contractor has to register on the CCEP Portal so that e-MB system can be adopted for recording measurements for allowing payment under contract otherwise no payment shall be processed.

57. The following jobs have been clubbed under this tender which have been scattered under AOR of AGE B/R-II and AGE E/M (East) the contractor advised visit the site of work before quoting their rates with prior appointment with Engr-in-Charge and no any extra payment on account of any misunderstanding etc will be entertained by the department, irrespective of the fact whether they have actually visited the site or not:-

(a) Provn of three single offrs accn at Air Defence Officers Mess 8 Parade Road under GE (East) Delhi Cantt-10. **(DA/80/CW/2025-26).**

(b) Provn for minor work for addressing seepage & other connected items at AAD Offrs Mess in various place & Bldg under GE (East) Delhi Cantt-10 **(DA/15/MW/2025-26).**

Contractor

AGE (Contracts)
for Accepting Officer

SCHEDULE 'B'
LIST OF MATERIALS TO BE ISSUED TO THE CONTRACTOR
SEE CONDITION 10 OF IAFW-2249

Srl. No.	Particulars	Rates at which materials will be issued to the contractor		Place of issue of store	Remarks
		Unit	Rate		
1	2	3	4	5	6

-----NIL-----

AGE (Contracts)
Signature of Contractor for Accepting Officer

SCHEDULE 'C'

LIST OF TOOLS & PLANT (OTHER THAN TRANSPORT)
WHICH WILL BE HIRED TO THE CONTRACTOR

(See Conditions 14, 15, 34 and 35 of IAFW-2249)

Srl No	Quantity	Particulars	Details of MES Crew Supplied	Hire Changes per unit per working Day Rs	Stand by changes per unit per off Day Rs.	Place of Issue by Name	Remarks
1	2	3	4	5	6	7	8

-----NIL-----

AGE (Contracts)
Signature of Contractor for Accepting Officer

SCHEDULE 'D'

TRANSPORT TO BE HIRED TO THE CONTRACTOR

(See Conditions 16 and 35 of IAFW-2249)

Srl No	Quantity	Particulars unit per Working Day	Rate per unit per Working Day	Place of Issue by Name	Remarks
1	2	3	4	5	6

-----NIL-----

AGE (Contracts)
Signature of Contractor for Accepting Officer

TENDER

TO THE PRESIDENT OF INDIA

Having examined and perused for following documents: -

1. Specifications signed by AGE (Contracts)

2. Schedule 'A', 'B', 'C' and 'D' attached hereto.

3. MES Standard Schedule of Rates 2009 Part I (Specifications) and amendments Nos 1 to 3 for Part-I and SSR-2020 Part –II (Rate) and amendment Nos 1 to 122 for Part-II (here-in-after and in IAFW-2249 referred to as MES Schedule) together.

4. General conditions of contracts, IAFW-2249 (1989 Print) together with amendments 1 to 49 along with errata Nos 1 to 20.

5. WATER: - Refer Condition 31 of IAFW-2249 (General Conditions of Contracts). Water will be supplied by the MES shall be paid for by the contractor @ 3.75/Rs 1000.00 worth of work done.

Should the tender be accepted** I/We agree:-

*(a) That the sum of Rs…….(Rupees .....only) forwarded as earnest money shall either be retained on account of security deposit or refunded by the Government on receipt of an appropriate account as security deposit within the time specified in condition 22 of IAFW-2249.

(b) To execute all works referred to in the said documents upon the terms and conditions contained or referred to therein and as detailed in General Summary/BOQ and to carry out such deviations as may be ordered vide condition 7 of IAFW-2249 upto a maximum of **10% (Ten Percent)** and further agree to refer all disputes as required vide condition 70 of IAFW-2249 to the sole arbitration of a serving officer, having degree in Engineering or equivalent or having passed final/direct final examination of Sub Division II of Institution of Surveyors (India) recognized by the Govt. of India to be appointed by the Chief Engineer Delhi Zone, or in his absence the officer officiating as Chief Engineer if specifically delegated in writing by Engineer-in-Chief, Army Headquarters, New Delhi whose decision shall be final, conclusive and binding.

* To be deleted where not applicable

TENDER (Contd…/-)

Signature _____in the capacity of _____

Duly authorised to sign the tender for and on behalf of_____

_____(IN BLOCK LETTERS) DATED _____

Witness_____ Postal Address _____

Address_____ Telegraphic Address_____

_____ Telephone No _____

ACCEPTANCE

_____alterations have been made in these documents and as evidence that these alterations were made before the execution of the contract agreement; they have been initialed by contractor and AGE (Contracts). The said officer is hereby authorized to sign and initial on my behalf the documents forming part of this contract.

The above tender was accepted by me on behalf of the president of India, for lump sum of Rs._____(Rupees_____ _____Only) on _____day of_____2025.

Signature_____ dated this_____ day of_____)

Appointment:
Garrison Engineer (East)
Delhi Cantt -10
(For and on behalf of the President of India)

GENERAL CONDITION OF CONTRACTS

(IAFW-2249 – 1989 PRINT)

(ITEM RATE CONTRACT (IAFW-1779A)

(a) A copy of General conditions of contracts (IAFW-2249-1989) print together with amendments 1 to 49 along with errata Nos 1 to 20 has been supplied to me/us and is in my/our possession. I/we have read and understood the provisions contained in the aforesaid GENERAL CONDITION OF CONTRACTS before submission of this tender and I/we agree that I/we shall abide by the terms and conditions thereof as modified, if any elsewhere in these tender documents.

(b) It is hereby further agreed and declared by me/us that the General Conditions of contracts (IAFW-2249-1989 print) including Condition 70 thereof pertaining to settlement of dispute by arbitration containing 34 pages with errata 1 to 20 and amendments No 1 to 49 of IAFW-2249-1989 print contain 27 pages form part of these tender documents.

(c) My/Our signature hereunder as well as digitally signed BOQ is in token of my/our having accepted the aforesaid General Condition of contracts together with Errata and amendments (IAFW-2249-1989 Print) and the clause relating to Arbitration forming an integral part of this tender.

(The documents mentioned above can also be seen in the office of GE (East) Delhi Cantt-10 or in any other MES (CWE/GE) office during working hours.)

Contractor

AGE (Contracts)
for Accepting Officer

SCHEDULE OF MINIMUM FAIR WAGES

1. A copy of the latest Schedule of Minimum Wages has been supplied to me/us and is in my/our possession. I/ we have read and understood the provisions contained in the aforesaid Schedule of Minimum Wages before submission of the tender and I/we agree that I/we shall abide by the terms and conditions thereof, as modified, if any elsewhere in these tender documents.

2. Schedule of Minimum Fair Wages is not enclosed along with tender but the I/we deem to have full knowledge regarding the minimum wages payable to laborers as legally effective on date of submission of tender and this tendered rates shall be deemed to have been based on the same.

3. It is further agreed and declared by me/us that schedule of Minimum wages forms part of these tender documents. My/our signature hereunder as well as digitally signed BOQ amounts to my/our having signed the aforesaid documents forming part of tender.

Contractor

AGE (Contracts)
for Accepting Officer

SPECIAL CONDITIONS

1. General:

(a) The following special conditions shall be read in conjunction with General Conditions of contracts IAFW-2249 (1989 Print), including errata/amendments thereto. If there are any provisions in these special conditions, which are at variance with the provisions in the above-mentioned documents; the provisions in these special conditions shall be deemed to take precedence.

(b) General Rules, preambles, special conditions, method of measurements etc. specified in MES Standard Schedule of Rates shall be applicable unless specific provisions to the contrary are made in these tender documents.

2. Visit to Site (S) Of Work: The tenderers are advised to visit the site(s) of work with prior permission of GE and make themselves familiar with the local working conditions, accessibility to sites(s), availability of material and other relevant conditions which effect their pricing, workman like execution and entire completion of work. The tenderer shall be deemed to have visited the site(s) before tendering whether they actually visit or not.

3. Condition For Working In Restricted/ Unrestricted Area:

(a) The work lies in unrestricted area. However, the contractor, his agents, servants, work people and vehicles may pass through the unit lines, in which case the Engineer-in-Charge shall, at his discretion has the right to issue passes, control their admission to the site of work or any part thereof. The contractor shall on demand by the Engineer-in-Charge submit a list of personnel etc concerned and shall satisfy the Engineer-in-Charge as to the bonafide of such people. Passes shall be returned at any time on demand by the Engineer-in-Charge and in any case, on completion of work.

(b) The contractor and his work people shall observe all the rules promulgated by the authority controlling the area in which the work is to be carried out e.g. prohibition of smoking, lighting and fire precaution, restricted hours of working etc. Any person found violating the security rules laid down by the authority, shall be immediately expelled from the area without assigning any reasons whatsoever and the contractor shall have no claim on this account. Nothing shall be admissible for any man hours lost on this account.

OR

(c) **Condition for working in restricted area**: All works lie in Restricted Area. The restricted for entry to work site and conditions of working in restricted area shall be as under:-

(i) **Entry And Exit**:- The contractor/his agents/ representatives/workmen etc. and his materials, arts, trucks or other means of transport etc. will be allowed to enter through and leave from only such gate or gates and at such times as the GE authorities in charge of the restricted area may at their sole discretion permit to be used. Contractor's authorised representative is required to be present at the places of entry and exit for the purpose of identifying his carts, trucks etc. to the person in charge of the security of the restricted area.

(ii) **Identity Of Workmen**:- Every workman shall be in possession of an identity card. The identity card will be issued after thorough investigation of the antecedents of the labours by the contractor and attested by the Officer in Charge of the unit concerned in accordance with the standing rules and regulations of the unit. Contractor shall be responsible for the conduct and action of his workmen, agent or representative.

(iii) **Identity Card Or Passes**:- The contractor, his agents and representatives are required individually to be in possession of an identity card of pass which will be examined by the security staff at the time of entry into or exit from the restricted area and at any time or number of times inside a restricted area.

(d) **Search** :- Thorough search of all persons and transport shall be carried out at each gate and for as many times a gate is used for entry or exit. Search may also be carried out any time or any number of times at the work site within the restricted area.

SPECIAL CONDITIONS

(e) **Working Hours**:- The units controlling restricted areas usually work during six days in a week and remain closed on the 7th day. The working hours available to the contractor's labour and staff are however appreciably reduced because of the time of entry and exit during working hours. The exact working hours, working days and number of working days observed for these restricted area(s) where works are to be carried out shall be deemed to have been ascertained by the contractor before submitting his tender. The contractor's attention is invited to the fact that the number of working hours for a unit are prescribed in regulations and that these cannot be increased by the GE or authorities controlling the restricted area. The definition of "working days" as given under condition 1 (t) of IAFW-2249 does not apply in cases where the works are carried out in restricted areas.

(f) **Working On Holidays**:- The contractor shall not carry out any work in gazette holidays, weekly holidays and other non-working days except when he is specially authorized in writing to do so by the GE. The GE may at his sole discretion declare any day as holiday or non-working day without assigning any reasons for such declaration.

(g) **Fire Precautions**:- The contractor, his agents, representatives, workmen etc shall strictly observe the order pertaining to fire precautions prevailing within the restricted area. Motor transport vehicles, if allowed by the authorities to enter the restricted area, must be fitted with a fire extinguisher in working condition.

(h) **Female Searcher**: If the contractor desires to employ female labour on works to be carried out inside the Area of a factory, Depot, park etc and a female searcher is not borne on the authorised strength of the factory, Depot, Park etc at the time of submission of the tender he shall be deemed to have allowed in his tender for pay and allowances etc, for a female searcher (class IV servant /Gp 'D' servant) calculated for the period, female labour is employed by him inside that area. If more than one contractor employs female labour during any month and female searcher(s) has/have to be employed in addition to the authorised strength of the factory, Depot park etc, the salary and allowances paid to the additional female searcher(s) shall be distributed on an equitable basis between the contractors employing female labour taking into consideration the value and period of completion of their contracts. The GE's decision in regard to the amount recoverable on this account from any contractor shall be final and binding.

1. Contractor's Representatives/Agents And Workers: The contractor shall employ only Indian Nationals and verify their antecedents and loyalty before employing them on the work. He shall ensure that no person of doubtful antecedents and nationality is in any way associated with the work. As a proof that he has employed only Indian Nationals, the contractor shall render a certificate to this effect to GE within one month from the date of commencement of work. The GE shall have full power, and without giving any reasons to order the contractor immediately to cease to employ in connection with this contract any representatives, agent, servants and workmen or employees whose continued employment in his (GE's) opinion is undesirable. The contractor shall not be allowed any compensation on this account. Tenderer's attention is also drawn to condition 25 of IAFW-2249 in this connection. Verification of antecedents of the Contractor's representatives/ labour deployed at site in connection with execution of work under the Contract, as per security requirement of user unit/ installation shall be the responsibility of the Contractor and all expenses in connection with verification of antecedents by police authority/ security agency shall be born by the Contractor.

2. Hours Of Working:- The working hours shall be as notified by the authority controlling the area, which will be normally from 0800hours to 1700 hours. If, however, contractor wants to work beyond these normal hours he can do so if permitted by the GE in writing. Similarly no work will be allowed to be carried out on Sunday and gazetted holidays unless specifically permitted by the GE in writing.

3. Storage Of Materials, Workshop, Office And Accommodation For Labourers:

3.1 The contractor shall apply to Station Headquarters on prescribed Performa available in GE's office for storage purpose only. Site/ land measuring 20mx10m for storage shed shall be allotted for the duration of work but not exceeding one year, subject to approval of the same by the Station Headquarters. Contractor is required to give undertaking that he is not occupying any other temporary site for any other contract in Delhi Cantt-10 and site will be vacated immediately on termination of allotment period. Final payment to the contractor will be made after vacation of allotted site/ land. Same site/land once allotted will not be subjected to extension after expiry of allotment period but the contractor may apply for an alternative site, if required.

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3.2 Please Note that land shall be allotted only for storage purpose which depends upon availability and at the discretion of Station Headquarter. No claim whatsoever shall be admissible on account of non-allotment of land. However, the contractor shall make his own arrangement for storage in case of not allotment of land under the work. The land for purpose of accommodation of labour and erection of temporary workshop shall not be allotted. Contractor shall have to arrange the land for these purposes outside of defence land from his own resources and nothing extra shall be admissible on this account.

4. **Contractor's Plant/ Equipment At Site**: The contractor shall furnish the Engineer-in-Charge every morning a distribution return of the plant and/or equipment on the site of work required for incorporation in the work. The Engineer-in-charge shall record the particulars of plant/equipment supplied by the contractor in the works diary and send the return to the GE for record in his office.

5. **Use Of Existing Buildings Etc**: The contractor shall not use the existing latrines, baths and other existing buildings if any for use by his work people or for storage of his materials or for any other purpose unless permitted to do so in writing by Garrison Engineer.

6. **Water :** Refer condition 31 of General Conditions of contracts IAFW-2249. Water will be supplied by MES at the point(s) decided by the GE. As the water supply of MES is likely to be intermittent the contractor shall make his own arrangements for storing the water required for the works labour and workmen etc at his own expenses. Water charges will be recovered at the rate of Rs 3.75 per Rs 1000/- worth of work done priced at contract rates. No claim/ whatsoever shall be admissible on account of compensation for intermittent /insufficient water supply by the MES. However, any subsequent change in rate of recovery of water charges shall be deemed to be included in CA.

7. **Electricity** : Electric supply required for the work upto a maximum of 20KVA single phase 50 Hz shall be made available by the MES at points decided by GE. KWH meters to register the electric energy supplied and main switch shall be provided and installed by the MES. Contractor shall provide all necessary cables, fittings etc from the tapping point in order to ensure a proper and suitable supply of electricity for execution of work. The contractor will be charged for the electric energy consumed for execution of works at the rates of Rs 9 (Rupees Nine only) per unit for lighting and power. MES do not guarantee continuity of supply and no compensation whatsoever shall be allowed for supply becoming intermittent or for breakdown in the system. In case GE or his representative is not satisfied with device or electric lines provided by the contractor, shall be disconnected from the supply if so directed by him.

11. **Net Work Analysis (CPM)/Project Planning**

11.1 The time and Progress Chart to be prepared as per Condition 11 of General Conditions of contracts (IAFW-2249) shall consist of detailed network analysis and a time schedule. Contractor in consultation with GE soon after acceptance of contract shall prepare the entire projects in MS Project clearly defining Stages of completion, Resources, Constraints, and Activities involved. The entire project shall be divided into sub projects consisting of different buildings, and these sub projects shall be sub divided into further stages of completion e.g earthwork, foundation etc. The resources shall consist of following broad heads viz. Materials, Manpower, Equipment. These heads shall be further sub divided into details eg. Material shall consist of cement, steel, 12.5 mm aggregate, 20 mm aggregate, bricks etc. Man power shall include Engineer, Supervisors, Mason, Carpenter, Mazdoor, Equipment shall consist of Concrete mixer, Shuttering material, Trucks etc. The output of the resources shall be judicially decided keeping in view the realistic output under the circumstances. The critical path net work will then be derived from the project jointly by the GE and the contractor. The time scheduling of the activities will be done by the contractor so as to finish the work within the stipulated time. On completion of the time schedule, a firm calendar date schedule will be prepared and submitted by the contractor to the GE who will approve it after the due scrutiny. The schedule will be submitted in four copies within two weeks from the date of handing over the site.

11.2 During the currency of the work, the contractor is expected to adhere to the time schedule and this adherence will be apart of the contractor's performance under the Contract. During the execution of the work the contractor is expected to review and update the net work finalized in consultation with GE The contractor shall submit original CPM chart within 30 days of commencement of work. The delays period not attributable to contractor for grant of extension under condition 11 of IAFW 2249/updation of CPM, shall be considered as an activity preceding the succeeding activity, the CPM shall be considered revised CPM. If contractor fails to submit CPM chart original and revised no extension of time shall be granted. If the contractor fails to adhere to revised CPM chart, the contractor shall be put under slow progress in his WLR which may result to nonissue of further tender to the contractor till the contractor achieves desired progress as per revised CPM.

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11.3 These reviews may also be undertaken at the discretion of GE either as periodic appraisal measure or when the quantum of work ordered on the contractor is substantially changed through deviation orders or amendments. Any revision of the schedule as a result of the review will be submitted by the contractor to the GE within a week who will approve it after due scrutiny. The contractor will adhere to the revised time schedule thereafter. In case of GE disagreeing with the revised time Schedule, the same will be referred to the Accepting Officer whose decision shall be final, conclusive and binding. GE's approval to the revised schedule resulting in a completion date beyond the stipulated date of completion shall not automatically amount to a grant of extension of time. Extension of time shall be considered and decided by the appropriate authority mentioned in Condition 11 of IAFW-2249 and separately regulated.

11.4 The contractor shall mobilise and employ sufficient resources to achieve the detailed schedule within the broad frame work of the accepted methods of working and safety.

11.5 No additional payment will be made to the contractor for any multiple shift work or other incentive methods contemplated by him in his work schedule even though the time schedule is approved by the department.

12. **Material And Sample**

12.1 Refer Condition 10 of IAFW-2249.

12.2 The materials and articles, which have been specified from certain makers/manufacturers, shall be of makes/manufactures as specified in Appendix 'D' & 'E' here-in-after. If the manufacturers specified in tender documents make both ISI marked and conforming to ISI, the materials/articles shall be ISI marked.

12.3 The materials and articles, which have not been specified in tender documents by makes/manufacturers, shall be as under:

(i) If ISI marked materials are being manufactured the same shall be ISI marked. For list of ISI marked manufacturers refer website of BIS i.e. www.bis.org.in.

(ii) If ISI marked materials are not being manufactured, the same shall be conforming to IS specifications.

12.4 Materials of local origin shall be as specified and conforming to samples kept in GE's office. The tenderer is advised to inspect sample of materials which are displayed in the office of GE, before submitting his tender. The tenderer shall be deemed to have inspected the samples and satisfied himself as to the nature and quality of materials, he is required to incorporate in the work irrespective of whether he has actually inspected them or not.

12.5 The contractor shall not procure materials and articles unless the samples are first got approved from the Garrison Engineer.

13. **Damage To Existing Buildings, Roads And Drains Etc**: The contractor shall be responsible to reinstate and make good at his own expense, the damage if any caused by his workmen etc. to existing buildings, roads, drains, cables etc., while carrying out the work under this contract. Rectification, reinstatement, replacement, making good and touching up, etc, shall conform to the standard of workmanship and materials, originally used and the same shall be done to the entire satisfaction of Engineer-in-charge.

14. **Quarries And Royalty**: Reference condition 14 of IAFW-2249 General Conditions of contracts. Delete this condition in toto and substitute as "Neither quarries are available nor the quarrying shall be permitted on the defence land".

15. **Octroi And Other Duties** : The price quoted by the contractor shall be deemed to include all duties, taxes Octroi, sales tax, Service tax, surcharge, cess etc. The contractor shall not include any condition for fluctuation in price of labour and/or materials in his tender.

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16. **Approaches**: The contractor shall provide at his own cost all temporary approaches to the site (where not existing) for the use of his labour and transport of materials, tools and plant.

17. **Minimum Wages Payable To Labour & Contract Labour Regulations** : In terms of Minimum Fair Wages Act 1948 & Fair Wages Act, the contractor shall not pay wages lower than minimum wages for labour as fixed by the Govt. of India/State Govt. whichever is higher. The fair wage referred to in condition 58 of IAFW-2249 will be deemed to be the same as the minimum wages as referred to above. Condition 26 of IAFW 2249 shall be binding on contractor to the full extent the regulations mentioned therein and Appendix C to IAFW 2249. Statutory Labour Laws contained under, Contract Labour (regulation & Abolition) Act 1970, EPF & MP Act 1952, Payment of Wages Act 1936 shall be in force and binding as per law. The payment to labour engaged by the contractor shall be made as per Wages Act. No payments to the contractor /firms will be made unless proof of payment of wages is submitted by the contractor/firm. In case payments are made through current coins or currency notes, certificate will be submitted by the contractor along with undertaking by the concern workmen asking for payments other than by cheques or by crediting in the bank account of the employee/workmen. Adherence to these instructions will be checked during various inspection. Being Principal Employer, the GE shall ensure that the contractor fully comply with above provisions. The documents held by the contractor shall be subject to scrutiny by law enforcement agencies. The contractor shall also provide proof of monthly payment details under EPF & ESI with account numbers along with details of regular payments which shall be verified by GE

18. **Co-operation With Other Agencies** : The contractor shall permit free access and afford reasonable facilities to other Agencies or departmental workmen engaged by Govt. to carry out other work services (whether or not in connection with the work under the scope of this contract) on the site, simultaneously. The contractor shall not be allowed any extra payment on this account.

19. **Acceptable Quality Of Work And Finishes**: To determine the acceptable standard of workmanship, the Garrison Engineer may order the contractor to execute certain typical portion of works and services such as some length of wall, portion of different type of floors, plaster, painting and other finished/items, sanitary fittings, plumbing, electrical fittings etc under strict and constant supervision of MES staff and label these works as guiding sample so that further works shall be executed to confirm these samples.

20. **Time And Progress Chart** : The time and Progress Chart to be prepared as per condition 11 of General Conditions of Contracts (IAFW-2249) immediately after handing over the site shall consist of detailed network analysis and a time schedule. The critical path network shall be drawn jointly by the GE and the contractor soon after acceptance of the tender to complete the as per schedule. The Contractor shall mobilize and employ sufficient resources to achieve the detailed schedule within the broad frame work of the accepted methods of working and safety.

21. **Site For Execution Of Work**: Site for execution of work will be available as soon as the tender is accepted. In case, it is not possible for the department to make the entire site available at the commencement of work, the contractor will have to prepare the CPM net work as required in terms of Special Condition 21 to achieve the target. No claim whatsoever for not giving the entire site on commencement of work and for giving the site gradually will be tenable.

22. **Security Of Classified Documents**: Tenderers special attention is drawn to condition 2A & 3 of General Conditions of Contracts (IAFW-2249) for compliance. The contractor shall not communicate any classified information regarding the work either to sub contractor (s) or to any other person(s) without prior written approval of the Engineer-in-Charge.

23. **Record Of Consumption Of Cement**: The contractor shall maintain a properly bound register in the form approved by the Engineer-in-Charge showing the daily receipt of cement, quantity used in works and balance at the end of each day. This register shall be signed daily by the contractor's representative in token of correctness of the entries made and it shall be signed by JE every day and AGE once in week. The register shall be kept at site of work in the safe custody of the contractor. The register shall handed over to the Engineer-in-Charge for record with MES on completion of work.

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24. Cleaning Down: The contractor shall clean all floors, joinery, glass panes etc. and remove drops of cement/lime paint etc. and touch up all painter's work and carry out other items of work as are necessary to make and leave the premises clean and tidy before handing over the buildings and/or works.

The contractor shall not be allowed any extra payment on this account.

25. Validity Of Tender: The tender shall remain open for acceptance for a period of 60 days (Sixty days) commencing from the next day subsequent to last date of bid submission.

26. Proprietary Materials/ Articles And Record Of Materials: The proprietary materials/articles such as paint, bitumen, water proofing admixture, etc. when brought to site shall be inspected and approved by the Engineer-in-Charge. The quantity of all proprietary articles brought to site shall be recorded in measurement book (IAFW-2261) as 'NOT TO BE ABSTRACTED' and signed by the contractor and the Engineer-in-Charge. The contractor shall ensure that the materials are brought to site, in original sealed containers/ packing, bearing manufacturer's marking except in the case of requirement of material (s) being less than smallest packing. Original purchase vouchers/bills including manufacturer's test certificate (where applicable) in respect of the following item shall be provided by contractor to the Engineer-in-Charge along with RAR against claim for material lying at site raised in the RAR:-

Water proofing compound	GI pipes and fittings	Steel windows /ventilators
Paint, distemper & cement base paints	Sanitary fittings and fixtures	PVC door
Galvalume Sheet.	Tiles, paver blocks	Aluminium doors/windows/ventilators
All grades of bitumen	False ceiling	Factory made panelled door shutter
Electrical fittings, fixtures, cables, conductors, equipments, LT panels, Transformer, DG Set, APFC Panel, Servo Stabilizer	Steel (TMT and structural)	GI/DI pipes and fittings, RCC pipes
Cement	MCB/ELCB/RCCB/MCCB, exhaust/ceiling fans, DB, Street light fittings, LED	lamps/fittings, Aviation light, Tube light fittings
High mast light, Steel tubular poles, Lightening arrestor, Insulators	Sealing compound, Water proofing membrane	CHS equipments and hot water piping

27. Purchase Voucher: Contractor shall produce vouchers/invoices from the manufacturers and/or their authorized agents for the full quantity of the following materials as applicable as a pre-requisite before submitting claims for payment for advance on account of the work done and/or materials collected in accordance with Condition 64 of General Conditions of Contracts (IAFW-2249):-

(i) Cement, steel, paints, water proofing compound, chemicals for anti-termite treatment and the like.

(ii) APP membrane, Bituminous primer, Bitumen 85/25 grade.

(iii) Cables, HT/LT panel, Electrical fittings i.e. LED light fittings for road and street light (outdoor) and LED light panel and luminaire (indoor).

(iv) Geyser, Water cooler, Air cooler, Refrigerator, Transformer, DG Set, Solar water heating system etc or any other equipment as directed by GE.

The contractor shall, on demand, produce to the GE, and original receipted vouchers/invoices in respect of the materials other than as stated in condition above. Vouchers/invoices so produced and verified shall be stamped by Engineer-in-Charge indicating contract number. The contractor shall ensure that the materials are brought to site, in original sealed containers/packing, bearing manufacturer's marking except in the case of the requirement of materials(s) being less than smallest packing.

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28 ADVANCE ON ACCOUNT OF MATERIALS (FOR WORKS WHOSE ESTIMATED COST AT MARKET RATE EXCEEDS RS 50 LAKHS):

28.1 Refer to condition 64 of IAFW-2249- Advance on Account.

28.2 The contractor may be paid advance on account to the full value of the undermentioned materials only, brought on the site, on his furnishing Bank Guarantee Bond(s) from a Scheduled Bank for the amount of the retention money which should otherwise be recoverable from him under contract:-

(a) Factory made door shutters.

(b) Factory made steel windows/Vents.

(c) Sanitary fittings.

(d) Iron Mongery

(e) Electrical fittings/fixtures

(f) Water supply pipes and pipe fittings/fixtures

(g) Steel.

(h) Cement

(j) SGSW Pipes and Reinforced cement concrete pipes

28.3 The Bank Guarantee Bond(s) shall be executed on a Stamp Paper for a period and on a form as directed by the Accepting Officer. The contractor shall further arrange to extend the period of Guarantee Bond(s) if and when necessary, as directed by the Accepting Officer or shall furnish fresh Guarantee Bond(s) of similar value in lieu.

29. REFER SPECIAL CONDITIONS 6 & 8: The contractor shall maintain site/area of land/buildings in a clean and tidy conditions at his own expenses and shall deliver up the same on/or before the date of completion of the works, or on termination of the contract, or in the event of the said areas or land/buildings required by the GE within one month of an order to that effect, in a clean and tidy state complete in every particular and to the satisfaction of Engineer-in-Charge. In addition to completion of all other stipulations under condition 49 of IAFW-2249, the completion certificate under condition 49 of IAFW-2249 shall be issued only after vacation of site/area of land/building allotted to the contractor. Payment of final bill shall be made to the contractor only after such areas of land/building are vacated by the contractor.

30. OFFICIAL SECRET ACT: The contractor shall be bound by the Indian official Secret Act 1923 particularly Section 5 thereof, all as per condition 2A of IAFW-2249.

31. SPECIAL CONDITION FOR CONCILIATION: The scope of conciliation shall be as per condition 71 of IAFW 2249.

(a) **Appointment of conciliator :** Disputes (other than those for which the decision of the CWE or any other person is by the contract expressed to be fixed and binding) shall be referred to the sole conciliator viz serving Officer not below the rank of superintending Engineer/ Superintending Engineer (QS&C) appointed by Chief Engineer Delhi Zone, New Delhi or in his absence the Officer officiating as Chief engineer.

(b) **Status & Effect Of Settlement Agreement** :- The settlement agreement signed by the parties as a result of conciliation proceedings shall be the same status and effect as it is an arbitral award on agreed terms.

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32. QUALIFIED TRADESMEN (Applicable for estimated cost one crore and above work as per NIT): In compliance with the condition 26 of IAFW-2249(General Conditions of Contracts) the contractor shall employ skilled/semi skilled tradesmen who are qualified and possessing certificate in particular trade from Industrial Training Institute (ITI)/Similar reputed and recognized Institutes by State/Central government to execute the works of their respective trade. The number of such qualified trades men shall not be less than 25% of total skilled/semi skilled tradesmen required in each trade.

33. APPLICABILITY OF TAXES & RE-IMBURSEMENT/REFUND ON VARIATION IN "TAXES DIRECTLY RELATED TO CONTRACT VALUE" :-

(a) The rates quoted by the Contractor shall be deemed to be inclusive of all taxes and cesses viz GST, labour welfare cess etc, duties, Royalties, Octroi & other levies payable under the respective Statutes. No reimbursement/refund for variation in rates of taxes, cesses, duties, Royalties, Octroi & other levies, and/or imposition/abolition of any new/existing taxes, duties, Royalties, Octroi & other levies shall be made except as provided in sub para (b) here-in-below. These taxes shall be levied on gross amount of work:-

(i) The taxes which are levied by State/Central Govt at certain percentage rates of Contract Sum/Amount shall be termed as "taxes directly related to Contract value" such as ,GST, Labour Welfare Cess/tax and like but excluding Income Tax. The tendered rates shall be deemed to be inclusive of all "taxes directly related to Contract Value" with existing percentage rates as prevailing on last due date of submission of bid. Any increase in percentage rates of "taxes directly related to Contract value" withreference to prevailing rates on last due date for submission of bid shall be reimbursed to the Contractor and any decease in percentage rates of "taxes directly related to Contract value" with reference to prevailing rates on last due date for submission of bid shall be refunded by the Contractor to the Govt/deducted by the Govt from any payments due to the Contractor. Similarly imposition of any new "taxes directly related to Contract value" after the last due date for receipt of tenders shall be reimbursed to the Contractor and abolition of any "taxes directly related to Contract value" prevailing on last due date for receipt of tenders shall be refunded by the Contractor to the Govt/deducted by the Govt from the payments due to the Contractor.

(ii) The contractor shall, within a reasonable time of his becoming aware of variation in percentage rates and/or imposition of any further "taxes directly related to Contract value", give written notice thereof to the GE stating that the same is given pursuant to this Special Condition, together with all information relating there to which he may be in a position to supply. The Contractors shall submit the other documentary proof/information as the GE may require.

(iii) The Contractor shall, for the purpose of this condition keep such books of account and other documents as are necessary and shall allow inspection of the same by a duly authorized representative of Govt, and shall further, at the request of the GE furnish, verified in such a manner as the GE may require, any documents so kept and such other information as the GE may require.

(iv) Reimbursement for increase in percentage rates/imposition of "taxes directly related to Contract value" shall be made only if the Contractor necessarily & properly pays additional "taxes directly related to Contract value" to the Govt, without getting the same adjusted against any other tax liability or without getting the same refunded from the concerned Govt Authority and submits documentary proof for the same as the GE may require.

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34. **BLASTING** : Blasting is prohibited.

35. **EMPLOYESS PROVIDENT FUND SCHEME**: Under the GoI Employees Provident Fund Act/Scheme 1952, the contractor shall have a EPF Code No and all labours employed by him or sub-contractor shall be enrolled as member of Provident Fund and should be given Universal Account Number (UAN). A certificate to effect that all working labours employed for execution of work are registered for EPF and due distribution have been credited into their account, shall be furnished by contractor along with running advances or final bill.

36. **DAMAGE TO EXISTING WORK** : Any damage done to the existing structure during the execution of work shall be made good by the contractor at his own cost and the site of work left clean and tidy on completion. Rectification/reinstatement etc. shall conform to the standard of materials originally used in the work and finished work shall match with existing work in all respects to the entire satisfaction of the GE. In case of any dispute on this account the matter shall be referred to the CWE, whose decision in writing shall be final and binding.

37. **SITE LABORATORY AND TESTING OF MATERIALS**

(a) **A LEVEL TESTS FOR WORKS COSTING Rs 100 LAKHS AND ABOVE:-** The contractor shall set up site laboratory of minimum size 3m x 4m with proper ventilation and lighting for testing of materials for A level tests as tabulated below. The contractor shall arrange all equipment's/ machines at his own cost with prior approval of GE. The cost shall be included in the costs quoted by the contractor. The contractor shall employ a competent technical representative as approved by the GE for the purpose of testing and all such tests shall be carried out in the presence of Engineer-in-Charge. The successful test results thereof shall be recorded and signed jointly by the contractor and Engineer-in-Charge. The charges of these tests i.e. 'A' level tests carried out in site laboratory of the contractor shall not be recovered. However in case a test which was specified to be carried out in site laboratory and which could not be carried out in site laboratory due to some reason, such test shall be carried out in Zonal/Command lab/ any other laboratory approved by GE and the recovery will be made at the rates given in Appendix for tests carried out in Zonal/Command lab or otherwise at the actual rates in case the tests are carried out in other labs. Testing charges will be deducted from the RAR's irrespective of the tests results for such tests.

(b) **A LEVEL TESTS FOR WORKS COSTING UP TO Rs 100 LAKHS:-** The contractor may set up site laboratory at his option for works costing up to 100 lacks. The other stipulations will be same as specified in preceding Para (a). However in case the contractor has not set up the site laboratory and tests are carried out in Zonal or any other laboratory approved/ set up by GE, the recovery shall be made at the rates given in table below for tests carried out in Zonal lab or otherwise as per actual rates in case the tests are carried out in other labs. Testing charges will be deducted from the RAR's irrespective of the test results for such tests.

(c) **B & C LEVEL TESTS**: For tests of 'B' and 'C' level as indicated in Appendix, the contractor shall provide all facilities for testing of materials at Zonal/Command Test Lab at his own cost. The rates quoted by the contractor shall deemed to be inclusive of these tests. For 'C' level tests, actual cost of testing shall be borne by the contractor and for 'B' level test, recovery shall be made at the rate of various tests conducted in Zonal/Command test lab of MES as indicated in table below.

(d) The recoveries on account of testing charges wherever applicable shall be effected from the running account payments due to the contractor payable after completion of the respective test or whenever the test is due whichever is earlier.

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Table of test and rate of recovery

S No	Material	Test	Method of Testing	Frequency of tests			Level of test	Rate of test Rs	Remarks
1	2	3	4	5			6	7	8
1.	Bricks	(i) Compressive strength	IS-3495 (Part-II)	As per IS 5454 as given under			'A'	330	Checks for visual and dimensional characteristics shall also be carried out as per IS-5454
		(ii) Water absorption	IS-3495 (Part-II)	Lot size	Sample Size	Permissible Nos of Defective Bricks	,A,	330	
		(iii) Efflorescence	IS-3495 (Part-I)	1001 to 10000	5	0	'A'	330	
				10001 to 35000	10	0			
				35001 to 50000	15	1			
2.	Coarse aggregate	(i) Sieve analysis	IS-2386 (Part-I)	One test for every 15 Cum of aggregate or part thereof brought to site			'A'	660	
		(ii)Flakiness Index	-do-	-do-			'A'	250	
		(iii) Estimation of deleterious materials	-do-	One test for every 100 Cum of aggregate or part thereof.			'A'	600	
		(iv) Organic impurities	-do-	One test per source of supply			'B'	275	
		(v) Moisture content	-do-	Regularly as required.			'A'	330	
		(vi) Specific Gravity.	--do— (Part-II)	One test for each source of supply			'B'	330	
3.	Fine aggregate	Sieve analysis	IS-2386 (Part-I)	One test for every 15 Cum of FA or part when brought to site			'A'	660	

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		(ii) Test for clay, silt and impurities	IS-2386 (Part-II)	--do--	'A'	500	
		(iii) Specific gravity.	-do-	One for each source of supply	'B'	330	
		(iv) Moisture content.	-do-	Regularly as required subject to 2 tests/day when being used.	'A'	330	
		(v) Tes for organic impurities	-do-	One test for each source of supply	'C'	275	
4.	Cement	(i) Setting time	IS-4031-63 AFFIRMED 1980	One for each consignment or as and when required	'B'	500	
		(ii) Soundness	-do-	-do-	'B'	550	
		(iii) Compressive Strength	-do-	-do-	'B'	550	
		(iv) Fineness	-do-	-do-	'B'	275	
5.	Structural Concrete (M-25 and Grade above)	(i) Slump Test (OR)factor compacting test (OR) Vee-Bee time	IS-1199	The minimum frequency of sampling of concrete of each grade shall be as under :-	'A'	300	Random sampling shall be carried out to cover all mid units.
		(ii) Compressive Strength	IS-516	Qty of concrete in the work (Cum) / No of sample			Refer IS 456-2000 clause 14 for frequency of sampling.
				1-5 / 1	'A'	300	
				6-15 / 2	'A'	300	
				16-30 / 3	'A'	300	
				31-50 / 4			
				51 and above 4 +1 for each additional 50 Cum or part thereof			
6.	(a) PCC blocks for walling (Hollow block)	(i) Compressive strength	IS 2156-1984 (Appx B)	8 Blocks out of 14	'A'	900	Sample : 14 blocks from consignment of every 5000 block or part thereof.
		(ii) Water absorption	do (Appx 'E')	3 Blocks out of 14	'B'	330	
		(iii) Density	Do (Appx 'A')	-do-	'B'	330	

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	(b) PCC solid block for walling	(i) Compressive strength	-do-	12 Block out of 18	'A'	900	Sample : 18 blocks from consignment of every 1000 block or part thereof.
		(ii) Water absorption	-do-	03 Block out of 18	'B'	330	
		(iii) Density	-do-	03 Block out of 18	'B'	330	These blocks to be checked for dimension and weight.
7	Cement flooring tiles/Tarrazzo tiles	(i) Water absorption	IS-1237 (Appx-'D')	06 tiles out of 18	'B'	330	Sample :18 tiles form each source of supply selected at random
		(ii) Wet transverse strength	IS-1237 (Appx-E)	06 tiles out of 18	'B'	660	
		(iii) Resistance to wear	IS-1237 (Appx 'F')	06 tiles out of 18	'C'	1000	
8	Burnt clay roofing tiles 9handmade) as per IS-2690 (Part-II)	(i) Water absorption	Is-3495 (Part-II)	06 tiles out of 12	'B'	216	Sample ;12 tiles from each source of supply selected at random.
	Length 150 mm to 250 mm Width 100 mm to 200 mm Thickness 25 mm to 50 mm	ii) Compressive strength	Do (Part-I)	-do-	'A'	180	
9	Mangalore pattern roofing tiles	(i) Water absorption	IS-654 (Appx-'A')	06 tiles out of 32	'B'	180	Sample ; 32 tiles from each consignment of 3000 tiles of part thereof. These tiles shall be checked for dimension
		(ii) Breaking load	IS-654 (Appx-'C')	06 tiles out of 32	'B'	120	

SPECIAL CONDITIONS

10	Timber	(i) Specific gravity and weight	IS-1708	Minimum 03 sample a lot of 04 CUM or 250 pieces of seasoned timber.	'B'	120	
		(ii) Moisture content	-do-	-do-	'A'	120	
11	Water for construction purpose	(a) Test for acidity	IS-456 & 3015	Once at the stage approval of source water	'B'	500	Also refer clause 4.3 of IS : 456 & its subsequent sub clauses regarding suitability of water
		(b) Test for alkalinity	-do-	-do-	'B'	500	
		(iii) Test for solid content	-do-	-do-	'C'	500	
12	Welding of steel work	Visual Inspection test	IS-822 Clause 7.1	100% by visual inspection	Work site	360	Specials tests, their method and frequency to be decided on consideration of their importance by the Accepting Officer.
13	Timber paneled and glazed door/wooden shutters (including factory made shutters)	(a) Dimensions sizes, workmanship and finish	IS 1003 (Part –I)	Frequency of sampling form each lot shall be as under	'A'	180	
				Lot Size / Sample Size 26-50 / 5 51-100 / 8 101-150 / 13 151-300 / 20 301-500 / 32 501-1000 / 50 1001 and above / 80			
		(b) Strength test (i) Slamming	IS-1303	From each lot 5% of the Factory made shutters shall Be manufacture tested for strength tests.			

SPECIAL CONDITIONS

		(ii)Impact Indentation	-do-				
		(iii) Shock Resistance	-do-				
		(iv) Edge Loading	-do-				
14	Ply wood 9IS:303)	Moisture content	IS:41734 (Part-I)	Six test pieces cut from each of the boards selected a s per table shall be subjected to tests.	'C'	240	Sampling shall be as per IS:7835-tables.
15	Wood particle board (Medium density IS-3097)	(a) Density	IS 2360 (Part-III)	Three test specimens from sample (Size 150x75 mm)	'A'	60	Sampling shall be as per IS 3487-with moisture meter
		(b) Moisture content	-do-	-do-	'A'&'B'	60	
		(c) Water absorption	Do Part-16	Do (size 300x300mm)	'A'	60	
		(d) Swelling due to surface absorption	Do Part-17	Do (Size 125 x 100mm)	'A'	60	
		(e) Swelling in water	-do-	Do (Size 200 x100mm)	'A'	60	
		(f) Modulus of rupture	-do- Part-04	Three test specimen as per IS;2380	'B'	90	
		(g) Screw withdrawal strength	-do- Part-04	Do as pe IS:2385	'C'	120	
16	Reinforcement Steel	i)	Physical tests upto 16 mm dia (Normal mass, tensile elongation, bend and rebend)		'B'	2500	
		ii)	More than 16 mm dia		'B'	2750	

Legend : A :Site Lab, B: Zonal Lab/CTL, C: Approved Test House / Engineering College.

Note :- (i) All 'B' level test shall be conducted in CTL in case there is no functional Zonal Lab in the adjoining area, with testing charges to be recovered as per contract conditions.

(ii) 'C' level test for which facilities are available in CTL shall be conducted in CTL with testing charges to be recovered from contractors as per Contract Conditions Balance 'C' level test shall continue to be conducted in SEMT/Govt Approved Lab/Lab of Govt Engg. College, as the case be under contractor's own arrangement and actual cost of testing shall be borne by the contractor.

(iii) Various random/independent testing for material/products required to be conduct in addition to routine tests as stipulated in contracts, be referred to CTL to get verification of quality of material/product.

Contractor

AGE (Contracts)
for Accepting Officer

(PARTICULAR SPECIFICATIONS)

1. GENERAL

(a) Work under this contract shall be carried out in accordance with Schedule `A' (BOQ), Particular Specifications, drawings if any, General Specifications and other provisions contained in MES Schedule Part-I (2009) and Part-II (2020) read in conjunction with each other. Where specifications for any item of work are not given in MES Schedule or in these particular specifications, specifications as given in relevant Indian Standard Code of Practice shall be followed. In addition to makes indicated in the BOQ, a list of manufacturer is enclosed at Appx 'A' to this PS. The contractor shall procure the material from the manufacturer indicated therein.

(b) **MATERIALS:** Unless specific makes/manufacturers are specified in the tender documents all the materials to be procured by the contractor for incorporation in the work under this shall be with ISI certifications mark. For materials of specific make/manufacturer's names refer Appendix enclosed with tender here-in-after. ISI marking shall not be mandatory for such items where makes/ manufacturers have been specified unless the items of makes/manufacturers specified are available both with IS marking and without ISI marking. If any material is not manufactured with IS certifications mark in the country, it shall be confirming to relevant IS and other specifications specified elsewhere and shall also conform to samples displayed in CE's/CWE's/GE's office as applicable. The makes specified in Schedule 'A' (BOQ) shall take precedence over the makes specified in Appendix to these particular specifications.

(c) **LOCAL MATERIALS:** Irrespective of actual distance involved, local materials such as stones, aggregates, sand, road metals etc shall conform to or superior to the specifications given here in after and to the samples kept in the office of CEs/CWEs/GEs. The tenderers are advised to inspect these samples. The tenderers shall be deemed to have inspected the samples and have full knowledge thereof whether they inspect them or not before quoting their tender.

(d) **SAMPLES OF MATERIALS:** Refer condition 10 of IAFW-2249. Specific requirements regarding dimensions, strength, weight and finishes, as per IS, MES SSR and the particular specifications given hereinafter vis-à-vis actual properties check, tests carried out, reference to test certificates and markings, etc based on which samples of each materials are approved as Conforming to relevant specification shall be recorded in the sample approval register.

(e) **MATERIAL PROCUREMENT:** The contractor shall not procure materials unless the samples are first got approved from the Garrison Engineer. All material costing more than Rs 1.00 Lakh shall be procured directly from the manufacturer. The contractor shall ensure that the materials incorporated in the work are identical with the approved samples. If, the same is not identical with the approved samples, the same shall be replaced by the contractor to the entire satisfaction of GE without any extra cost of the Govt. All material incorporated shall comply with relevant Indian Standard.

(f) **RECORD OF MATERIALS:** Refer Special condition 27. The quantity of all the proprietary materials shall be recorded in measurement books and signed by the contractor and the Engineer-in-Charge.The contractor shall procure all the materials (where specific makes/manufacturer's specified) directly from their manufacturer's or from their authorised dealers only. The contractor shall produce to the GE original printed and machine numbered purchase vouchers/invoices including manufacturer's test certificate for all the materials mentioned here-in-before in Special Conditions.

(g) **STANDARD OF WORKMANSHIP**. The workmanship shall be as per standard engineering practices and as specifies and directed. All relevant Indian Standard shall be applicable to entire CA items and complied with.

(h) **SCOPE OF WORK**. The scope of work under this contract comprises of full, final and entire completion of works and services described in different parts of schedule `A' (BOQ) and in Sch 'A' notes as specified in these particular specifications & general specifications.

(j) **PREPARATORY WORK**. Before commencing the construction, contractor shall carry out the preparatory work such as site clearance, construction of storage accommodation etc to the entire satisfaction of Engineer-in-Charge. The cost of such work as may be necessary shall be deemed to be included in the contractor's quoted lump sum.

(k) In case certain items in the scope of work/ BoQ are not having corresponding makes list as per Appendix 'A'. Accepting Officer shall decide on the makes to be incorporated for such items. The makes so selected shall be ISI marked and in case there is no BIS certification system/ Indian. Standard for such materials / products. These shall be certified by accredited (NABL) laboratories to conform to other slandered of repute, like British Standard / EU Standard / American Standard

(PARTICULAR SPECIFICATIONS(Contd.....)

2. EXCAVATION AND EARTH WORK:

(a) **EXCAVATION**: Unit rates for items of Schedule 'A' Part-I shall include for excavation and earth work in soft soil. If rock (soft/hard/disintegrated) is met at site, contractor shall immediately notify the fact to the GE in writing, who shall after due verification, regularize the change through a proper deviation order.

(b) **FILLING IN TRENCHES/UNDER FLOORS:** The approved earth obtained from excavation in foundations shall be used for filling in trenches, under floors and any other situation. Filling around pipes after the pipes are laid and tested shall be earth-free from pebbles/stones. Filling under floors/sides of trenches shall be in layers not exceeding 250mm thick, and each layer shall be watered and well rammed. Surplus soil shall be removed and spread at places as directed by the Engineer-in-Charge, to a distance exceeding 250 meters but not exceeding 500 meters, and leveled as directed by Engineer-in-Charge.

(c) **TRENCHES FOR FOUNDATIONS AND PIPES :**The excavation shall be restricted to dimensions shown on the drawings. Excavation made, if any, in excess of required depth/width shall be made good by the contractor with cement concrete mix (1:5:10) without extra cost to the Govt.

(d) **BAILING/PUMPING OF WATER:** All water that may accumulate in trenches of foundation, column footings, rafts etc, during the progress of work from rains, sub soil water, spring etc shall be bailed/pumped out or other wise removed all as per clause 3.17 of MES SSR Part-I and nothing extra.

(e) **HARD CORE:** Unless otherwise specified, the hard core shall be of hard broken stone or boulders broken to gauge not exceeding 63mm. The material of hard core shall be well graded for providing dense and compact sub grade. Hard core shall be deposited, spread and leveled in layers not exceeding 15cm thick and watered and well rammed to a true surface and compacted with sufficient fine material. The thickness of hard core specified or indicated in the drawing, is the thickness after consolidation. Where thickness has not been mentioned it shall be 150mm consolidated.

3. ANTI TERMITE TREATMENT:

a) Pre-construction and post construction anti termite treatment shall be done as per Para 3.26 & 3.31 respectively of MES Schedule Part-I and IS: 6313 Part-1 and Part-2:2013/2016(latest).

b) Anti-termite treatment shall be got done through approved specialist agency which is registered by the Pest Control Association of India (PCAI) / Indian Pest Control Association (IPCA). Persons employed to do the anti-termite treatment in this work shall be qualified as per rule 10 of the rules framed under the inspection rule 1971. The specialist agency may be any one from list approved by E-in-C Branch/Headquarter Chief Engineer Western Command/ Headquarter Chief Engineer Delhi Zone (Delhi Cantt), which are located and active in the area with prior approval of GE.

c) Chemical for ATT shall be Chloropyriphos EC 20% (as per IS: 8944-2005) or imidacloprid 30.50 SC (as per IS: 16131) all as specified in Table 1 on page 2 of IS 6313 (Part 2) : 2016. The chemical shall be brought in original sealed containers & should be purchased from valid BIS license holder (CML) or their authorized agents and any other firm manufacturing ISI marked chemical after taking written approval from GE .The Contractor shall submit test certificates of the chemical from manufacturers. The contractor shall be required to get the chemical tested from any laboratory approved by GE, if so desired by him for its properties and the cost of chemical for testing, testing charges and any other charges such as transportation charges etc. shall be borne by the contractor irrespective of its test results.

3.4 Concentration of the chemical as emulsifiable concentration is indicated on the sealed containers. For obtaining specified concentration, chemical shall be diluted with water in the required quantity before it is used. Graduated containers shall be used for the dilution of the chemical.

(PARTICULAR SPECIFICATIONS(Contd.....)

3.5 **GUARANTEE FOR ANTITERMITE TREATMENT:-**

3.5.1 Should the GE at any time during construction or reconstruction or prior to the expiration of period of Seven years after the certified date of completion of work as per contract finds that buildings have been infested with termites, the contractor shall on demand in writing from the GE, specifying the building(s) complained or not with standing the same may have been in advertantly passed, certified and paid for forthwith under take to carry out such treatment as may be necessary to render (the said building) free from termite infestation at his own expense for period of ten years from the certified date of completion and in the event of failing to do so with in a period to be specified by the GE in his demand aforesaid, the GE shall under take such treatment at the risk and expenses in respect of the contractor. The liability of the contractor under this condition shall not extend beyond the period of ten years from the certified date of completion unless the GE has previously given notice to the contractor. Conditions of contract (IAFW 2249) shall be deemed to be amended to the extend as mentioned above.

3.5.2 An amount of security deposit equal to 2.50% (Two point five zero percent) on the amount of anti termite treatment at contract rates for the buildings for which treatment to be carried out shall be retained out of the final bill amount towards security deposit for ant termite treatment and will be refunded to the contractor after expiry of the period of seven years from the certified date of completion by the GE provided there are no termite infestation in the buildings. Conditions 10,46 and 68 of General Conditions of Contracts (IAFW 2249) shall be deemed be to amended to the extent mentioned above.

4. **CONCRETE WORK**

4.1 **MATERIALS FOR CONCRETE** :-

(a) **LIME** Lime shall be eminently hydraulic lime (Class 'A') and shall conform to IS : 712 as specified in clause 4.2 of SSR (Part I).

(b) **CEMENT** Cement shall not be issued under Sch 'B'. The contractor shall procure cement from manufacturer as specified here-in-after. Cement shall be stored in dry a place covered all round and no stacks shall be more than 15 bags high.

(c) **COARSE AGGREGATE** : Coarse aggregate shall be crushed stone aggregate and shall Conform to specifications given in clause 4.4 on Srl page 52 of SSR (Part I). However in case of controlled quality concrete the provisions made in IS-456 shall be applicable. Where due to less thickness of concrete the use of 20mm graded is not possible, the aggregate shall be of 12.5mm grading as specified in clause 4.4.7.1 on Srl Page 53 of MES Sch (Part I) without any price adjustment.

4.2 **Cement** :- Cement to be used in all types of works shall be ordinary Portland cement Grade 43 conforming to IS: 8112-1989 (Latest Revision) or Portland Pozzollana cement confirming to IS 1489-1991. The Contractor shall procure cement for all sections of Schedule 'A' required for this contract as the work proceeds at site.

4.2.1 Cement shall be procured by the contractor from the main producers of cement enumerated as under:-

Srl No.	Company Name	Address	Type of Cement
1.	The Associated Cement Companies Ltd Brand "**ACC**"	414-421, Splendor Forum (4th Floor), 3, District Centre, Jasola, New Delhi-110044	All
2	M/s Ultra tech Cement Ltd Brand:" **ULTRATECH**	'B' Wing 2nd Floor Mahakali Caves Road, Andheri (East), Mumbai-400093 Ph- 022-66917800	All
3.	The India Cement	Dhun Building, 827, Anna Salai, Chennai-600002	All
4.	M/s Dalmia Cement (Bharat Ltd) Brand" **DALMIA**" "**KONARK**"	Dalmiapuram Distt-Truchirappalli, Tamil Nadu-621651	All

(PARTICULAR SPECIFICATIONS(Contd.....)

5.	Century Cements Brand "**Century**"	Industry House, 159 Chruch gate Reclamation, Mumbai-400020 Ph-022-22023936	All
6.	Saurashtra Cement Brand " **SAURASHTRA**"	Gala No. A-1 Ground Floor, Udhyog Sadan No 3 MIDC, Central Road, Andherei (East), Mumbai-400093 Ph-022 32955557/67	All
7.	The Ramco Cements Ltd (formerly Madras Cement)	Auras Corporate Centre, 98-A, Dr Radhakrishnan Salai, Mylapore, Chennai-600004, Ph-044-28478666	All
8.	Mangalam Cement Ltd. Brand "**MANGLAM**"	PO Adityanagar,Morak,Dist-Kota, Rajasthan-326520 Mob-9351468076	All
9.	Birla Corporation Ltd Brand "**BIRLA**"	Birla Building (3rd & 4th Floor) 9/1 R.N Mukherjee Road Kolkata-700001 Ph-033-30573700	All
10	Orient Cement Brand "**ORIENT**"	5-9-22/57/D, 2nd and 3rd Floor, GP Birla Centre, Adarsh Nagar, Hyderabad-500063 Ph-04423688600	All
11.	Nuvoco Vistas Corporation Ltd(Formerly Lafarge Cement) Brand **"NUVOCO"**	Equniox Business park, Tower-3, East Wing, 4th Floor, LBS Marg, Kurla (west), Kurla, Mumbai, Maharashtra-400070	All
12.	Shree Cement Brand "**SHREE**"	Bangur Nagar, Beawar, Distt-Ajmer, Rajasthan-305901 Ph-01462228101/06	All
13.	JK Cement Brand "**JK**"	Kamla Tower, Kanpur-208001	All
14.	JK Lakshmi Cement Ltd Brand "**JK LAKSHMI**"	Jaykaypuram, Dist-Sirohi, Rajashthan Ph-02971 244409/10	All
15.	Jaypee Rewa Cement Brand "**JAYPEE**"	Jaypee nagar, P.O Jaypee Nagar, Rewa-486450	All
16.	Ambuja Cement Ltd. Brand "**AMBUJA**"	Kodinar, PO-Ambujanagar, Taluka-Kodinar, Distt-Junagadh, Gujrat-362715 Ph-02795237000	All
17.	M/s Sagar Cements Limited Hyderabad, Brnad "**SAGAR**"	M/s Sagar Cements Limited Plot No 111, Road No 10, Jubli Hills, Hyderabad-500033 Tele-+91-40-23351571, 2335672. Fax-+91-402335673 Email:info@sagarcements.in Website: www.sagarcements.in	(a) OPC 43 (b) OPC 53 (c) PPC (d) PSC (Valid upto 24 Aug 25)
18.	M/s Chettinad Cement Corporation Ltd. Brand "**CHETTINAD CEMENT**"	4th Floor, Rani Seethai hall Building, 603, Anna Salai Hall, Chennai-600006 Tele No. 044-42951800, Fax No. 044-28291558 Email: info@chettinalcement.com Website : www.chettinadcement.com	(a) OPC 43 (b) PPC (Valid upto 13 May 27)
19.	M/s My Home Industries Ltd. Brand "**MAHA CEMENT**"	9th Floor, Block-3, My Home Hub, Madhapur, Hyderabad-500081, Telangana, Ph-040-66929696 Fax-+91-40-23118000 Email: corp@myhomegroup.in Website : www.myhomeindustries.com	(a) OPC 43 (b) PPC (c) PSC (Valid upto 13 Oct 27)
20.	M/s Wonder Cement Ltd Brand : **"WONDER CEMENT"**	17, Old Fatehpura, Seva Mandir Road, Udaipur-313004, Rajasthan (India), Tele:- 91-294-33991133, Fax-91-294-3006333, Email : corp.office@wondercement.com Website: www.wondercement.com	(a) OPC 43 (b) OPC 53 (c) PPC (Valid upto 19 Jan 30)

(PARTICULAR SPECIFICATIONS(Contd.....)

The particulars of the manufacturer / supplier of cement alongwith the date of manufacture shall be produced by the contractor for every lot of cement separately. The documents in support of the purchases of cement shall be produced before the Engr-in-Charge for verification by the GE.

4.2.1.1 The following conditions shall have to be satisfied before PPC can be allowed to be used in the work:-

(a) PPC should meet the strength criteria as per IS-8112-1989.

(b) The minimum period before striking off formwork while using PPC shall be as given in PS Clause 3.8 hereinafter.

(c) In cold climate regions where temperature is lower than 15 ^{o} C only OPC shall be used.

(d) Mixing of OPC & PPC shall not be allowed in a work. However with the strict control of ground executives different buildings can have different types of cement.

(e) While procuring PPC, the following requirements are to be ensured and certificate to the effect is to be obtained by the contractor for each batch from the manufacturer and the same shall be submitted to the GE for approval to procure PPC.

(i) The quality of fly ash is strictly as per IS-1489 (Part-I) – 2002 Fly ash is inter-ground with clinker and not mixed with cement.

(ii) Dry fly ash is transported in closed containers and stored in Silos. Only pneumatic pumping should be used.

(iii) The fly ash received from thermal power plants using high temperature combustion above 1000°C should be used.

4.2.2 **TESTING:** The contractor shall submit the manufacturer's test certificate in original alongwith the Test Sheet giving the result of each physical test as applicable and the chemical composition of the cement or authenticated copy thereof, duly signed by the manufacturer with each consignment clearly bringing out lot No. The Engineer-in-Charge shall record these details in the Cement Supply & Acceptance register as given in Appx 'E' here-in-after after due verification. The GE shall also organize independent testing of random samples of cement (both physical and chemical properties) drawn from various lots for each consignment to cement brought out by the contractor before incorporation in the work from the National Test House, SEMT, Regional Research Laboratories, Government approved laboratories, as per IS: 3535 (Method of sampling Hydraulic cement), IS: 4031 (Method of physical test for Hydraulic Cement) and IS: 4032 (Method of chemical analysis of Hydraulic cement.)

4.2.3 Following mandatory tests shall be carried out for cement procured by the Contractor:-

(i) Initial and final setting time.

(ii) Soundness test.

(iii) Compressive strength test at 3, 7 & 28 days as specified in relevant IS code.

4.2.4 The cement shall conform to chemical requirements and physical requirements as specified in respective IS. The tests carried out as per provisions of IS codes specified herein before shall be the criteria for acceptance of cement by Engineer-in-Charge. If samples from a lot/lots are not within the acceptance limits of Indian standard the lot/lots shall be rejected without any claims or compensation to the contractor for the lot/lots purchased. The contractor shall replace the lot/lots with the fresh one, which shall be tested again for acceptance. The cost of all tests carried out on cement before acceptance for incorporation in the work shall be borne by the contractor whether the results are acceptable or not.

(PARTICULAR SPECIFICATIONS(Contd.....)

4.2.5 **STORAGE** :- Cement shall be stored over dry platform at least 20cm high in such a manner as to prevent deterioration due to moisture or intrusion of foreign matter. In case of storerooms, the stock should be at least 20cm above from floors and 60 cm away from walls in addition to precautions specified in clause 4.3.1 on Srl Page 51 & 52 of SSR (Part-I). Inspections shall be carried out once a day by the Engineer-in-Charge. It shall be ensured by the Engineer-in-Charge that tested and untested cement are segregated and stored separately with distinct identification. The cement godown shall be provided with two locks on each door. The key of one lock at each door shall remain with the Engineer-in-Charge or his representative and that of the other lock with the contractor's authorized representative at site of works so that cement is removed from the godown only according to daily requirement with the knowledge of both the parties.

4.2.6 **DOCUMENTATION:** The contractor shall submit original vouchers from the supplier for the total quantity of cement supplied under each consignment to be incorporated in the work. All consignments received at the work site shall be inspected by GE alongwith the relevant documents before acceptance. The original vouchers and the Test Certificates shall be defaced by the Engineer-in Charge and kept on record in the office of GE duly authenticated and with cross-reference to the control number in the cement Supply & Acceptance Register. The cement supply & Acceptance Register will be signed by Supdt B/R Gde-I/ JE (Civil), Engineer-in-Charge, GE and the Contractor. The Accepting Officer may order a Board of Officers for random check of cement and verification of connected documents. The entire quantity of all types cement shall also be suitably recorded in the Measurement Book for record purpose not to be abstracted before incorporation in the work and shall be signed by the Engineer-in-Charge and the contractor.

4.2.7 **SCHEDULING OF SUPPLY:** Schedule of procurement of cement shall be finalized by the contractor with GE and shall be incorporated in the CPM chart so that procurement is in accordance with the progress contemplated in the CPM prepared. The complete requirement of cement shall be worked out before making any RAR payment and procurement of cement by the contractor shall be completed sufficiently in advance of the execution of work.

4.2.8 **FINE AGGREGATE (SAND):** Fine aggregate for all concrete work shall be Badarpur sand and /or Ghaggar river sand (from near Chandigarh area or any other Sand) conforming to samples, complying with the requirement as specified in clause 4.4.7 (2) for grading zone I of SSR (Part I). Sand in the zone II grading may also be permitted provided mix design is done and the requisite strength of the concrete is achieved.

> **Note:** Wherever Badarpur sand has been specified in the tender documents, the Contractor shall be permitted to use Ghaggar sand (from near Chandigarh area) at his own cost, if he so desire due to inadequate supply/non availability of Badarpur Sand. No price adjustment shall be made on this account. Sample of sand shall however, be got approved from the GE before incorporation in the work. Other provisions in the respective clauses shall remain unchanged.

4.3 All mixes of concrete and mortar mentioned in this tender document shall be by volume except design mix concrete conforming to IS-456 of 2000 for which all the ingredients are proportioned and mixed by weight.

4.4 **PLAIN AND REINFORCED CEMENT CONCRETE:** Plain and reinforced cement concrete shall be all as specified in clauses 4.11 of SSR Part-I.

4.5 **MIX OF CEMENT CONCRETE** Cement concrete wherever shown specified on drawings shall be of the following mix unless otherwise mentioned elsewhere:-

(a) PCC lean concrete in foundations of raft slab, column Footing, brick walls, brick pillars under brick steps, Dwarf wall and lean concrete not mentioned elsewhere	PCC (1:4:8) type D-2 (Nominal Mix)
(b) Sub base of floors except as mentioned in the above(a)	PCC (1:4:8) type D-2 (Nominal Mix)
(c) Lean concrete under floors for cycle shed & ramp	-do-
(i) Plinth protection including toe	PCC (1:3:6) type C-2 (Nominal Mix)
(ii) PCC bed blocks, PCC bedplates and concrete around pipes, PCC block (Other than for hold fast).	PCC (1:3:6) type C-1 (Nominal Mix)
(iii) PCC blocks for hold fasts, PCC cills, PCC kerbs, coping, benching and any other PCC item not specifically mentioned elsewhere	PCC (1:2:4) type B-1 (Nominal Mix)
(iv) All RCC works	M-25 Design Mix

(PARTICULAR SPECIFICATIONS(Contd.....)

4.5.1 Minimum cement content and max water cement ratio for PCC nominal mix of various grades shall be as per Table 5 and 6 of IS-456-2000.

4.5.2 All RCC work shall be M-25 (Design Mix).

4.6 **DESIGN MIX CONCRETE**

4.6.1 Where the concrete is specified by grade i.e. M-25, M-30, M-35 and M-40 the same shall be design mix. The design concrete mix shall be as per IS 456: 2000, IS-10262-1982, SP-23 (S &T) 1982 and as specified hereinafter. Design Mix M-25 & M-30 shall be carried out on recommended guide lines for concrete mix design with following stipulations:-

(a) Grade of Concrete	: M-25/M-30
(b) Characteristic compressive strength Compliance requirement	: As per IS-456: 2000 Table 11
(c) Type of cement pozzolona cement for	: Ordinary Portland cement 43 grade or Portland RCC Work
(d) Maximum size of aggregate	: 20mm (Crushed aggregate) for sections more than 75 mm and 10mm (Crushed aggregate) for sections less than or equal to 75 mm
(e) Type of exposure	: Moderate: as per IS-456: 2000 Table 3
(f) Maximum water cement ratio	: 0.50
(g) Degree of quality Control	: Very Good(Refer table 40 of SP23-1982)
(h) Degree of workability	: As per IS-456:2000 with slump for pumped and non pumped concrete
(i) Minimum quantity of cement	: As per IS 456: 2000 Table-5.

4.6.2 At the time of tendering, the contractor after taking into account the type of aggregate, plant and method of laying, he intends to use, shall allow for the water cement ratio which he consider will achieve the strength requirement specified and will produce workability which will enable concrete to be properly compacted for its full depth.

4.6.3 Actual proportion of cement, aggregate and water to obtain the required strength shall be determined by preliminary tests as specified hereinafter. The water cement ratio may be reduced in the trial mix design but not increased.

4.6.4 As soon as possible, after receiving the order to commence the work, the Contractor shall procure sufficient quantities of the aggregate and cement samples for testing and mix design. The optimum mix to achieve the target mean strength shall be determined in the laboratory Conditions and the Concrete shall conform to relevant IS Specification. Design mix shall got done from National Test house, IIT or any other Govt approved laboratory approved by GE. Cost of materials and transportation and testing charges are to be borne by the Contractor and same shall deem to be included in the rates quoted by the Contractor. Details/results of trial mixes from the engineering establishment along with their recommendations regarding quantities of cement, water, sand and coarse aggregates to be used in the work to achieve requisite strength in the field at 28 days shall be submitted by the contractor to the GE for final approval. Govt approved laboratories for testing Mix Design for concrete and all type of building materials:-

(PARTICULAR SPECIFICATIONS(Contd.....)

(a) Shri Ram Inistitute for Industrial Research (NABL certified) 19 Cavalary Line University Road, Delhi-110007.

(b) Bharat Test House (NABL certified) 454/2 Ist floor Timber market Near Azadpur Chowk, Delhi-110033.

4.6.5 The approval of the GE will not absolve the contractor of his responsibilities to obtain the required minimum strength of concrete at any stage.

4.6.6 Contractors unit rates shall include for the above provisions and as brought out here-in-after. Further, in case there is any change in source of local materials viz sand/aggregates or grade of cement etc, the procedure as in Para 3.6.5 & 3.6.6 here-in-before shall be repeated.

4.6.7 In proportioning design mix concrete, the quantity of both cement and aggregate shall be determined by mass and all in accordance with clause 9.2 and sub clauses thereon of IS-456-**2000** as applicable. All the materials shall be tested at a frequency as laid out in relevant IS. Use of weigh batcher is compulsory for all grades of concrete.

4.6.8 **MIXING OF CONCRETE:** The contractor shall use power driven mechanical concrete mixer of adequate capacity with hopper of approved type of mixing. The coarse and fine aggregate for mixing of cement concrete shall be put in the hopper after measuring the same by weigh batching, the drum shall be loaded with about the half of the coarse aggregate, then with the fine aggregate, then with the cement and finally with remaining coarse aggregate at the top. Water shall be added immediately before the rotation of the drum is started. The period of mixing shall be not less than 2 minutes after adding all the materials in drum and shall continue till the resulting concrete is uniform in appearance.

4.6.9 **WEIGH BATCHING**:-

(i) In design mix concrete, the quantity of cement and aggregate shall be determined by weight. In no case volume batching of aggregates based on weight- volume conversion in the mix design shall be allowed. Water shall be either measured by volume in calibrated tanks or weight. Due care shall be given to adhere to maximum free water cement ratio indicated here in before. All measuring equipment shall be maintained in clean serviceable condition and their accuracy periodically checked. In this connection also refer clause 4.11.3.1 to 4.11.3.5 of SSR Part-I.

(ii) Quantity of concrete mixed in any one batch shall not exceed the rated capacity of the mixer. The whole of the mixed batch shall be removed before adding new batch. Concrete mixed above shall not be modified by addition of water or otherwise in order to facilitate handling or for any other purpose.

(iii) On cessation of work and other stoppages exceeding 20 minutes, the mixer and other plant used for the wet mix shall be thoroughly washed with clean water. Pick up and throw over blades in the drums of the mixer, which are worn down 20mm or more in depths shall be replaced with new blades.

4.7 **SAMPLING AND ACCEPTANCE CRITERIA FOR CONCRETE**: The criteria for sampling and acceptance of the concrete shall be as per Para 15.1 to 16.1 of IS-456:2000, (Code of practice for plain and reinforced concrete). The minimum frequency for the work tests shall be as specified therein and as specified here in after.

4.8 **PACKING AND TRANSPORTATION OF SAMPLES** The Contractor must, at his own expenses, properly pack the samples. The Contractor shall bear the cost of transportation of the samples required to be tested from site of work to the laboratory of Chief Engineer Delhi Zone or any other laboratory.

4.9 **TESTS:** The following tests shall be carried out during the execution of work. Tests shall be carried out in accordance with IS-516 and IS-1199. The contractor shall provide all facilities and equipment for casting and curing of test cubes.

(PARTICULAR SPECIFICATIONS(Contd.....)

(a) **SLUMP TEST:**
The frequency of the test shall be decided by the GE. The slump for the vibrated concrete shall be as specified in IS-456. The GE, however, reserves the right to vary the limit which will be ascertained at the time of deciding the mix design for each grade. Any batch from which a slump test is being made shall not be transferred to the places of laying until the slump in excess of the required for the particular batch is not removed from the site.

(b) **COMPRESSIVE STRENGTH TEST:**
(a)Three cubes shall be tested at 7 days and three at 28 days for crushing and the remaining three cubes shall be kept as reserved exclusively for Govt use for subsequent testing (if so desired by Govt) and preserved for one year from certified date of completion of work under the contract. The contractor cannot object testing of other cubes for the purpose as a matter of right.

(c) For the purpose of subsequent identification of the work test cubes, the concrete to which these pertains shall be cross referred and record of this maintained and signed by the Engineer-in-Charge and contractor or his authorized representative.

4.10 **TRANSPORTING, PLACING, COMPACTION and CURING:**

(i) It should be as per clause 13 of IS 456:2000. Concrete shall be transported without delay and incorporated in works at the position of laying within 20 minutes from the time of discharge from the mixer.

(ii)Mixed concrete shall be deposited in final position and solidly packed around reinforcement, carefully poured and consolidated by means of portable vibrators or mechanically operated and of the kind as suitable for a particular situation as directed by the GE. Care shall be exercised that no voids or honey comb pockets are formed. The concrete shall not be laid in position for more than 1 metre in height in one concrete operation. The compaction of concrete in RCC roof and floor slabs shall be done with plate vibrators.

(iii)Exposed surfaces of concrete shall be kept continuously in a damp or wet Condition by pounding or by covering with a layer of sack, canvas, Lenin and kept constantly wet for at least 14 days. The concrete shall be protected from premature drying for at least 8 days after pouring and shall be cured as directed by the Engineer-in-Charge for a period of not less than 14 days.

4.11 **CURING AND MINIMUM PERIOD FOR STRIKING OF FORM WORK WITH POZZOLANA CEMENT USED:**

4.11.1 Where pozzolana cement is used for RCC works in various locations, the minimum period for curing and striking off form work shall be as under in lieu of that specified in clause 4.11.6.3 of MES Schedule Part-I :-

(a)	Curing :- Structural RCC work, PCC work, Plastering etc.	: 14 days
(b)	Striking off Form Work Walls, columns and vertical sides of beams	: 3 to 4 days
(c)	Slabs (Props left under)	: 8 days
(d)	Beams soffits (Props left under)	: 9 days
(e)	Removal of props to slab/beam	
	(i) Spanning upto 6 M span	: 16 days
	(ii) Spanning beyond 6 M span	: 22 days
(f)	For cantilever portion of slab form work with support shall be retained until the completion of the casting of the entire frame work of the building.	

4.11.2 In case of bad weather, periods mentioned above may be revised at the discretion of the Engineer-in-charge. The contractor shall be deemed to have considered the above provision before tendering and quote lump sum accordingly. No claim will be entertained if longer periods required for striking off form work and curing and all such effected matters is looked into upon the use of pozzolana cement as a result being issued to the contractor in full or in part or for bad weather.

(PARTICULAR SPECIFICATIONS(Contd.....)

4.11.3.Curing shall be carried out using pump of suitable capacity at no extra cost to the Govt. all as directed by Engineer-in-Charge.

4.12 **PLINTH PROTECTION:** Plinth protection shall be 75 mm thick in PCC (1:3:6) type C-1 using 20mm graded crushed stone aggregate over 75mm thick consolidated broken stone hard core over rammed earth surface and shall be filled with mastic filling comprising of one part of heated bitumen 85/25 or 80/100 grade and 3 parts of sand (all by weight).

5. **FORM WORK:**

5.1 Formwork shall comply with the requirement specified in para 11 of IS-456 & IS 14687. Use of deformed and dented steel sheets shall not be permitted for use as form work.

5.2 All formwork including props shall be of Steel of adequate strength. Props shall be adjustable type, to avoid slant props. However, the form work to be used for surfaces specified in these specifications to be plastered shall be such that after application of the specified thickness, fair finished surface is achieved as specified herein. The contractor may at his discretion use timber or steel or ply wood form work for which no price adjustment will be admissible & in such cases he will have to properly roughen the concrete surfaces for receiving plaster finish. Wall thickness shall not be made use of as formwork. Wall shall be built after the columns are cast. In the event of deviation, rates in MES Sch for clean sawn form work will be applicable. In this connection also refer to clause 14.14.3 of SSR (Part I). In case of bad weather stripping time for form work as specified may be increased as directed by Engineer- in – charge without additional payment to contractor.

6 **BRICK WORK**

6.1 **BRICKS** :- Bricks shall be kiln burnt, available locally or in the neighboring region, best quality conforming to the samples kept in CWE/GE's office with sub class 'B' old size bricks having minimum crushing strength of 75 Kg/Sqcm. Water absorption of bricks shall not exceed 20% when tested in accordance with IS-3495. Sampling and testing of bricks shall be carried out as per IS-5454 and IS-3495. The nominal size of brick shall be 230mm x 115mm x 75mm. The tolerance in the dimensions shall be ± 8 percent.

6.2 **CEMENT** Refer clause under ‘Concrete’ here-in-before.

6.3 **SAND**. Sand for mortar shall be as specified in clause 5.4 of MES Schedule Part-I.

6.4 **WORKMANSHIP**. Workmanship shall be all specified under relevant clause of SSR Part I.

6.5 **FLY ASH BRICK WORK**: Bricks to be used in the Sch ‘A’ shall be Fly Ash cement bricks of size 230 x 115 x 75 mm with frog size 100x40x10-20mm having minimum compressive strength 75 Kg / Cm2, best locally available and shall comply with the requirements as specified in IS- 12894: 2002 for all aspects except that the word lime shall be replaced with the word ‘cement‘ in the above code. Cement shall be OPC 43 Grade and shall be minimum 10 % by weight. However, Bricks to be used in below ground level shall be best available kiln burnt clay bricks of size 230 x 115 x 75 mm with frog size 100x40x10-20mm having minimum compressive strength of 75 Kgs per sqcm and shall comply with the requirements of Sub Class ‘B’ as specified in IS-1077. Fly ash to be used in the manufacturer of fly ash cement bricks shall conform to Grade 1 or 2 of IS-3812-1981. Minimum percentage of fly ash to be used in the fly ash cement bricks shall not be less than 25%. Manufacturers test certificate and independent testing conforming chemical and physical requirement /characteristics and proportion of fly ash shall be produced by the contractor to the GE for approval. Fly ash shall be procured from coal / lignite based thermal power plants. Water absorption for the fly ash cement bricks shall not be more than 20% of weight. Efflorescence when tested according to IS-3495 (Part III) 1976 shall have the rating of efflorescence not more then 12.5%. Testing of fly ash cement bricks shall be carried out in accordance with procedures listed in IS-13757-1993. Frequency of test shall be as per IS-1077 and IS-5454. Handling and storage of bricks shall be as per clause 5.6.9 of SSR Part I.

(PARTICULAR SPECIFICATIONS(Contd.....)

7. WOOD AND JOINERY WORK

7.1 WOOD WORK

7.1.1 Timber to be used in various situations shall be as mentioned in the respective items of Sch`A` unless otherwise specified. All wood work shall be wrought on all faces and shall be of second class hard wood (Hollock).

7.1.2 Maximum permissible moisture contents for various purposes shall be as stipulated in Para 7.7 of SSR part I except for form work. Contractor shall get the timber kiln seasoned to ensure that permissible moisture contents stated in this clause is not exceeded. The timber which exceeds the permissible moisture content limit even though kiln seasoned shall be rejected.

7.2 JOINERY

7.2.1 Timber to be used for joinery in various situations (except flush shutters, veneered particle board, black board, plywood) shall be second class Hardwood (Hollock).

7.2.2 All shutters for doors shall be factory made.

7.2.3 All other timber required for joiners work not mentioned above or elsewhere in particular specification shall be second class Hard Wood (Hollock).

7.2.4 For seasoning of timber for joiners work refer clause 7.1.2 here-in-before.

7.2.5 Provision of clause 7.1.2 mentioned here-in-before regarding moisture contents shall apply to timber for the factory made shutters mentioned under clause 7.2.2 here-in-before.

7.3 **PARTICLE BOARD:** Particle board shall conform to the requirement of IS-3097(duly ISI marked) exterior grade (grade-1), specifications for veneered particle board (medium density). Particle board shall be of uniform thickness and density throughout and bonded with phenol formaldehyde synthetic resin (BWR type). Both the faces of particle board shall have sanded smooth finish. Where thickness of particle board has been shown as 20mm in the drawings / else-where, the same shall be amended to read as 18 / 19mm.

7.4 **PLYWOOD :**All plywood where indicated shall be BWR exterior grade conforming to IS-303. Decorative plywood where indicated shall conform to IS-1328.

7.5 PRELAMINATED PARTICLE BOARD: Pre-laminated particle board where indicated shall be factory impregrated exterior grade One or both side decorative type lamination as approved by GE. In case of one side decorative lamination, other side shall be balancing white prelamination; shall confirm to IS-12823 Gde-I Type-I. Thickness of the board shall be 18/19 mm if not indicated in drawings.

7.6 FACTORY MADE PANELLED, GLAZED AND GAUZED SHUTTERS

7.6.1 Panelled, glazed and gauzed shutters shall be of 2nd class hard wood (Hollock).

7.6.2 All fully/partly panelled, glazed and gauzed shutters (but not chowkats) shall be factory made conforming to IS : 1003 (Part I) The frames (rails etc) of panelled, guazed and glazed door shutters shall be of timber as specified in clause 7.1.2 here-in before. The inserts (except where wire gauze/glass) shall be of 12mm thick veneered particle board exterior grade solid core, general purpose type with commercial veneer facing on both sides and bonded with phenyl formal dehyde synthetic resin and conforming to IS-3097 unless otherwise specified. Factory made shutters shall be procured from any of the make as specified in Appendix “A “hereinafter.

7.7 FLUSH DOOR SHUTTER

7.7.1 Flush door shutter shall be provided complete all as specified in Schedule ‘A’ (BOQ) and other as specified in para 8.25 of MES Schedule Part-I and as per IS 2202 (Part : I -1999).

(PARTICULAR SPECIFICATIONS(Contd.....)

7.7.2 Flush door shutter shall be solid core types with block core, and shall conform to IS 2202 (Part I) - 1999, Specification for wooden flush door shutters (solid core type) Part I.

7.7.3 All timber used shall be well seasoned and chemically treated. Adhesive shall be phenol formaldehyde synthetic resin BWP type specified in IS:848-2006. All dimensions shall be finished dimension and Manufacture's Test Certificate for test specified in IS: 2202 (Part I) shall be rendered.

7.8 **SCANTLING**: Unless otherwise indicated, the Contractor shall supply scantlings in one piece in lengths upto 3 m. For lengths over 3 m the Contractor shall be allowed to use more than one piece. One for every extra 3 m or part thereof; and shall connect the same with proper scraf or other lengthening joints, as directed by the EIC. Timber shall be of good quality, well seasoned, fairly uniform in colour and texture and free from blemishes, hollow pockets and loose knots. Coniferous sawn timber (soft wood) shall be free from spiral or twisted grain, warp, any kind of decay or live insect attack and cup shakes. Non-coniferous sawn timber (hard wood) shall be free from bow, any kind of decay, live insect attack, spiral or twisted grains, splits across the grain, spring, warp, cup and shake. Work shall be carried out all as specified in the relevant clauses of SSR Part-I.

7.9 PVC DOORS AND DOOR FRAMES :

7.9.1 PVC doors and door frames shall be provided of dimension as per BOQ. PVC door frame (Chowkhats shall be of size 50 mm x 47 mm with a wall thickness of 5 mm made out of cotruded 5 mm rigid PVC foam shocked mitre cut at two corners and jointed with 2 Nos 150 mm long brackets of 15mm x 15 mm MS square tube. The two vertical door profiles shall be reinforced with 19 mm x 19 mm square tube of 19 gauge Rubber gasket weather seal shall be provided through out the frame. The door frames shall be fixed with wall using 65/100 mm long MS screws all as per manufacture's instructions and as directed by Engineer-in-Charge.

7.9.2 Factory made solid panel PVC door shutters 30 mm thick consisting of frame made out of MS tubes of 19 gauge thickness of size of 19 mm x 19 mm for stiles and 15mm x 15 mm for top and bottom rails MS frame shall be covered with 5 mm thick heat moulded PVC channel of size 30 mm x 50 mm forming stiles and 5 mm thick 75 mm wide PVC sheet for top rail, lock rail and bottom rail on either side. 10 mm thick (2 x 5 mm). 20mm wide cross PVC sheet shall be provided as gap insert for top bottom rails . Panel insert of 5 mm thick PVC sheets shall be fitted in the MS frame welded/scaled to the stiles and rails with 30mm wide x 5 mm thick PVC sheet beading on either side and jointed together with solvent cement adhesive. The size of stiles and rails shall be 75 mm and 100 mm respectively. The door shutter shall be fixed to PVC door frame as per manufacturer's instructions.

8. BUILDERS HARDWARE

8.1 **GENERAL:-** All articles of builder's hardware shall bear ISI marking. In case ISI marked articles are not manufactured, those shall conform to the relevant IS specifications given in the MES schedule for the relevant item. Finish of articles shall be as specified in clause 9.2.4 of the MES Schedule Part-I.Screws used for fixing the articles of builders hardware shall be as specified in clause 9.2.6 of the MES Schedule (Part-I). Article shall be as under:-

(i) **BUTT HINGES** : Butt hinges shall be of MS steel medium weight all as specified in clause 9.7.1 & 9.7.2 of MES Schedule Part-I.

(ii) **CONTINUOUS (PIANO) HINGES:** Continuous (Piano) Hinges shall be finish as specified in clause 9.7.6 of MES Schedule Part-I. The width of the flap shall be to suit the thickness of the shutter.

(iii) **BARREL TOWERBOLTS**: Aluminiumanodised barrel tower bolts of extruded section shall be as per clause 9.3 of SSR Part I.Size of barrel tower bolts shall be pae BOQ. Where size of barrel tower bolts not indicated, the same shall be 150 mm for doors and 100 mm for windows.

(iv) **HANDLES:** Extruded aluminium alloy handles fabricated type, anodised shall be as per clause 9.11 of SSR Part-I.Size of handles shall be as per BOQ. Where size of handles not indicated, the same shall be 150 mm for doors and 125mm for windows.

(PARTICULAR SPECIFICATIONS(Contd.....)

(v) **HASP AND STAPLES** : Hasp and staples, safety type, aluminium anodised with mild steel hinge pin shall be as per clause 9.10 of SSR part I.Size of Hasps and staples shall be as per BOQ. Where size of Hasps and staples not indicated, the same shall be 75 mm .

(vi) **DOORS SPRINGS :**Ordinary rat tail rod pattern MS spring including roller and roller plate shall be as per clause 9.7.8 of SSR part I. Size of door spring shall be as per BOQ. Where size of door spring not indicated, the same shall be 150 mm.

(vii) **HYDRAULIC DOOR CLOSER: H**ydraulic door closer shall be hydraulically regulated designation No 2, Universal type aluminium body suitable for door shutter of weight between 36 to 60 Kg shall be provided as specified in clause 9.16 of SSR Part-I. One number hydraulic door closer shall be provided to each shutter in all Aluminium doors.

(viii) **TOWEL RAILS** :Towel rails shall be tubular, mild steel, chromium plated, 22 gauge, min 18/19 mm bore, 'D' shaped with flanged ends for fixing. In case length is not mentioned in drawings, it shall be 60cm long. Wherever aluminium towel rails are indicated in the drawings, these shall be deemed to read as chromium plated mild steel rails as specified in this clause.

(ix) **WIRE CLOTH :**Wire cloth to be used for wire gauzed shutters shall be stainless steel with average width of aperture 1.18mm and nominal dia of wire 0.457 mm. In case of deviation, rates given in MES schedule part II for wire cloth of nominal dia of wire of 0.56 mm shall be applicable. Where SS wire mesh is specified in the BOQ, it shall be 0.36 mm thick and 1.40mm aperture.

(x) **KNOBS:**Shutters and drawers of small size built in cabinets, switch boxes and built in furniture shall be provided with aluminiumanodised knobs of not less than 38mm dia.

(xi) **ALUMINIUM DOOR STOPPER/WOODEN CLEATS**:Aluminium wooden door stoppers/wooden cleats of suitable size shall be provided to each open able shutter of window, ventilators and doors.

(xii) **HELICAL DOOR SPRING:** Where shown on drawings, wire gauged door shutters shall be provided with mild steel helical door spring 150mm long. MAKE: RITZ or Equivalent.

(xiii) **SLIDING BOLT:**Aluminium anodized sliding door bolts (type 3) 250 mm long with hasps and staple (bolttype) and fixing clips of sheet and bolt of extruded section of aluminium alloy, 16mm dia and shall be provided in all type of dwelling units. These shall be complied with IS 2681-1993 and with fixing bolts.

(xiv) **TIE HANGER:**The tie hanger in cupboards shall be of mild steel chromium plated pipe of 10mm internal diameter, 300mm long with standard mild steel chromium plated brackets as approved by GE. Wall thickness of mild steel chromium plated pipe should be more than 1mm.

(xv) **HANGING ROD:**Hanging rod of 19 mm (±1 mm) dia G.I. pipe light grade with appropriate socket for hanging rod in cupboards shall be provided as shown on the drawings.

(xvi) **LOCKING BOLTS:**Aluminium alloy (anodized) locking bolts (10 mm dia, 100 mm size) shall be ISI marked (IS 2681) and be provided to each shutter (one each)

(xvii) **HOOK &EYES:**Hook and eye shall be of aluminium anodized and matching with window and of size as shown on drawings. If the size is not specified in drgs, the same shall be of 125mm size.

(xviii) **MAGIC EYE:** Magic eye shall be provided to each entrance door (one per DU) as per locations shown on drawings and shall be as per sample approved by the GE. It shall be of brass body, with both sides lenses, and of approved make.

(xix) **BALL CATCH :**Aluminium magnetic catcher of heavy duty of standard make shall be provided in all type of Dwelling Units.

(xx) **ALDROP BOLT:**Aldrop bolt shall be aluminium anodized aldrop bolt (sliding door bolt) type 3, bolt of extruded section 16mm dia, ISI marked (IS-2681).

(xxi) **LOCK FOR CUPBOARD:** Locks for cupboard shall be of Grade -2, size 50 mm 4 levers unless otherwise specified and wherever shown in drawings shall be of brass all as specified in clause 9.9.4 of MES SSR Part-I. Make of locks : Godrej/Harrison/link. In case of Aluminium windows/CSWs, the edges of wire gauge shall be bent over Aluminium beads and the beads pressed well into the angle of Aluminium section to hold the gauge on two faces as shown on drawings.

(PARTICULAR SPECIFICATIONS(Contd.....)

9. STEEL AND IRON WORK.

9.1 All items of steel and iron work including steel for reinforcement required for incorporation in this contract shall not be issued under Schedule "B" . Items of steel and iron brought to site by the contractor for incorporation in the work shall be new and free from defects all as specified in clause 10.4.3 and 10.17.6 of SSR Part-I and shall be conforming to IS specifications. Other parameters are us under:-

(a) **Quality Standards:**

(i) **STRUCTURAL STEEL**. Gde E-250 (Fe410W) quality 'A' or 'B' ISI marked (IS-2062).

(ii) **STEEL FOR REINFORCEMENT**. High strength deformed steel bars produced by Thermo Mechanical Treatment process; TMT steel bars of Grade Fe 500/500D meeting all requirements of IS-1786.

(iii) **MILD STEEL**. Mild Steel as in fencing posts, Hold fasts, grills, guard Bars,steel door ,railing, steel gate, members of steel windows , doors and the like; Gde E-165 (Fe-290) ISI marked (IS-2062).

(iv) **Fabric**. Fabric reinforcement for concrete ; Conforming to IS-1566.

(v) **Steel Sheet**. Black/Galvanised steel sheets; Conforming to IS-277. Galvanised steel sheets (plain & corrugated) shall have Class 8 coating.

(b) **PROCUREMENT**

(i) **TMT STEEL**. TMT Steel bars of all sizes to be supplied by the contractor shall be procured from primary/main producers of steel namely STEEL AUTHORITY OF INDIA LTD (SAIL), TATA IRON AND STEEL CO. (TISCO), RASHTRIYA ISPAT NIGAM LTD (RINL), M/S SHRI BAJRANG POWER & ISPAT LTD RAIPUR, M/S GALLANTT ISPAT LTD, GUJRAT (GALLANTT TMX), M/S RASHMI METALIKS LTD, W.B (RASHMI TMT), M/S JAI BALAJI INDUSTRIES LTD, W.B. (BALAJI SHAKTI), M/S SHYAM STEEL INDUSTRIES LTD W.B. (SHYAM), M/S STEEL EXCHANGE INDIA LTD A.P. (SIMHADRI TMT), M/S JINDAL STEEL AND POWER LTD, HARYANA (JINDAL PANTHAR), M/S REAL ISPAT AND POWER LIMITED, CHHATTISHGARH (GK TMT), M/S SUPER SMELTERS LIMITED KOLKATA (SUPER SHAKTI), M/S MSP STEEL AND POWER LIMITED, CHHATTISHGARH (MSP TMT 500D), M/S SPS STEEL ROLLING MILLS LTD, W.B. (ELEGANT TMT), M/S SUGNA METALS LIMITED, HYDERABAD, (SUGNA TMT), M/S SHYAM METALICS AND ENERGY LTD, W.B. (SEL TIGER), M/S TULSYAN NEC LIMITED, CHENNAI (TULSYAN TMT), M/S ELECTROTHERM (INDIA) LTD GUJRAT (ET TMT). However authorized conversion agents of main producers can supply the steel to the contractors provided main producers have given such directions in a specific sale/purchase order conversion agents are producing TMT steel using Thermax or Tempcore or Evcon Turbo Technology or Siemen's HYQST/TMT. All such transactions shall be with main producers only and the contractor shall not procure any steel section (TMT steel bars) directly from the conversion agents. Purchase vouchers and Test certificates shall be from main producers.

(c) **STRUCTURAL STEEL (E-250, Fe-410W**. Structural steel sections shall be procured from primary/main producers of steel namely STEEL AUTHORITY OF INDIA LTD (SAIL), TATA IRON AND STEEL CO. (TISCO), RASHTRIYA ISPAT NIGAM LTD (RINL**)**. In case of non-availability of structural steel from all of the main producers mentioned above the same can be procured from secondary producers mentioned in clause here-in-after who manufacture structural steel out of ISI marked billets and are having BIS certifications of ISI marking on their products and approved by Engineer-In-Chief, with necessary minus price adjustment shall be @ 5% (Five percent) of the quoated rate by the contractor of structural steel. However in case of the desired section of structural steel is not manufactured/rolled by primary producer their shall be no price adjustment in use of structural steel procured from approved secondary producer. Secondary producers of Structural steel are;M/s KL Steel Pvt Ltd, M/s Shri Badrinarain Alloys and Steel Ltd, M/s Pushpak Steel Industeries Pvt Ltd, M/S Amba Shakti Ispat Ltd, SRMB Udyog Ltd, M/s Shree Parashnath Re-Rolling Mills Ltd, M/s Tata Steel Structure, M/s KL Concast Pvt Ltd, M/s Karam Steel Corp**.** The correct address and contact No of secondary producers shall be obtained from GE before placing order.

(PARTICULAR SPECIFICATIONS(Contd.....)

(i) Steel sections for railings, gates, fencing. Guard bars, grills , steel chowkhats , hold fasts etc. which do not constitute structural members, may be procured from main producers / secondary producers / BIS marked manufacturers or their authorized dealers at the option of contractor without any price adjustment . Testing of these articles is optional at the discretion of GE.

(ii) **GALVANISED STEEL SHEETS AND FABRIC REINFORCEMENT FOR CONCRETE**. Galvanised steel sheets and Fabric reinforcement for concrete shall be procured directly from main producers namely STEEL AUTHORITY OF INDIA LTD (SAIL), TATA IRON AND STEEL CO. (TISCO), RASHTRIYA ISPAT NIGAM LTD (RINL). Reinforced steel, structural steel and Galvanised steel sheets & fabric reinforcement for concrete may be procured from authorised dealers of main producers in case the total requirement of steel is less than 5 tones. Steel sections for railings, gates, fencing, guard bars, grills, steel chowkhats ,holdfasts etc, (E-165) which do not constitute structural members can be procured from main producers/BIS marked manufacturers of their authorised dealers at the option or Contractor without any minus price adjustment. Tests shall not be insisted upon for such steel sections. Galvanized steel sheet, plain or corrugated shall comply with the requirement of IS 277-2003, Specification for galvanised steel sheets, plain or corrugated. All other specifications shall be as per SSR Part-I Clause 10.29.

(d) **DOCUMENTATION.** The contractor shall submit original purchase vouchers and Test Certificates from the manufacturer for the total quantity of steel which shall be defaced by the Engineer-in-Charge and kept on record. The Steel Acceptance Register shall be signed by JE, Engineer-in-Charge, GE and contractor.

(e) **TESTING OF STEEL**. The GE/CWE shall also organize independent testing of random samples of steel drawn from various lots as applicable and from a National Test House , SEMT wing CME Pune, Regional Research Labs and NABL approved Labs, Government approved labs, Engineering college, Zonal Labs, Command Labs etc as required. The elongation shall not be less than 18%. In order to undertake Department testing, requisite facilities shall be organized by the contractor. The record of such checks shall be maintained in the steel test register. Thermo mechanically treated bars shall be tested for tensile test, bend test and re-bend test as per clause 8 of IS-1786-1985. Test specimen shall be taken as per clause 10 of IS-1786-1985. Mild steel bars shall be tested for tensile test and bend test as per clause 10 of IS-432(Part-I) 1982. Minimum yield stress shall be 250 N/Sqmm. The structural steel (E-250 and mild steel section E-165) shall be tested for tensile test and bend test as per IS-226/ IS-2062/IS-1977 as applicable. Minimum yield stress shall be 250 N/Sqmm. Chemical test shall be carried out to ascertain chemical composition, which shall conform to the norms laid down in relevant ISI codes. For TMT bars, chemical composition shall be as given in relevant Appendix 'C' attached to these particular specifications.

(f) **TOLERANCE**. Rolling and cutting tolerance of structural steel products shall be as specified in IS-1852 of 1985 Tolerance on size and weight of reinforcement bars shall be as specified in clause 10.17.4 and 10.17.5 of SSR part-I. Steel with less weight than permissible shall not be used.

(g) **SCHEDULE OF SUPPLY**. The contractor shall ensure that procurement action is taken immediately after acceptance of contract so as to adhere to schedule of activities as per CPM network.

(h) **STORAGE, ACCOUNTING PRESERVATIONS AND MAINTENENANCE**. Structural steel/bars of different classification, section, sizes and lengths shall be stacked separately. Steel shall be marked with distinct painting marks for easy identification. Steel shall be recorded in MB for record purpose only. All steel shall be stored on Dunnage at-least 15cm above ground level. In case of long time storage, suitable protective measures like cement coating, provision of temporary shelter etc to save it from rusting/distortion etc shall be taken. Any item of steel that has deteriorated or corroded or considered defective by Engineer-in-Charge shall not be used in the work and shall be removed from the site of work.

9.2 **WELDING**. Welding wherever required shall be by metal arc process in accordance with IS-816 and IS-823.

9.2A **GRILL/GUARD BARS ONLY FOR WINDOWS & VENTS OTHER THAN ALUMINIUM WINDOW/VENTS**. Mild steel Grill/Guard bars of respective type as directed shall be provided to all external steel windows and ventilators. Size and type/pattern of grill/guard bars shall be as directed by Engt-in-Charge.

9.3 **HOLD FASTS/LUGS**. Flat iron hold fasts/lugs shall be provided by welding except those, to be provided to wooden chowkhats which shall be fixed with bolts/nuts as per details. Holes in wooden chowkhats shall be plugged with hard wood plugs.

9.4 **STEEL WINDOW AND VENTILATORS**. Steel windows and ventilators where specified shall be fabricated as specified. Steel doors, windows and ventilators shall comply with IS 1038-1983, Specification or steel doors, windows and ventilators; except with regard to sizes, which shall be as indicated; and shall be of approved make. Rolled steel sections for fabrication shall conform to IS 7452-1990.

9.5 **STEEL DOOR.** Steel door shall be fabricated as per details given by Engr-in-Charge. The size of door shall be as directed and thickness of MS sheet shall be 2 mm . All iron and steel work shall be given two coats of synthetic enamel paint over a coat of red oxide primer.

9.6 **PRESSED STEEL DOOR FRAME**

9.6.1 Pressed steel frames shall be provided complete all as specified in Schedule ‘A’ (BOQ) and shall confirms to IS-4351 and as specified in clause 10.27 of SSR Part I. The tolerance over the profile size shall be +2 mm. The thickness of sheet shall be 1.6mm irrespective of whatever is shown on drawings and it shall be bent to shape as per drawing. The inner surface of the frames shall be treated with a coat of tar. Shop coat and red oxide shall be applied outside before erecting in position. Mild Steel Tie Bar of 16 mm dia Round Bars shall be provided at the bottom of frames by welding to the frame and this tie bar shall be embedded and left inside floor as per drawing/IS. Cold rolled steel hinges as per IS shall be bolted to pressed steel frame.

9.6.2 Mild Steel Conduit 20 mm dia of 16 gauge closed at one end shall be welded to inner face of frame for receiving shoot of bolt. M.S. sockets for housing shoot of tower bolt shall also be provided as directed by Engineer-in-Charge.

9.6.3 The fixing lugs and inner surfaces of door frames coming in contact with concrete shall be treated with two coats of bituminous paint. The door frame shall be thoroughly cleaned, made free from rust, dirt, oil, etc.

9.6.4 The fixing of Pressed Steel Door Frames shall be done as per Clause No. 5.34.5(A) of MES Schedule Part -I.

9.7 **PRESSED STEEL WINDOW/ VENTILATOR**:

9.7.1 Pressed steel windows/ventilators shall be provided complete all as specified in Schedule ‘A’ (BOQ) and as directed by Engr-in-Charge and as specified following :-

(a) Steel required for steel window shall be of cold reduced low carbon steel sheet of designation CR2 and grade D as per IS 513 :2008. MAKE of sheet : **SAIL, Rashtriya Ispat Nigam Ltd, IISCO, TISCO, M/S Jindal Steels and Power Ltd.**

(b) While supplying, steel windows shall have one shop coat of anticorrosive paint (Red oxide zinc chrome primer).

(c) Window shall be given another coat of anticorrosive priming and two coats spray painting before fixing of glass and wire mesh.

(d) Steel windows shall be fixed as per IS-1081 embedding in cement concrete block of appropriate size.

(e) All steel windows, ventilators shall be factory made and of approved make and shall be procured from any of the approved firms listed in Appendix `B' to Particular Specification.

(PARTICULAR SPECIFICATIONS(Contd.....)

(f) Pressed steel frame shall conform to IS 4351.

(g) Glass and frame shall be joint with 6x6 mm silicone polymer base sealant. The fitting shall be provided as following **:-**

(i)	70 mm long MS Box type hinge with two nos stainless steel washer(each end) all as specified in clause no 6.2.1.1 of IS 1038	:	02 nos each shutter (both glazed shutter and gauged (fly proofing) shutter)
(ii)	250mm long 1.5 mm thick MS telescopic stay with lock screw as shown on drawing	:	01 no each gauged (fly proofing) shutter.
(iii)	MS spring Latch	:	01 no each glazed shutter
(iv)	75 mm long 'C' section MS handle	:	01 no each shutter gauged (fly proofing) shutter.
(v)	75 mm long MS tower bolt	:	02 no each gauged (fly proofing) shutter
(vi)	Any other fitting as per manufactures requirement		

9.7.2 **FABRICATION** :- Refer Clause No. 10.25.2 of MES Schedule Part-I and relevant Clause of IS-1038.

9.8 **ALUMINIUM WORKS (DOORS, WINDOWS, VENTILATORS, PARTITIONS, ETC.)**

9.8.1 **ALUMINIUM SECTIONS & BEADS, ETC.**

9.8.1.1 Refer Clause No. 10.37 of MES Schedule Part-I as applicable and as approved by GE.

9.8.1.2 Specification for aluminium door and window shall be 15 micron anodized (colored) aluminum section having embossment of the name of the manufacturer on each section at every metre and thickness of colour anodizing shall be not less than 15 micron. The section particulars shall be provided as per drawing and clause 10.37 of SSR Part-I, 2009. In case of non availability of any particular section mentioned the next available section (having higher per Kg/Rm weight) shall be provided without any price adjustment. The test certificate from Govt authorized testing laboratory shall be produced by the contractor to conform thickness of anodizing.

9.8.1.3 The contractor shall submit the manufacturer's test Certificate in original along with the test sheet giving the results as per relevant IS and duly signed by the section manufacturer with each consignment.

9.8.2 **ALUMINIUM FITTINGS, ETC**: Aluminium Doors and Windows shall be provided with the Standard Aluminium Fittings as specified in relevant Clause No. of MES Schedule Part-I, bearing ISI Mark and as approved by GE

9.8.3 **SHEET GLASS**:-Refer to relevant Clause No. of MES Schedule Part-I and as approved by GE.

9.9 **ALUMINIUM DOORS/WINDOWS/VENTS**

9.9.1 Aluminium Doors/Windows/Vents shall be fabricated, erected and fixed in position and as per the details/ size shown in TD drawings. These shall be in conformity with the standard aluminium fittings conforming to the specifications of relevant IS.

9.9.2 The cleats for mechanical, vertical/ horizontal joints shall be of extruded aluminium section so as to avoid any play between jointed members.

9.9.3 Neoprene crest glazing gasket shall be provided with snap or beveled glazing beads. Glass shall be fixed with PVC/rubber gasket so as to avoid metal to glass contact and no projection at any bending shall be visible thereby making it flush glazed.

9.9.4 All aluminium work shall be deemed to include in various items to complete work including Fittings, Fixtures, Stays, Locks, Handle, Special Hinges, Floor Springs, Neoprene/Rubber Linings, Gaskets, Bushes, Rollers, Sealants, etc. as directed and approved by the GE.

(PARTICULAR SPECIFICATIONS(Contd.....)

9.9.5 The items of Aluminium Glazing, Windows, Vents and Louvers Window Work shall include the provision of mullions, coupling bars, as required to join various units of Glazing, Windows and Louvers etc to form larger glazing, windows vents or louvers cum window/glazing area as per drawings. Also the necessary jointing through Cleats, Glazing Clips, Rubber Packing, Anodized Aluminium Snap Beading, Screws, etc. complete all as specified. No extra payment shall be made for Coupling Members, Mullions, etc.

9.9.6 Aluminium Sections used for windows shall be of weight as specified on drawing. This shall be suitable for use to meet Architectural designs of relevant works and shall be subject to approval of GE who would require to be satisfied about their being appropriate on Technical, Structural Functional and Visual Considerations.

9.9.7 All joints shall be accurately fabricated and be hair line in appearance. The finished surface shall be free from visible defects.

9.9.8 Each aluminum frame shall be Tailor Made as per openings at site. No cutting and making good of concrete surfaces shall be permitted.

9.9.9 Frames shall be fixed to concrete or brick work with approved dash fastners. Method of fixing shall be got approved by the GE before installation.

9.9.10 All hardware items shall conform to the relevant IS Specifications. Design, quality type, number and fixing of hardware shall be got approved by the GE.

9.9.11 No visual variation in shade shall be permitted. The fabricator shall clearly indicate the shade variation tolerance as measured by standard equipment.

9.9.12 The requirements, provisions, for all aluminium work shall conform to requirements and specifications and Clause No. meters given in this tender drawings / instructions and shall at least provide for or conform to fabrication, finishing, erection installation etc.

10. **CEILING :**

(a) **GYPSUM BOARD & GI SHEET CEILING WITH METAL FRAME WORKS** :- Gypsum board false ceiling shall be provided with 12 mm mm thick Gypsum board plain manufactured by 'Everest/Armstrong/Gypsum India' over GI grid frame of size 600mmx600mm and GI perforated sheet ceiling shall be 600x600 mm, 0.50 mm thick weight not less than 3.92 Kg/Sqm (with concealed GI frame work) of configuration as per 'Armstrong' Model No L2SC060605M11TW or equivalent in other approved makes. Entire ceiling system shall be obtained from manufacturer and shall be fixed as per manufacturer's instructions through their applicators. As indicated in BOQ, Gypsum boards shall be finished with two coats of oil bound distemper before installing it on to the concealed GI frame work.

(b) **E-BOARD CEILING WITH ALUMINIUM FRAME WORK:** E-Board ceiling shall be provided with 6mm mm thick E-board (Reinforced multipurpose board) manufactured by 'Everest/Bison/Ramco' of size 605mmx605mm. Aluminium framework shall be of the same manufacturer and shall be fixed as per manufacturer's instructions. Suspension hanger shall be of 4mm dia GI wire with adjustment screw. GI hold down clips shall be used. D-board shall be finished as per Schedule after installing it on aluminium frame work. If finishes not specified then it shall be two coats of oil bound distemper.

(c) **NON ASBESTOS CEILING WITH ALUMINIUM FRAME WORK:-** Board ceiling shall be provided with 4mm mm thick non asbestos by 'Everest/Bison/Ramco' of size 605mmx605mm. Aluminium framework shall be of the same manufacturer and shall be fixed as per manufacturer's instructions. Suspension hanger shall be of 4mm dia GI wire with adjustement screw. GI hold down clips shall be used. Board shall be finished as per Schedule after installing it on aluminium frame work. If finishes not specified then it shall be two coats of oil bound distemper.

(PARTICULAR SPECIFICATIONS(Contd.....)

(d) **POLYCARBONATE SHEET** : The sheets shall confirm to IS-14443. The colour, texture and other parameter shall be got approved from GE. The thickness shall be10 mm.

11. **ROOFING**.

11.1 **GALVALUME SHEET ROOFING** Galvalume sheet shall be 0.50mm thick (total coated thickness) of profile and colour as specified of 550Mpa minimum, yield strength. It shall be coated with hot dip alloy of 55% Aluminium 43.5% Zinc 1.5% Silicon and finished with resin coat on both surfaces @ 150 gm/Sqm of coating (total both surfaces) and shall be fixed using hot dip galvanized, self-drilling and self-tapping, screws neoprene and EPDM washers. Penetrations and laps in sheet shall be sealed by using proper sealant profile. HDPE fillers shall be provided wherever required to close voids between sheets, sheet & fasteners etc. The sheeting shall be fixed with side lap and end lap as recommended by manufacturer. Ridges and hips shall be covered with Galvalume sheets of 0.5mm thickness. These shall be fixed to roof as per manufacturer instructions and standard Engineering practice. The entire fixing process of Galvalume sheets shall be executed through a specialist agency identified by the supplier and as approved by GE. The contractor shall submit the original manufacturer's test certificate in connection with chemical composition of the steel, tensile/yield strength of steel sheet, corrosion resistance (salt spray) test, dimensional (thickness of both base metal and coating), coating adhesion (180 degree bend test) etc and also submit manufacturer's guarantee certificate of product confirmation of standard. GE shall also carry out independent test as per the test mentioned above from random samples through Govt approved laboratories/command/zonal laboratories. The cost of testing, transportation of materials etc shall be borne by the contractor irrespective of the results of testing and no extra claim whatsoever shall be admissible. The sheets shall be protected with guard film. The material shall be procured directly from manufacturer.

11.2 **FIBRE REINFORCED TRANSLUCENT ROOF SHEETS**: This shall be provided as mentioned in the BOQ and relevant Indian Standard.

11.3 **EAVES BOARD:** Eaves board shall be provided of MS Sheet 1.40mm thick and 250mm wide fixed with angle iron frame 30x30x4mm in lieu of wdn eaves board if shown on drawings. Exposed surfaces of eaves board shall be painted with two coats of synthetic enamel paint over a coat of primer. Rear/hidden surfaces of eaves board shall be treated with one coat of red oxide primer.

11.4 **RAIN WATER PIPES: R**ain water pipes and fittings shall be of dia as specified UPVC pipe (SCH 80). The pipes and fittings shall be ISI marked (IS-13592) and 6 Kgf/Sqcm pressure rating, procured from any of the manufacturers as per the approved list of makes/ agencies. Pipes shall be fixed on walls with suitable PVC holder bat clamp and shall be embedded in PCC 1:3:6 type C-1 block of size 150 x 150 x 150 mm and as directed by GE.

11.5 **OUTLET PIPES / SPOUTS:** Length of outlet pipes / spouts shall not be less than 300 mm projected from the outer face of the wall where length is not shown, it shall be taken as 600 mm overall. The outlet pipes /spouts shall be of galvanized iron pipes of medium grade, 50 mm bore unless otherwise shown on drawings. The balconies shall be provided with 50 mm dia GI pipes.

12. **FLOORING**

12.1 **GENERAL**. Ordinary Portland cement as specified in clause 3.2.1 here-in-before shall be used for all flooring. Provisions contained in clauses 13.25, 13.27, 13.32 & 13.38 of MES Schedule (Part-I) are to be adopted for laying floors and pavements. Floors shall be laid to levels or to falls as shown on drgs and as directed by the Engineer-in-Charge. Floor finish shall be extended over dwarf walls, doors and other openings. Floor topping of cast-in-situ cement concrete shall be finished even and smooth without using extra cement as specified in clause 13.32.8 of MES Schedule Part-I. Cast-in-situ terrazzo shall be in grey cement (OPC) without pigment unless otherwise indicated. In location where terrazzo with white cement has been indicated, it shall be casted with white cement and pigment. White cement shall be Birla/L&T/JK Cement. Marble aggregate (Chips) shall be of size 6mm to 12mm for terrazzo floors. The colour of marble chips shall be as shown on drawings or as approved by GE, when not indicated in drawings. Surfaces finished in terrazzo shall be grinded and wax polished as specified in clause 13.38.6 to 13.38.8 of MES Schedule Part-I. The grinding by hand may be allowed in WC/Bath, if grinding by machine is not feasible in such situations. These situations are in addition to those as specified in clause 13.38.6 of MES Schedule Part-I. Sub base shall not be laid in panels.

(PARTICULAR SPECIFICATIONS Contd.....)

12.2 **PCC** 40/25 mm thick PCC (1:2:4) type B-1/B0 (using 20/12.5 mm graded crushed stone aggregates) finished even and smooth without using extra cement laid in alternate bays. 4mm thick glass dividing strips shall be provided only where mentioned in BOQ.

12.3 **KOTA STONE FLOORING**. 18 to 20mm thick Kota stone slabs (Machine cut) of best quality as approved by GE, jointed with grey cement slurry, with pigment to match the shade of the slab shall be provided over 15-18mm thick bedding layer of cement mortar (1:3) using cement slurry @ 4 Kg/Sqm. The floor shall be grinded and polished to mirror finish all as specified in MES schedule Part-I.

12.4 **PRE-POLISHED KOTA STONE.** Pre-polished Kota stone shall be of 18 to 20mm thick pre mirror polished hand cut of best quality set and jointed with grey cement slurry with pigment to match the shade of slab using cement slurry @ 4 Kg/Sqm all as directed.

12.5 **GLAZED CERAMIC TILES**

12.5.1 Material and workman ship of Glazed Ceramic Tiles shall be as specified in clause No 13.14, 13.40, 13.40.1 to 13.40.2 of SSR Part – I unless otherwise specified.

12.5.2 Tiles shall be from any of the make as specified in Appendix “A “hereinafter.

12.6 **NON SKID CERAMIC TILES FLOORING:** Non skid ceramic tiles flooring shall be Ist quality provided with mat superior finish ceramic tiles of size specified in the BOQ and Coloured/ Pattern, Make/Brand as specified conforming to IS 15622-2006/ISO-3006 (Grade B-II (a)) Shall be laid over screed as mentioned in the BOQ using cement slurry @ 3 Kg/Sqm over PCC 1:2:4 type B-0 (using 12.5 mm graded crushed stone aggregate, over 75 mm thick PCC.

12.7 **VITRIFIED PORCELAIN TILES** Material and workman ship of vitrified Tiles shall be as specified in clause No 13.15, 13.41, 13.41.1 and 13.41.2 of SSR Part – I unless otherwise specified. Make as specified in Appendix “A “hereinafter.

12.8 **TERRAZO FLOORING TILES :-**

12.8.1 Material and workman ship of cement concrete and terrazzo flooring tiles shall be as specified in clause No 13.17, 13.17.1, 13.17.1, 13.39, 13.39.1 TO 13.39.5 of SSR Part – I unless otherwise specified.

12.8.2 Cement concrete and Terrazzo flooring tiles shall be from any of the make as specified in Appendix “A hereinafter.

12.9 **PLASTERING, POINTING & SKIRTING/DADO**. Plaster and skirting/dado shall be returned in jambs, soffits of lintels and windows etc. All plastered surfaces shall be trowelled to even and smooth surfaces without using extra cement. Thickness of cement plaster mentioned is exclusive of dubbing. Dubbing may, however, be done in one operation with plaster and no separate payment for dubbing shall be admissible. All corners, angles, junctions shall be truly vertical or horizontal as the case may be and shall be carefully finished. Corners around joints openings and junction of walls shall be rounded to minimum radius of 5mm. For specification of refer clause on concrete herein before. The sand shall comply with clause 14.5.2 of SSR Part-1.

12.10. **KOTA STONE SKIRTING.** The kota stone tiles in skirting shall be 12 to 15mm thick. The shade of the tiles shall be same as of floors. Tiles shall be set in neat cement paste over 10 mm thick rendering in CM (1:3) and jointed in the cement paste of the same shade. Grinding and polishing shall be done as specified for floor but by hand.

12.11 **INTER LOCKING PAVING TILES/ BLOCK.** The thickness and grade of factory made cement concrete interlocking paver block shall be as mentioned in BOQ. The paver blocks shall be made by block making machine with strong vibratory compaction of approved size and design/ shape in required colour and pattern. The joints shall be filled with the material as specified, complete all as per the direction of Engineer-in-charge/ GE. Precast concrete Paver blocks shall conform to IS 15658:2006. Paver blocks shall be sound and free from cracks or other visual defects. The tolerance on length or breadth of paver blocks shall be +2mm and tolerance on thickness of tiles shall be +3mm. Water absorption shall not be more than 6 percent by mass. Shapes shall be triangular, Zigzag, Haxagon or other shape as indicated. The paving tiles/block shall be tested before incorporation in the work from any NABL certified lab or IIT or Govt. Engineering College or any other Govt approved lab as approved by GE as per frequency indicated in relevant IS and all as directed by Engr-in-Charge. Cost of test shall be borne by the contractor.

(PARTICULAR SPECIFICATIONS(Contd.....)

12.12 **CHEQUERED CEMENT CONCRETE TILES**. Chequered cement concrete tiles, shall be of colour conform to IS 13801-1993 specification. Chequered tiles shall be with the centre to centre distance of chequers not less than 25mm and not more than 50mm, the groove in chequers being uniform and straight with the depth of grooves not less than 3mm. Tiles shall be manufactured by pressure process. Thickness of wearing layer measured from top of the chequers shall not be less than 6mm. Unless otherwise directed, the tiles shall be supplied with initial grinding and grouting of the upper layer. The upper layer of the tiles shall be free from projections, depressions and cracks, holes, cavities and other blemishes. The edges of the tiles may be rounded. All angles shall be right angles and all arises shall be sharp and true. The colour and texture of the wearing layer shall be uniform throughout its thickness. The size of Chequered Tiles shall be 300 x 300mm x 25mm thick as indicated. The tolerance on length or breadth of tiles shall be +1mm and tolerance on thickness of tiles shall be +5mm. In addition the difference in thickness between the thickest and thinnest tile in the sample shall not exceed 3mm. Laying of tile shall be as per SSR Part I clause 13.39.

13. **WHITE/COLOUR WASH, DISTEMPERING AND ACRYLIC PAINTS ETC**

13.1 **WHITE WASH**

13.1.1 The contractors quoted rates shall be deemed to include for working at any height of the buildings/structures and all the necessary scaffolding required and nothing extra on this account shall be admissible.

13.1.2 Surface to be white (lime) washed shall be first prepared by broming down and all as specified in clause 15.12.2 of MES Schedule (Part-I).

13.1.3 White (Lime) wash shall be prepared and applied as specified in Clause 15.12.3 & 15.12.5 of MES Schedule (Part-I) respectively.

13.1.4 After the surface has been prepared as specified above a coat of white wash shall be applied over the patches where required. Then the specified number of coats shall be applied on the entire surface to present a uniform finish. The contractor's quoted rates shall be deemed to include the above and nothing extra shall be Admissible on this account.

13.2 **COLOUR WASH**

13.2.1 Surfaces to be colour washed shall be prepared all as specified for white (Lime) wash hereinbefore.

13.2.1 Colour wash shall be prepared and applied as specified in clause 15.12.4 and 15.12..6 and 15.12.6.1 of MES Schedule (Part-I).

13.2.2 After the surface is prepared as above a coat of colour wash shall be applied over patches and specified number of coats of colour wash shall then be applied to present uniform finish. The contractor's quoted rates shall be deemed to include the above and nothing extra shall be admissible on this account.

13.2.3 Lime colour of approved make and shade (as approved by the Engr-in-Charge) shall be incorporated in the colour wash, irrespective of any other materials specified elsewhere.

13.3 **DISTEMPERING WITH OIL EMULSION DISTEMPER**

13.3.1 Distemper Oil emulsion shall be of quality conforming the relevant Indian Standard Specifications (ISS) as mentioned in the MES Schedule. The distemper shall be mixed in the proportion (by wt) of four parts of paste to one part of cold water (or in the proportion if any, specified by the manufacturer this proportion shall be strictly adhered to) so as to form uniform mixture suitable for applying by brush and shall be carried out as specified in MES Schedule.

13.3.2 The oil emulsion distemper shall be any of the make as specified in Appendix 'A' "hereinafter and brought in the sealed container.

(PARTICULAR SPECIFICATIONS(Contd.....)

13.4 **CEMENT BASE PAINT**

13.4.1 Surface to be painted shall be prepared by washing and stopping as specified in clause 15.15.2 of MES Schedule (Part-I) Cement based paint shall be prepared and applied on prepared surface all as specified in clause 15.15 of MES Schedule Part-I Brand tint/shade shall be got approved from GE.

13.4.2 Contractor shall use/incorporate cement paint from any of the make as specified in Appendix "A" hereinafter. Before placing bulk order in any of above manufacturers/brands of paint, contractor shall obtain written permission from GE for the brands of cement paint be intends to used and get the samples approved thereof before their use in the work under this contract.

13.5 **WALL PAINTING WITH ACRYLIC EMULSION PAINT (INTERIOR WALLS**): It shall be done as under:-

(i) The Acrylic emulsion paint shall be of manufacturer as given in Appendix and of premium quality. The paint shall be 100% Acrylic and semi Acrylic paint shall not be used in the work.

(ii) The surface to be applied with Acrylic emulsion paint shall be cleaned to remove loose dirt or dust, lichen, algae, fungi or any organic growth by use of stiff brush. Then the surface shall be washed well and allow water to dry. All cracks, voids and minor damages shall be patched/repaired with white cement putty or with plaster of paris and surface shall be prepared with white cement based putty all as specified in clause 15.19 of SSR Part-I.

(iii) Over this prepared surface, a coat of acrylic primer shall be applied as per manufacturer's instructions. This shall be allowed to dry for 4 to 6 hours before application of subsequent coat.

(iv) Acrylic paint shall be applied in atleast two coats as per manufacturer instructions. The Acrylic paint shall be prepared as per manufacturer's instructions by adding water in proportions as suggested. No other thinner shall be used. Paint shall be applied with brushes or roller and shall be allowed to dry for 3 to 4 hours between subsequent coats.

13.6 **DRY DISTEMPER** Dry distemper shall be in two coats over a coat of primer as per Clause 17.1.3 of SSR (Part-I) manufactured by reputed firms mentioned for paints conforming to IS-427 shall be provided over prepared surface as specified in clause 15.13 of SSR (Part – I). DDL (Fevicol) synthetic resin in the preparation as recommended by the manufactures shall also be added in Distemper solution.

13.7 PLASTIC EMULSION Refer clause 17.6 of SSR Part-I (2009).

13.8 ACRYLIC EXTERIOR WEATHER RESISTANT PAINT (WEATHER PROOF PAINT):- Refer clause No 17.17 of SSR Part-I 2009. Preparation of surfaces shall be got approved from Engr-in-Charge before applying finishes.

13.8.1 Newly plastered surfaces shall be scrapped to remove dust, loose or flaking materials, grease & dirt to ensure a smooth and clean surface. Paint shall be thinned down as per manufacturer's instructions. Application shall be done using brush or roller. The recoating time between two coats shall be around 1 to 6 hours.

13.8.2 The period of Guarantee for the acrylic exterior weather resistant paint (weather proof paint) treatment shall be 5 (Five) years from the certified date of completion for treatment on newly plastered surface of brick or concrete and treatment after complete removal of existing treatment on wall/ ceiling . The contractor shall be responsible for effectiveness of the acrylic exterior weather resistant paint (weather proof paint) during this period and any damage/ discolour noticed therein during this period shall be made good by him at his own cost.

13.8.3 The contractor shall furnish guarantee in favour of the Garrison Engineer for the efficacy of the acrylic exterior weather resistant paint (weather proof paint) during the guarantee period.

(PARTICULAR SPECIFICATIONS(Contd.....)

13.8.4 An amount of security deposit equal to 2.5% (Two point five zero percent) on the amount of acrylic exterior weather resistant paint (weather proof paint) at contract rates for the building for which paint to be carried out shall be retained from the contractor's final bill as security deposit for the acrylic exterior weather resistant paint (weather proof paint) which shall be released to the contractor only after expiry of guarantee period. The contractor may alternatively furnish a fixed deposit receipt in favour of GE in lieu of sum to be retained as security deposit for acrylic exterior weather resistant paint (weather proof paint) from any approved Schedule Bank.

13.8.5 Condition 46 of the General Conditions of Contracts (IAFW 2249) shall be deemed to be amended to the extent mentioned above.

14. **GLAZING**. All glazing shall be with sheet glass of ordinary quality and shall conform to IS -2835. Glass shall be free from all defects viz bubbles, waviness etc. and shall be of uniform thickness. In case thickness not mentioned elsewhere glazing to windows/doors shall be with 5mm thick sheet glass. Glass used shall be plain sheet glass except in baths/WC/Toilets/Lav/WC as applicable where it shall be pinhead glass. The glazing to steel frames, shall be fixed with special spring glazing clips with putty and glazing to wooden joinery shall be fixed with wooden beading with putty all as specified in clauses 16.5 to 16.10.2 of MES Schedule Part-I. Glass shall be fixed to Aluminium windows /vents with aluminium snap beading alongwith PVC/neoprene gasket.

15. **PAINTING**. All synthetic enamel paints and emulsion paints shall be of 1st quality manufactured by the standard firm. The contractor shall inform the GE, within three weeks of the acceptance of the tender, the brand/ names of the manufacturers of paint proposed to be used in the work and submit samples thereof to GE before their use in work and obtain prior approval of GE. Paints for priming coat, under coat and finishing coat shall be of the same brand. Tint of paint, if not mentioned in drgs/schedule of finishes, shall be as approved by the GE. Pink primer shall be used on wooden surface and red oxide primer shall be used on steel surface. Concrete surface shall be applied with cement primer. The workmanship shall be as under:-

(a) All surfaces required to be painted shall be smoothened size and knotted and then applied with priming coat, stopping and filling [filler coat] shall be done after priming coat and surface rubbed down to a level and smooth surface and thereafter under coat and finishing coat shall be applied all as specified in clause 17.6 of MES Schedule Part-I. Steel and iron work shall be painted in the manner as specified in clause 17.8 of MES Schedule Part-I.

(b) All steel and wooden surfaces including wooden based surfaces such as plywood, particle board etc shall be given two coats [under coat and finishing coat] of synthetic enamel paint. Aluminium painted shall be given two coats of aluminium paint conforming to IS-2339 over a coat of appropriate primer.

16. **WATER PROOFING TREATMENT TO RCC ROOFS**

16.1 The Water Proofing Treatment to RCC Roof shall be provided as per Sch'A':-

(i) Taking down existing roof treatment from roof and terrance, roughening old concrete surfaces by picking, sweeping, cleaning and washing down with water to remove dust, etc.

(ii) Roof surface shall be plastered with 10 mm thick in CM 1:3 mixed with water proofing compound as per manufacturers instruction, finished the surfaces fair and even.

(iii) Curing of plastered surfaces by ponding with water shall be done all as specified.

(iv) Cleaning of plastered roof surface and applying one coat of cold bituminous primer @ 0.25 Kg per Sqm conforming to IS: 3384-1986.

(v) A layer of APP based polymeric membrane, 3mm thick (minimum), weighing 3.5 kg/Sqm reinforced with polyester non woven fabric matt (weighing not less than 160 gms/Sqm) shall be applied over bituminous primer as specified above. Application of polymeric membrane shall be by torch application. The overlaps shall be minimum 100mm on the transverse direction and 75mm on longitudinal direction. Applying one coat of aluminium paint @ 0.11 Litre per Sqm over APP membrane.

(PARTICULAR SPECIFICATIONS(Contd.....)

NOTES:-

(a) **ATACTIC POLYPROPYLENE POLYMER MODIFIED WITH NON –WOVEN POLYESTER MATT PREFABRICATED MEMBRANE (APP)** :- The APP modified membrane shall be 3mm thick APP (Atactic Polypropylene Polymer) modified prefabricated five layer, weight not less than 3.50 kg/Sqm water proofing membrane, black finished reinforced with non woven polyester matt and weight not less than 160 gms/Sqm. APP modified membrane shall be as specified in para 11.44 (including sub paras) of MES Schedule Part-I.

(b) In no case damage / puncture of APP modified proofing membrane shall be permitted.

(c) The entire work of water proofing treatment to RCC roof slabs shall be carried out by engaging authorised applicator of the manufacturer. For this purpose, contractor shall submit the details of authorised applicator alongwith supporting documents to GE well in time and the work shall be executed only after written approval by GE, all materials used for water proofing treatment shall be of same manufacturer. However, the main contractor shall give guarantee for efficiency of water proofing treatment for a period of Ten Years as specified here-in-before.

(d) Random samples of Polymeric Water Proofing membrane, as selected by GE if so desired by him shall be got tested from National Test House, SEMT Wing CME, Regional Research labs, IITs, National Institutes of Technology, NABL laboratories, Govt Engineering Collage or Any Govt laboratories where such facilities exists before incorporation in the work.

(e) In case the test result does not meet the specifications given above, the entire lot shall be rejected by the GE. In such a case, the contractor shall bring the fresh lot of desired quantity of APP modified Water Proofing Membrane and the same procedure of testing repeated till the time satisfactory test results are obtained. Nothing extra will be paid to contractor on this account.

(f) The work of fixing membrane shall be got executed through the authorized applicator of the manufacturer of the membrane.

16.2 **TESTING OF WATER PROOFING TREATMENT**

16.2.1 After the water proofing treatment is completed, leak proof tests shall be done by ponding. For this purpose, in roof, mud, mortar, fillet ponds shall be made longitudinally one meter a part over entire treated surface of roof to form pond of suitable size as directed by the Engineer-in-Charge. (These ponds shall be filled with potable water so that an average 50 mm (minimum 25 mm and maximum 75 mm) height of water is maintained during the test period.

16.2.2 Test shall be carried out continuously for a period of 48 hours. Any seepage notified shall be rectified by the contractor and making good the defective portion to entire satisfaction of the GE, who will pass this stage.

16.2.3 All mud fillet bends shall be removed and surface made clear and tidy after completion of satisfactory testing.

16.3 **GUARANTEE FOR WATER PROOFING TREATMENT**

16.3.1 The period of Guarantee for the water proofing treatment shall be **10 (ten) years** from the certified date of completion. The contractor shall be responsible for effectiveness of the water proofing treatment during this period and any leakage noticed therein during this period shall be made good by him at his own cost.

16.3.2 The contractor shall furnish guarantee in favour of the Garrison Engineer for the efficacy of the water proofing treatment during the guarantee period.

(PARTICULAR SPECIFICATIONS(Contd.....)

16.3.3 An amount of security deposit equal to 2.5% (Two point five percent) on the amount of water proofing treatment at contract rates for the building for which treatment to be carried out shall be retained from the contractor's final bill as security deposit for the water proofing treatment which shall be released to the contractor only after expiry of guarantee period. The contractor may alternatively furnish a fixed deposit receipt in favour of GE in lieu of sum to be retained as security deposit for water proofing treatment from any approved Schedule Bank.

16.3.4 Condition 46 of the General Conditions of Contracts (IAFW 2249) shall be deemed to be amended to the extent mentioned above.

16.4 **WATER PROOFING TREATMENT TO SUNKEN SLABS**

16.4.1 Water proofing treatment to sunken slabs shall be provided as per Schedule 'A'.

16.4.2 Testing of all traps & joints and sunken slab shall be as per IS-1742 before filling of sunken slabs, including smoke test.

17. **GUNITING**: Guniting shall be done as indicated in Sch 'A' BOQ. The steps to be taken while carrying out the gunting are: Taking down cement plaster, raking out joints, hacking for key. The parts of brick work which is intact shall be roughened to needful gunite. Removal of dust from joint and keys by sand blasting and wire brushing as directed. Applying polycrete for concrete and mixture. Providing Guniting with pressure 2.2 to 2.50 kg/Sqcm with mixture of 1 cement and 2 of sand (1:2) of thickness as indicated including stripping, floating with wooden stripper for proper dimensional control including necessary form work, scaffolding complete. Dismantling of concrete member by chiseling from RCC member (Column, Beams, Slab, Chajjas, facia etc). Exposing the existing reinforcement without disturbing the core of RCC member, cleaning the surfaces with wire brushes, providing additional reinforcement as directed at site, where required, binding with 75x75x2mm thick (dia) welded mesh with binding wire and tack welding using adhesive, plaster create for concrete admixture. The unit rate quoted by tenderer shall be deemed to be inclusive of the above provision. Gunited surfaces shall be cured by the contractor for seven days at his own cost and arrangement. The contractor shall make arrangement at his own cost for removing the rubbish etc before and after the Guniting work.

18. **SANITARY APPLINCES AND PLUMBING WORKSANITARY APPLIANCE**

18.1 **GENERAL**

18.1.1 Refer Clause No.18.32.1 and 18.32.1.1 of MES Schedule Part-I.

18.1.2 These shall be first quality, white, vitreous china & ISI marked.

18.2 **SQUATTING PAN (ORISSA PATTERN)** :-

18.2.1 Squatting pans water closet shall conform to IS-2556 (Part-III) and their pattern shall be Orissa pattern size as mentioned in Schedule 'A' (BOQ) and shall be as per MES schedule Part-I.

18.3 **(EUROPEAN TYPE WC (PEDESTAL PATTERN)**:-

18.3.1 Wash down water closet shall conform to IS-2556 (Part-II) and their pattern shall be Pattern – 4 (Pedestal WC with independent cistern and concealed S-trap or P-Trap), white, 410 mm high (minimum).

18.3.2 Refer Clause No. 18.32.2 of MES Schedule Part-I.

18.4 **UNIVERSAL WATER CLOSET (ANGLO-INDIAN TYPE)** :

18.4.1 Universal Water Closet (Anglo-Indian Type) shall conform to IS-2556 (Part-15) and their pattern shall be Size shall be as specified in Schedule 'A' (BOQ) with S-trap or P-Trap, and shall be as per MES schedule Part-I.

18.5 **PLASTIC WATER CLOSET SEATS AND COVERS**

18.5.1 Refer Clause No. 18.36 of MES Schedule Part-I.

(PARTICULAR SPECIFICATIONS(Contd.....)

18.5.2 Plastic water closet seat and cover shall be ISI marked on ISI 2548 recessed bottom made out of moulded poly propylene.

18.6 **LOW LEVEL FLUSHING CISTERN**

18.6.1 Refer Clause No. 18.34 and 18.87 of MES Schedule Part-I.

18.6.2 The flushing cistern shall be of PVC "feather touch" with symphonic fittings ISI marked moulded from vergin poly propylene.

18.7 **URINAL**

18.7.1 Refer Clause No. 18.32.7 and 18.88.8 of MES Schedule Part I.

18.7.2 These shall be with grating and unions, waste pipe 32mm dia nominal bore GI medium grade of required length.

18.8 **PVC FLUSHING CISTERN FOR URINAL**

18.8.1 Refer Clause No. 18.34, 18.88.2 and 18.88.3 of MES Schedule Part-I.

18.8.2 Flushing cistern shall be PVC automatic flushing cistern.

18.8.3 The flushing cistern shall be of PVC "Feather Touch" operated bearing IS-7231, diaphragm bearing IS-13049 and detrim valve, float valve.

18.9 **WASH BASIN**

18.9.1 WHB shall be provided complete all as specified in Schedule ‘A’ (BOQ).

18.9.2 Refer Clause No. 18.32.5 and 18.89 of MES Schedule Part-I.

18.9.3 Wash hand basin (WHB) shall conform to IS-2556 (Part-IV), white.

18.10 **PVC CONNECTOR**

18.10.1 Refer Clause No. 18.5 of MES Schedule Part-I.

18.11 **MIRROR**

18.11.1 Refer Clause No. 18.37, 18.38, 18.91 and 18.92 of MES Schedule Part-I.

18.11.2 The mirrors shall be of indigenous of make as per APPENDIX ‘A’, copper plated at the back free from ripples, spats, bubbles, and other defects with edges slightly rounded off around.

18.12 **TOWEL RAIL**

18.12.1 Refer Clause No. 9.24 of MES Schedule Part-I.

18.13 **STAINLESS STEEL SINKS WITH DRAINAGE BOARD**:

18.13.1 Refer Para 18.100 of MES SSR Part I. The sink with draining board shall be type ‘C’ and overall size as specified in Schedule ‘A’ (BOQ). The sink with drainage board shall be provided with plug, washer and waste and fixed with a pair of MS cantilever bracket. The sink shall be provided with waste pipe of 32mm bore light grade GI pipe with fittings, length not less than 1m, duly connected to union. The surface of the sink shall be plastic coated to make it safe and scratch free.

18.14 **STAINLESS STEEL SINK WITHOUT DRAINAGE BOARD**:

18.14.1 Stainless steel sink shall be made of sheet thickness 1mm (min). The overall size shall be as specified in Schedule ‘A’ (BOQ). The sink shall be provided with plug, washer and waste and fixed with a pair of MS cantilever bracket. The sink shall be provided with waste pipe of 32mm bore light grade GI pipe with fittings, length not less than 1m, duly connected to union. The surface of the sink shall be plastic coated to make it safe and scratch free. All other details shall be mentioned on SSR Part I clause 18.100.

(PARTICULAR SPECIFICATIONS(Contd.....)

18.15 **STAINLESS STEEL PLATE RACK**:

18.15.1 Stainless steel plate rack shall be made of sheet thickness 0.80mm (min). The size of plate rack shall be as specified in Schedule 'A' (BOQ). The surface of the plate rack shall be plastic coated to make it safe and scratch free. All other details shall be mentioned on SSR Part-I clause 18.100.

18.16 **GLASS SHELVES**:-Glass shelves shall be of 6.0mm thick plain glass (minimum).

18.17 **UNPLASTICIZED POLYVINYL CHLORIDE (PVC-U) PIPE FOR RAIN WATER SOIL/ WASTE/ VENT PIPES & FITTINGS**

18.17.1 Refer Clause No. 11.20, 18.67.2 and 18.67.2.1 of MES Schedule Part-I and IS 13592-2013.

18.17.2 **COLOUR OF PIPE**:- Surface colour of the pipes shall be dark shade of grey.

18.17.3 **RAIN WATER PIPE**:- Rain Water Pipe shall be unplasticized polyvinyl chloride (PVC-U) pipes, ISI marked.

18.17.4 **SOIL/WASTE/VENT PIPES & FITTINGS** :-

18.17.4.1 Soil/waste/vent pipe shall be unplasticized polyvinyl chloride (PVC-U) pipes. In case of non availability of any diameter (Nominal outside diameter) mentioned the next available diameter (Nominal outside diameter) as per IS 13592 shall be provided without any price adjustment.

18.17.5 The contractor shall submit the manufacturer's test Certificate in original along with the test sheet giving the results as per IS 13592 and duly signed by the manufacturer with each consignment.

18.17.6 **TESTING**: - After laying of PVC (SWR) pipes or any other type of pipes, smoke testing shall be done all as specified in Clause No 18.79.2 and water testing shall be done as specified in Clause No 18.79.3 of MES SSR Part-I (2009).

18.18 **GULLY TRAPS**

18.18.1 Gully traps shall be of unplasticized polyvinyl chloride (PVC-U) with `P' or `S' trap with square/round mouth conforming to IS-651. PVC Gratings shall be of heavy quality as approved by GE. Gully Traps shall be provided with two cast iron gratings of appropriate size.

18.19 **NAHANI TRAPS/FLOOR TRAPS**

18.19.1 Refer Clause No. 18.25 of MES Schedule Part-I and shall conform to IS 3989.

18.19.2 Nahani Trap/Floor Trap shall be long arm type so that the joint must come outside the external walls.

18.20 **(PVC-U) COWLS**

18.20.1 Refer Clause No. 18.24 of MES Schedule Part-I.

19. **TUBES AND FITTINGS**

19.1 GI Tube and Fittings shall be of Mild Steel Galvanised Medium Grade bearing IS-1239 mark. Fittings conforming to IS-1239 if not available readily may alternatively comply with the requirement of IS-1879 for malleable Cast Iron Fittings but these shall also be ISI marked. GI Pipe shall be laid in floor/ on surface/ concealed in Wall/Floor conforming to provision given in IS-2065 and IS-12183.

19.2 **PP-R PIPES AND FITTINGS**

19.2.1 Refer Clause No. 18.114 of MES Schedule Part-I.

19.2.2 PP-R Pipes and Fittings shall be as described in Schedule 'A' (BOQ). Pipes and Fittings shall be of make mentioned in APPENDIX 'B' as approved by GE.

(PARTICULAR SPECIFICATIONS(Contd.....)

19.3 C-PVC PIPES AND FITTINGS

19.3.1 Refer IS 15778 and Clause No. 18.114 of MES Schedule Part-I for fitting and laying.

19.3.2 PP-R/CPVC Pipes and Fittings shall be as described in Schedule 'A' (BOQ). Joint of pipes should be done executed through authorized applicator of the pipe manufacturer and performance certificate from the manufacturer shall be submitted by the contractor before execution of work and applicator shall be approved by the GE.

19.4 **CAST IRON SOIL, WASTE AND VENTILATING PIPE**:- Refer clause No. 18.23 of MES Schedule Part-I.

19.5 **CAST IRON COWLS**:- Refer clause No. 18.24 of MES Schedule Part-I.

19.6 **CAST NAHANI TRAPS**:- Refer clause No. 18.25 of MES Schedule Part-I.

19.7 **CAST FLOOR TRAPS**:- Refer clause No. 18.26 of MES Schedule Part-I.

19.8 **HDPE PIPES**:- Refer clause No. 18.6 of MES Schedule Part-I and shall be confirm to IS 16098-2013.

19.9 **BIB TAPS AND STOP VALVE** :- These shall be screw down type and shall bear ISI marking. Bib taps & Stop valve shall be polished bright.

19.10 **FANCY BIB TAPS AND STOP VALVES** :-They shall be cast copper alloy screwed down type and shall conform to IS-8931. They shall be approved by the GE and shall be nickel chromium plated as given in Schedule 'A' (BOQ).

19.11 **GATE VALVES AND GLOBE VALVES** :- These shall be standard quality Class I conforming to IS:778, Specifications and as approved by the GE. Valve shall be of Class I suitable for 1.0 MPa Working Pressure.

19.12 **BALL VALVE** :- Ball Valves for Water Tanks shall be high pressure type of brass with Polythene Ball conforming to IS-1703.

19.13 **SHOWER ROSE** :- Shower Rose shall be brass chromium plated as specified in Schedule 'A' (BOQ).

19.14 **BOTTLE TRAP** : Refer Clause No. 18.22.1 of MES Schedule Part-I.

19.15 **TOILET PAPER HOLDER** : Refer Clause No. 18.39 of MES Schedule Part-I.

19.16 **GLASS SHELF ASSEMBLY** : Refer Clause No. 18.37 of MES Schedule Part-I.

19.17 **WATER STORAGE TANK** :-

19.17.1 Refer Clause No. 18.60 on of MES Schedule Part-I.

19.17.2 HDPE water tanks shall be provided all as per manufacturer's instructions. Tank shall be of rotational moulded polyethylene water storage tanks in double layers otherwise specified in Schedule 'A' (BOQ) and cylindrical vertical type with manhole lids (closed top) with ISI marked and Make as given in APPENDIX 'B' here-in-after.

19.17.3 **JOINTING OF PIPES TO STORAGE TANKS** :- For jointing steel pipes to storage tanks, the end of the pipe shall be threaded, passed through a hole in the tank and secured by back nuts both inside and out side. The Pipe End shall be flush with the face of the inside back nut to obviate corrosion of the pipe threads.

19.17.4 **SEWAGE PIPE SGSW**: All salt glazed stone ware pipes shall be of Grade 'A' socketed pipe of required dia as per Sch 'A', conforming to IS 651 – 1980. Pipes shall be laid accurately to line and gradient. The joint shall be made by caulking two strands of tarred gasket of yarn in thick cement slurry maximum up to one forth (or 13mm) of the depth of the socket. The remaining of the socket shall be filled with CM (1:1) all as specified in clause 18.70 of MES Schedule Part I. Width of excavation of salt glazed stone ware pipes for purpose of payment shall be authorized width as defined in clause 3.2 of MES schedule Part-II. Concrete bedding for pipe shall be as per clause 18.68 on srl page No 18–26 of SSR Part-I 1991. The make of SGSW pipe shall be of M/S Perfect Potteries Jabalpur, Devraj Industries Gaziabad or any equivalent make as approved by GE.

(PARTICULAR SPECIFICATIONS(Contd.....)

19.17.5 **LLDPE SEPTIC TANK**: The Septic tanks shall be made of roto moulding process with good impact strength and stiffness, excellent internal and external finish. This shall have density not less than 938 Kg/Cum and shall be easy to install, chemical resistant, hygienically safe and free from corrosion and able to avoid root penetration. The tank shall have ability to ensure a high degree of settlement and sedimentation, thus providing a low level of suspended solids, suitable for discharge into any underground soak-away system. Septic tank shall be provided primarily for biological treatment to crude sewage produced in a normal domestic environment. "Produced shall stays on top of the effluent and escapes through the vent pipe. This effluent shall undergoes further anaerobic digestion in the subsequent chambers all along the long flow path without any turbulence inside the tank ensuring high degree of settlement & sedimentation.

19.17.6 **SFRC MANHOLE COVERS**: The size of SFRC cover & frames shall be as mentioned in BOQ. The SFRC cover & frames shall confirm to IS 12592 and shall be of size and design/ shape of required colour and pattern as approved by GE. SFRC cover & frames shall be fixed as per manufacturer instructions and complete all as per the direction of Engineer-in-charge/ GE. The technical specification of SFRC cover & frames shall be followed strictly as per relevant Indian Standard on product. **Makes :- Polycrete, A-One Tiles, Ultra Tiles Pvt Ltd, Nitco, Hindustan Tiles.**

20 ELECTRIFICATIONS

20.1 GENERAL REQUIREMENTS

20.1.1 **KNOWLEDGE OF WORK**:- The tenderers shall be deemed to have thorough working knowledge of the exact requirement of the job. In case of any doubt clarification may be sought from CESZ before tender is submitted. Tenderers are advised to visit the site and get themselves conversant with site conditions. No claim what so ever shall be entertained/accepted in case the tenderer is coming up with an excuse of having incomplete knowledge of work involved.

20.1.2 **WORK IN ACCORDANCE WITH SPECIFICATION** :- Work under this contract shall be carried out as per brief description given in Schedule 'A' (BOQ), Particular Specifications. Work shall also be in conformity with IE Rules Latest Edition, relevant IS specification and MES Schedule which are forming part of the contract. The entire work shall be carried out properly with due care and under the supervision of competent technical personnel.

20.2 GENERAL

20.2.1 Refer Clause No. 19.2 of MES Schedule Part-I and requirements of Clause No. shall be strictly followed.

20.2.2 Supply and Installation of Electrical Equipments and Accessories shall strictly comply with the provisions contained in the Latest Edition of India Standard Code of Practice and/or IS Specification as applicable, Works except where such Regulations and Rules are modified by these Particular Specifications.

20.2.3 All Electrical Work shall be carried out in conformity with the requirements of the Indian Electricity Act 1910, Indian Electricity Rules 1956 framed there under, Indian Standard Codes of Practice, Fire Insurance Act as applicable and also the relevant Regulations for Electric Supply and Installations of all Electrical Equipments, Fittings and Accessories as amended from time-to-time.

20.2.4 The supply and installation of all Electrical Equipments and Accessories shall strictly comply with the provisions contained in the latest edition of Indian Standard code of Practice IS Specification as applicable to above mentioned work except where such regulations and rules are modified by these specifications.

20.2.5 **QUALIFIED ELECTRICAL SUPERVISORS / ENGINEERS** :- Refer Condition 25 of General Conditions of Contracts, IAFW–2249. All Electrical Works shall be carried out by properly Skilled and Licensed Electricians under the supervision of qualified Electrical Supervisors/ Engineers. The Contractor shall on demand shall produce such evidence of qualifications of his Workmen/ Supervisors/Engineers, either at the commencement of Work or at any time thereafter/during currency of the Contract. The entire Work shall be High Class with the best Workmanship and to the entire satisfaction of Engineer-in-Charge/GE.

(PARTICULAR SPECIFICATIONS(Contd.....)

20.3 INTERNAL WIRING

20.3.1 **MATERIALS**

20.3.1.1 All the Electrical Accessories incorporated in the work shall conform to the relevant IS and shall bear ISI mark. The steel conduit shall be of maximum 16 SWG thickness and rigid PVC conduit shall be suitable for heavy mechanical stresses as given in IS and will be ISI marked.

20.3.1.2 Work under this part will cover the point wiring and fixing of various light fittings in the building as per drawing. Work will be carried out as per provision of IS 732-1989. IE Rules as per MES Schedule and as per Sound Engineering Practices.

20.3.1.3 The general layout of light fittings is shown in drawing. The location of fitting may be altered by Engineer- in-Charge to suit the site requirement in which case the contractor shall have no claim for any extra amount on account of such changes in point wiring.

20.3.1.4 No price adjustment in rates shall be made on account of changes in location of fittings and accessories shown on drawings.

20.3.1.5 Recessed terminal boxes for housing switches, socket outlets, power plug, regulator etc. shall be provided flush with the walls.

20.3.1.6 The internal wiring shall be in concealed PVC conduits. The PVC conduits and its fittings shall conform to the requirements of IS-9537-1983 and IS-3480-1966. PVC conduit shall be suitable for heavy duty as asked in IS. The size of wiring cable shall be as stipulated in Schedule 'A' (BOQ). The PVC unsheathed cable should conform to IS-694-2010 and suitable for 650/1100 volt grade. All cables and switches in work shall have ISI mark.

20.3.1.7 It shall however be noted by the contractor that layout of the fittings shall be completed to match the casted structural members and cost of extension/supports considered unavoidable and essential by the tenderer shall deemed to be inclusive in the unit rate quoted for wiring.

20.3.1.8 The switch boards shall be located in such a manner that they are easily accessible and such that there is adequate working space around the switch boards. The tentative location shall be actually marked on the walls in chalk and written approval of Engineer-in-Charge shall be obtained before proceeding with the erection of the switch boards. The Engineer-in-Charge and the contractor shall associate themselves during the civil work so as to finalize the firm locations of switch boards, DBs, etc in consultation with departmental staff while executing the civil work.

20.3.1.9 All conductors used in point wiring/ sub-main wiring shall be FR-LSH (Flame retardant low smoke and halogen) Copper conductor conforming to IS 694-2013 and shall have ISI mark. Insulated wire used in point wiring will have colour code matching with phase i.e. wire starting from 'R' phase will have red colour and similarly with 'Y & B' phase will have yellow and blue respectively. In neutral, only black colour wire will be used. No joints in PVC sheathed Copper conductor for phase and neutral or in earth wire is permitted. All stranded wires will be crimped with suitable Copper lug before they are connected to MCB. Green colour wire shall be used in earth wire along with other wires.

20.3.1.10 The cable and the flexible cords shall be stranded Copper conductor. Same colour code suggested in above said para shall be followed.

20.3.1.11 The surface wiring for points outside the building shall run on internal face of wall as far as practicable. Only at terminal points it shall be taken out by puncturing the wall and connect with light fitting/switches.

20.3.1.12 The maximum number of point wiring of light/socket 5/6 Amp on an independent circuit from DB shall not exceed 8 Nos in any case and not more than two power points in an independent circuit from DB.

20.3.1.13 Each circuit will have its independent neutral and looping is only permitted with in the circuit only. For example if five circuits are going out from one distribution board then five neutral wire will also go along with phase from neutral bus-bar.

(PARTICULAR SPECIFICATIONS(Contd.....)

20.3.1.14 For concealed wiring, it is advisable not to lay conduit horizontally in wall but all horizontal running of conduit should be done in RCC slab. As far as possible, only vertical run of conduit should be laid in the wall.

20.3.1.15 Wherever more than two conduits are running together vertically concealed in wall, tenderer is advised to wrap the PVC conduits in wire mesh and then do plaster otherwise cement plaster may not stick to the PVC conduit and it is likely to come out. Cost of wire mesh used is deemed to be included in the quoted cost of wiring.

20.3.1.16 Wherever Tube Light Fitting has been shown suspended by Conduit from Soffit of Roof Slab, to fix Conduit, Contractor shall fix Mild Steel Flat of suitable size in Slab while casting the Roof Slab/Floor Slab.

20.3.1.17 PVC Conduit used in Point Wiring for flushed Florescent Fitting and Ceiling Fan fixed at False Ceiling Area, can be laid on False Ceiling.

20.3.1.18 Mild Steel Conduit used for suspension of Tube Light Fittings shall conform to IS-9537.

20.3.2 **PVC CONDUIT**:

20.3.2.1 PVC conduit to be used for point wiring/sub main wiring shall be 'Heavy Grade' and all as specified in Para 19.29 of MES Schedule Part-I.

20.3.2.2 PVC circular box and bend used for PVC conduit shall also be conforming to IS-3419.

20.3.2.3 PVC terminal box shall be ISI Marked.

20.3.3 **FLEXIBLE CABLE** Flexible cable shall be with tinned annealed stranded copper conductors of 1 Sqmm 3 runs (phase, neutral and earth) with PVC insulation.

20.3.4 **CEILING ROSES**: Ceiling rose shall be surface type and shall comply with IS-371 having three terminals plate and as specified in Para 19.32 of SSR 2009 Part – I.

20.3.5 **EARTH WIRE**

20.3.5.1 Copper PVC insulated wire conforming to IS-694 green in colour of size given in Schedule 'A' (BOQ) will be continuously laid along with other colour coded wire of the each circuit as asked in Schedule 'A' (BOQ).

20.3.5.2 The cables shall be with multistranded copper conductor PVC insulated unsheathed suitable for 1100 volts, for PVC conduit as specified. For point/sub main wiring color code of cables as per IE rule shall be maintained.

20.3.5.3 Earthing shall conform to IS - 3043. Min. distance between Two Earth Pits shall be kept 4.5 Metre.

20.3.6 **MCB AND DISTRIBUTION BOARD**

20.3.6.1 Extra length of cable shall be kept inside the distribution board for connection with MCB. All cable terminals with MCB shall be provided with suitable crimping lugs. All MCB and their distribution board shall comply with IS 8828-1978 and shall have minimum rupturing capacity as 10.0 KA.

20.3.6.2 All MCB, DBs shall be located at Load Centre (as per as possible) irrespective of whatever is shown on Drawings.

20.3.6.3 All MCB, DBs shall be same manufacturers.

SWITCH BOXES:-The Switch Box to accommodate Switches for Light, Fan and Socket shall be fabricated with 16 SWG sheet conforming to IS-5133. The box will have cover of minimum 3 mm thick Laminated Sheet of suitable shade approved by GE. The Switch Boxes shall be of adequate sizes to house all Switches/Regulator/ 3 Pin Plug. If space inside the box is found to be insufficient due to number of Wires then Two Switch Boxes shall be provided side by side. Before fixing the box, outer and inner surface will be treated against corrosion.

(PARTICULAR SPECIFICATIONS(Contd.....)

20.3.7 **FIXING OF FITTINGS AND FANS**:-All tube light fittings if suspended will be fixed at 2.60 Metre from finished floor level or as directed by Engineer-in-Charge. All ceiling fans will be kept at 2.70 Metre from finished floor level or as directed by Engineer-in-Charge. Air circulator will be fixed on wall with proper MS bracket grouted properly either with MS bolt or split bolt. Cost of MS brackets is deemed to be included in the quoted cost of air circulator.

20.3.8 **ELECTRICAL TEST**

20.3.8.1 On completion of wiring, following tests shall be carried out conforming to IS 732.

(a) Insulation Test.

(b) Polarity Test

(c) Testing of Earth Continuity

20.3.9 **LED LIGHTS/LUMINARIES**:

20.3.9.1 All LED lights/luminaries shall be meet the provision laid down in IS No-16101, IS No -16102 Part I/II, IS No -16130 Part I, IS No -15885 Part II/ Sec 13, IS No -16104, IS No -16105, IS No - 16106, IS No - 16108: 2012 and LM 79 & LM 80.

20.3.9.2 All LED light products shall be replacement warranty of complete LED luminaries fitting shall not less than three years from the date of completion and Life expectancy of LED's minimum 50,000 burning hrs. The product will replace free of cost in the following cases :-

(i) Manufacturing defects.

(ii) Failure due to mechanical and electrical impact.

(iii) Drop in lumen (or lux at lm) below 90% of claimed values of lumen (or lux at 1m) of the LED/Luminaries.

20.3.9.3 The. above warranty shall not include natural disaster, sudden surge in voltage recorded by digital electronic meter or due to wrong handling of light fittings

20.3.9.4 Manufacturer's test certificate of LED light fitting shall be submitted by the contractor before any payment.

20.3.9.5 Guarantee certificate of LED light fitting shall be submitted by the contractor before any payment. Contractor shall be responsible for replacing at his cost any parts which are rendered defective and unserviceable during the above period.

20.4 INSULATION TEST

20.4.1 The insulation resistance shall be measured by applying between earth and the whole system of conductor or any section thereof with all fuses in place and all switches closed, and except in earthed concentric wiring, all lamps in position or both poles of installation otherwise electrically connected together, a DC voltage of not less than twice the working voltage, provided that it does not exceed 500 volts for medium voltage circuits. Where the supply is derived from three-wire (AC or DC) or a poly-phase system, the neutral pole of which is connected to earth either direct or through added resistance the Working Voltage shall be deemed to be that which is maintained between the outer or Phase Conductor and the Neutral.

20.4.2 The insulation resistance in megohms of an installation measured as in Particular Specification here-in-above shall not be less than 50 divided by the number of points on the circuit, provided that the whole installation need not be required to have an insulation resistance greater than one megohm.

20.4.3 **POLARITY TEST**:- Each Switch of Light Fitting, Ceiling Fan, Exhaust Fan and other Fixtures shall be connected with phase and not with neutral. For this Each Point should be tested by opening Switch Boxes and recorded in the Test Sheet.

(PARTICULAR SPECIFICATIONS(Contd.....)

20.4.4 **EARTH CONTINUITY TEST**

(a) The Earth Resistance of Each Electrode shall be measured.

(b) The Earth Resistance of Earthing Grid shall be measured.

(c) All Electrode shall be connected to the Grid and the Earth Resistance of the entire Earthing System shall be measured.

(d) These Test shall preferably be done during the Summer Months.

20.4.5 **CABLES**:-Cable installation shall be checked as follows :-

(a) It shall be ensured that the Cables conform to the relevant Indian Standards. Tests shall also be done as laid down in IS:1255-1983. The insulation resistance before and after the Tests shall be checked.

(b) The Insulation Resistance between each Conductor and against Earth shall be measured. The Insulation Resistance varies with the type of insulation used and with the type of insulation used and with the length of Cable.

(c) The following empirical Rule gives reasonable guidance

$$\text{Insulation resistance in megohms} = \frac{10 \times \text{voltage in KV}}{\text{Length in KM}}$$

20.4.5.1 The Test Results shall be recorded and signed by Engineer-in-Charge and the Contractor.

20.4.5.2 Electrical gadget (tools/devices), material & labour required to carry out the above test shall be the responsibility of the contractor.

20.5 ELECTRONIC BALLAST

20.5.1 Manufacturer of Electronic Ballast should have tested its proto type at any of the Electronic Regional Test Laboratory, Deptt of Electronics.

20.5.2 This should meet the standard laid out in IS-1534-1977 and IS-13021 Part I & II 1991.

20.5.3 In case of failure of the lamps connected across the ballast, the circuit should be designed to switch OFF to prevent repeated attempts at reignition. The ballast should switch on automatically after replacement of failed lamps without the need or any manual resetting. There should be provision of fuse within the ballast to prevent any Short Circuit from affecting the Main Supply. Power Factor of the system when used with filter coil should be > 0.96.

20.5.4 Ballast will have filter coil either fixed inside or outside the ballast. Ballast will offer min 32 Watts of power to fluorescent tube and tube should offer same lumen output as with 36 Watts TLD. The ballast should ensure a low stroboscopic effect of less than 4% meeting the requirement of IS-13021 Part II.

20.6 CABLE WORK

20.6.1 **GENERAL**:- Refer clauses 19.19 & 19.73 to 19.96 of MES Schedule Part-I. Where the cables are to be laid in the trenches, 80 mm thick cushion of clean dry sand shall be provided below the bottom of cable. The cable shall be covered by a further layer of clean dry sand above the bottom of the Cable, to the height as mentioned in MES Schedule. The Cable Cover shall be laid dry and side by side to protect the cable from Mechanical Injury, the cables shall be looped at ends for furthers joints. The above additional length of cable as directed by Engineer-in-Charge shall also be measured and paid for under the respective items of cable in Schedule ’A' (BOQ). Laying Sand and Bricks at places where Cables are laid, shall be done as directed by the Engineer-in-Charge shall also be Measured and Paid separately for under the respective Items of in Schedule ‘A' (BOQ).

20.6.2 **SAND CUSHIONING** :- The sand shall be local sand as approved by GE. For laying of Cable in trenches, provision of cushioning shall be carried out all as specified in Clause No. 19.75 and 19.77 of MES Schedule Part-I.

(PARTICULAR SPECIFICATIONS(Contd.....)

20.6.3 **LAYING OF CABLES**

20.6.3.1 The LT underground cables shall be of 1100 volts, grade PVC armoured, heavy duty underground cable, conforming to IS-1554.

20.6.3.2 The cable should be laid at a depth of not less than 460 mm.

20.6.3.3 LT cable shall bear ISI certification mark.

20.6.3.4 Both ends of pipe protecting the cable shall be sealed with wooden bushes cut into two halves of appropriate size.

20.6.3.5 The cable shall be protected by means of 15 cm of Sand Cushion on top and under the cable as described in MES Schedule Part-I. On top of Sand Cushion, Pre-cast Cover shall be provided as protection.

20.6.3.6 Earth Work, Sand Cushioning and Cable Protection shall be Measured and Paid separately.

20.6.3.7 The scope of work will include proper laying of Cable in trenches/in pipes/on wall, fixing Identification Tags, crimping with Crimping Lugs, fixing Compression Glands at Termination Point in Panel/Marshalling Box. The above shall be deemed to be part of the installation work and the Contractor shall not be eligible for any Extra Charges other than those specifically mentioned in Schedule 'A' (BOQ).

20.6.3.8 LT Cable shall be aluminium conductor PVC insulated inner sheathed armoured and PVC sheathed conforming to IS-1554 suitable for 1100 Volt grade. Cable shall be laid as per provision of IS-1255- 1983. The cable drum and the cables shall be handled in the correct manner so as to ensure safety for personal and obviating kinks or other damage to the cable insulation, armouring or cable sheath.

20.6.3.9 All LT Cables shall be tested at manufacturer works as laid down in relevant Indian Standard Specifications including Physical Tests, Insulation Resistant Test and Voltage Tests. The Test Certificate shall be submitted to the Engineer-in-Charge.

20.6.4 **TEST**:- After laying the HT and LT Cables, each core of HT/LT cables will be subjected to High Voltage Tests before commissioning. High Voltage Value for HT and LT Cable shall be as given in MES Schedule Part-I vide Clause No. 19.94. Each core of cable shall be subjected to high voltage for a duration of 15 Minutes. The results will be recorded and submitted to Engineer-in-Charge. Testing shall be done in the presence of Department Representative. Contractor is to make arrangements for High Voltage Test of cable at no Extra Cost to the Department. Apart from this Test, IR Value Test will also be conducted as given in IS-1255 and in MES Schedule Part I. The Contractor shall note that these Tests shall be carried out before connecting Cable to Switches.

20.7 DISTRIBUTION / FEEDER PILLARS BOX

20.7.1 Refer clauses 19.21 of MES Schedule Part-I. , IS 5039 and TD drawing

20.7.2 The feeder pillars box shall be factory made and CPRI approved. Feeder pillars box shall be obtained from the specified manufacturers and shall be epoxy powder coated factory made and tested as per IS. All feeder pillars boxes shall have extra space for installing one MCCB in future.

20.7.3 The contractor shall design & submit drawings of panel board before bringing same to the site.

20.7.4 The cable chamber of the board shall be with adequate space for housing cable and connecting for safety of working and for avoiding accidental contacts with live parts by providing barriers.

20.7.5 In case nothing is mentioned in Schedule 'A' (BOQ) bus bar shall be copper.

20.7.6 In case mounting of panel board has not been mentioned in Schedule 'A' (BOQ) the same shall be floor mounted and the fixing arrangement shall be provided by the contractor and the unit rate quoted shall be deemed to inclusive for the same.

20.8 **LT SWITCH GEAR PANEL**:-

20.8.1 Refer Para (including sub paras) 19.100 of MES Schedule Part-I and IS 2147.

20.8.2 The panel board shall be factory made and CPRI approved. Panel boards shall be obtained from the specified manufacturers and panels will be epoxy powder coated factory made and tested as per IS. All LT panels shall have extra space for installing one MCCB in future.

(PARTICULAR SPECIFICATIONS(Contd.....)

20.8.3 The contractor shall design & submit drawings of panel board before bringing same to the site.

20.8.4 The cable chamber of the board shall be with adequate space for housing cable and connecting for safety of working and for avoiding accidental contacts with live parts by providing barriers.

20.8.5 In case nothing is mentioned in Schedule 'A' (BOQ) bus bar shall be copper.

20.8.6 In case mounting of panel board has not been mentioned in Schedule 'A' (BOQ) the same shall be floor mounted and the fixing arrangement shall be provided by the contractor and the unit rate quoted shall be deemed to inclusive for the same.

20.9 **MOULDED CASE CIRCUIT BREAKERS (MCCBS)**:-

20.9.1 Moulded case circuit breakers (MCCBs) shall be suitable for operational voltage of 415V AC, 50 Hz, 3 phase, 4 wire system for a rated current and ultimate breaking capacity and it shall conform to IS : 13947 (Part 3, 1993) and IEC-60947 (Part 2).

20.9.2 MCCBs shall be provide earth leakage module type.

20.9.3 MCCBs shall be of compact and elegant design suitable for reversible load and line terminations without affecting its performance. MCCBs shall be suitable for fixing flush on the panels and shall be provided with handle operating mechanism including Rotary Handle vari-depth type.

20.9.4 The insulating case and cover of MCCBs shall be made of high resistant and flame retardant thermosetting insulating materials. The switching mechanism shall be quick make, quick break and trip free. The position of the operating knob/handle shall clearly indicate ON, OFF and TRIP position.

20.9.5 Each pole shall be provided with a pair of contacts, which shall open at a high speed over a large distance under short circuit faults. The special designed arc chutes of insulating materials shall be provided to contain the arc by providing effective arc quenching device.

20.9.6 The tripping mechanism shall be hydraulic type or Electronic release or thermal magnetic release for protection for over load and short circuit as per the details given below. –

Capacity Of MCCBs	Ultimate breaking Capacity	Overload release	Short circuit release
Upto 125 Amps	16 KA	Adjustable Thermal release (0.7 to 1.0 In)	Fixed magnetic type suitable for 10.0 In.
160 or 250 Amps	36 KA	Adjustable Thermal release 0.64 to 1.0 In	Adjustable magnetic release (3.5 to 10.0 In)
400 and 630 Amps	36 KA	Electronic release with over load Zone of adjustment 0.4 to 1.0 In	Electronic release with Short circuit Zone of adjustment 1.5 to 10.0 Ir. (Adjusted current) with time.

20.9.7 Under voltage trip, mechanical interlocks etc shall be provided as per standard practice and ISS. Under voltage trip shall be designed to operate when the control voltage drops below a tripping threshold i.e. 20% to 70% of rated voltage and shall be suitable for operation on 230V/415V AC.

20.9.8 The terminals shall be suitable for both copper terminations.

20.9.9 In case number of poles of MCCB are not specified in Schedule 'A' (BOQ) then 4 pole shall be provided.

20.9.10 After erection at site and before being connected to the power supply, all items of electrical equipment, all circuits shall be fully tested to prove correct connection, defects in installation pointed out/noticed shall be rectified immediately by repairing or replacing the defective part of the equipment at no extra cost. All instruments appliances, oil, grease and other materials etc, required for carrying out the test shall be provided by the contractor at his own cost.

(PARTICULAR SPECIFICATIONS(Contd.....)

20.9.11 The insulation and other safety measures will be inspected and checked by electrical inspector before commissioning. The electrical inspector shall be nominated by the Accepting Officer.

20.9.12 If owing to storage or other cause, the electrical insulation resistance has deteriorated, the equipment shall be thoroughly dried out or replaced or other steps taken to restore proper insulation resistance before connection it to the power supply.

20.9.13 After installation of the equipment and laying and jointing of cables, necessary tests including relay co-ordination as per relevant ISS shall be carried out by the contractor to the satisfaction of GE.

20.9.14 After connection the power supply, the whole of the installation shall be tested to demonstrate its ability to operate satisfactorily.

20.9.15 The installation shall be deemed to be complete only after satisfactory completion of all tests. Approval by the Engineer-in-Charge for materials, workmanship etc, during manufacture of items at site will not relieve the contractor from his obligations to comply with all the requirements of the contract. The contractor shall supply schematic diagrams (Three copies) of the equipment installed by him on completion of the work.

20.9.16 One month before expiry of the defects liability period, the contractor shall arrange an experienced Engineer together with any additional labour required to attend at the site to open up the equipment to check the various connections clearances etc. The detail checks shall be done by the contractor in the presence of the GE before the end of the defects liability period.

20.10 EARTHING

20.10.1 Earthing shall be carried out by Earth Electrodes complete all as specified in Schedule 'A' (BOQ) and all as shown in Electrical Plate No. 3 of MES Schedule Part-I and as specified in IS-3043 including upto date amendments therto. The Earth Electrodes shall be placed all as shown on Drawings and 2.0 Metre away from the Main Structure. The surplus soil obtained from the excavation shall be removed to distance not exc. 50 Metre. The Concrete for Chamber shall be PCC 1:3:6 (using 20 mm graded stone aggregate) and exposed surfaces need to be finished fair. Funnel shall be prepared out 3.15mm thick PGI Sheet by welding and shall be leak proof. It shall be provided with wire gauge with 1.18mm average width of aperture and 0.45mm average dia of wire soldered to Funnel Cover. Dust shall be used for Medium Grade 20mm with 12mmdia holes drilled staggered @ 300mm C/C through out Pipe. The bottom of Pipe shall be layed at an angle of 45º.

20.10.2 The following Tests shall be carried out to the entire satisfaction of the Engineer-in-Charge :-

(a) Earth Resistance Test : Resistance of each Earthing shall be measured individually and shall not exceed 0.5 Ohm in ordinary soil and 3 Ohm in Rocky situation.

(b) In case the Resistance is more than as specified, the Contractor shall increase the depth of the Bore hole, quantity of Charcoal, Salt and length of Electrodes to bring the Resistance of Earth to the Value specified above, without any Extra Cost to the Government.

20.11 GUARANTEE/ PRODUCTION OF TEST CERTIFICATE

20.11.1 After completion of work the contractor shall guarantee that materials and workmanship of the equipment, power system and connected items offered and installed by him are new and best quality in every respect and he will make good the defects, damages which are not attributed to normal wear and tear or misuse for which decision of the GE is final and binding. Contractor shall be responsible for repairing and or replacing at his cost any parts which are rendered defective and unserviceable during the defect liability period.

20.11.2 Manufacturer's test certificate in respect of HT/LT Cables, Geyser, Water Cooler, Air Cooler, Refrigerator, Transformer, DG Set, Solar Water Heating System other equipment as directed by GE are to be submitted to the GE by the contractor before any payment from Department.

(PARTICULAR SPECIFICATIONS(Contd.....)

21 **DEMOLITION/ DISMANTLING**:

21.1 **PRECAUTIONARY MEASURE**: Precautionary measure prior to demolition/dismantling as described in SSR Part-I shall be adopted and work shall be executed as per Schedule 'A' (BOQ).

21.2 **REMOVAL/SITE CLEARANCE**: Old demolished materials considered under Schedule of Credit shall be the property of the contractor and will remove by the contractor at his custody. Debris and other materials if any shall be disposed off and sites to be cleared to the entire satisfaction of the GE.

Contractor

AGE (Contract)
for Accepting Officer

APPENDIX 'A'

APPROVED LIST OF MANUFACTURERS/ MAKES

SER NO	NAME OF PRODUCT	MAKES/ BRANDS/ MANUFACTURER
B/R ITEMS		
1	PRESSED STEEL DOOR FRAMES /CHOWKHATS FOR DOORS/ CUPBOARD AND WINDOW FRAMES	SHAKTI, MADHU INDUSTRIES, NCL SECCOLOR, SHIVAM METAL SHAPERS, ASHISH INDUSTRIES, TRISHUL INDUSTRIES
2	STEEL WINDOWS/VENTS	SHAKTI, NCL SECCOLOR, MADHU INDUSTRIES, SHIVAM METAL SHAPERS, ASHISH INDUSTRIES
3	ROLLING SHUTTER & ROLLING GRILLS	PARSHOTAM STEEL IND (PSI) RATNAKAR STEEL WORKS
4	FACTORY MADE PANELLED DOOR SHUTTERS	EVEREST, JAINDOORS PVT LTD (SONIPAT) , JAINWOOD INDUSTRIES (JAYNABOARD) DELHI
5	PVC DOOR FRAME/ SHUTTERS	ACCUCEL, POLYLINE, ECOCELL, RAJSHRI PLASTIWOOD, GIZA, BLACK COBRA
6	PVC PANEL DOOR/WINDOW	ECOCELL, BLACK COBRA, RAJSHRI, SINTEX,
7	UPVC WINDOWS/DOORS	YASHPOLY, ACCUCEL, APARNA VENESTER/ OKOTECH, MADHU INDUSTRIES, NCL VEKA, H2O SOLUTION, FENESTA, ENCRAFT, SUPREME, RAJSHRI PLASTIWOOD, ASHISH INDUSTRIES
8	ANODISED ALUMINIUM DOORS/ WINDOWS/ FRAMES PARTITIONS	AXSYS, JINDAL, KALCO, ALPURE, INDAL, HINDALCO, ASHISH INDUSTRIES
9	GALVANISED STEEL SHEETS & COLOUR COATED GALVANISED STEEL SHEETS	JSW, TATA, JINDAL, KAMDHENU
10	TERRAZZO TILES/ PPC TILES / PPC CHEQUERED TILES	SURYA KIRAN, MOHINDRA INDUSTRIES, SITCO OR ISI MARKED AVAILABEL IN DELHI NCR AS APPROVED BY GE
11	INTERLOCKING TILES/ EXTERIOR TILES / PAVER BLOCKS	SURYA KIRAN, SOMANY, SITCO, POOJA CONCRETE, MAHINDRA IND, K B SPUN, SARAL IMPEX OR ISI MARKED AVAILABEL IN DELHI NCR AS APPROVED BY GE
12	ROOF TILES, KERB STONE	MOHINDRA INDUSTIRES, RN TILES AND PAVER, SWASTIC ALW OR ISI MARKED AVAILABEL IN DELHI NCR AS APPROVED BY GE
13	GLAZED CERAMIC/NON SKID WALL/FLOOR TILES	AGL, NITCO, JOHNSON, VITERO TILES, KAJARIA, VARMORA, SOMANY, HINDWARE
14	VITRIFIED TILES (NORMAL / ANTI SKID)	AGL, VITERO, CERA, NITCO, KAJARIA , JOHNSON, VARMORA, HINDWARE
15	HDPE & LLDPE STORAGE WATER TANKS	PLASTO, SPL, VECTUS, SINTEX, SUPREME
16	STAINLESS STEEL KITCHEN SINK WITH OR WITHOUT DRAINING BOARD, PLATE RACK	JAINKO, CERA, PRAYAG, DIAMOND, NIRALI
17	BALL VALVES (HORIZONTAL PLUNGER TYPE)	PRIMA, JAINKO, PRAYAG, FLOW JET
18	BRASS VALVES	PRAYAG, ZOLOTO, L&T VALVES, AMCO, VIP VALVES
19	SHOWER ROSES CHROMIUM PLATED (SWIVEL TYPE)	MARC, JAQUAR, KOHLER
20	CP BATH ROOM FANCY FITTINGS AND ACCESSORIES/ CP BATH FITTINGS	MARC, JAQUAR, KOHLER

(APPROVED LIST OF MANUFACTURERS/ MAKES (Contd…)

21	BIB TAPS AND STOP VALVES (CP) FANCY FITTINGS	MARC, JAQUAR, KOHLER
22	PTMT FITTINGS	PRAYAG, PRINCE, POLYTUF, SYMET, POLYPLAST
23	READYMADE BATHROOM CABINETS	PRAYAG, CERA, PARRYWARE
24	HDPE PIPES / SWR PIPES (WATER/SEWAGE)	PLASTO, SUPREME, UTKARSH, DURALINE, SFMC, PRINCE, RELIANCE
25	SAND CAST IRON SPIGOT & SOCKET SOIL WASTE, VENT PIPES & FITTINGS	SKF, RPMF, SRIF, SINGHAL IRON FOUNDARY, NECO (JAYASWAL)
26	RCC HUME PIPES	KB SPUN, INDIAN HUME PIPE CO
27	CEMENT BASED PAINTS	NEROLAC, SNOWCEM, ASIAN, BERGER, DULUX, JK MAXX PAINTS
28	INTERNAL ACRYLIC EMULSION PAINT	NEROLAC, ASIAN, BERGER, DULUX, JK MAXX PAINTS
29	PLASTIC EMULSION PAINT	ASIAN (ROYALE, APCOLITE ADVANCE) BERGER (SILK, RANGOLI EASY CLEAN) NEROLAC (IMPRESSION), JK MAXX PAINTS
30	HIGH BUILT TEXTURE & WALL FINISHES (EXT & INT)	ASIAN, BERGER, SNOWCEM, DULUX ,NEROLAC, JK MAXX PAINTS
31	SYNTHETIC ENAMEL PAINT	ASIAN, BERGER, DULUX ,NEROLAC, JK MAXX PAINTS
32	RED OXIDE ZINC CHROMATE PRIMER AND PINK PRIMER	ASIAN, BERGER, DULUX , NEROLAC
33	FRP DOOR FRAME & SHUTTER	FIBERWAYS, SPC, ASHISH INDUSTRIES
34	PVC WALL PANELING/ PVC CEILING	POLYLINE, ECOCELL, DECOINN, SUNBEAM, RAJSHRI PLASTIWOOD, BLACK COBRA
35	HDF/PVC DOOR FRAME SUTTER MOULDED	POLYLINE, ECOCELL, HR ENTERPRISES, ECHON, ALCAD, DECOINN, SINTEX, RAJSHRI PLASTIWOOD LTD.
36	PVC SHEET CLADDING	POLYLINE, ALCAD, DECOINN
37	ACOUSTIC PANELS	HIMALYAN ACOUSTICS, JYPTECH
38	UPVC CEILING & PANELLING	YASHPOLY, ACCUCEL, HR ENTERPRISES
39	PRE FAB PANEL BOARDS/WALL LINNING CEILING TILES	DEXUNE, AEROLITE, HI-STEEL, RAJSHRI PLASTIWOOD
40	FALSE CEILING TILES/ BAORD (GYPSUM BAORD/POP/MINERAL FIBRE)/JOINTING & FINISHING MATERIAL	DEXUNE, HI-STEEL, USG BORAL, GYPCORE, ARMSTRONG, RAJSHRI PLASTIWOOD
41	FALSE CEILING / METAL FALSE CEILING / METAL FRAMING	DEXUNE, HI-STEEL, USG BORAL, GYPCORE, ARMSTRONG, RAJSHRI PLASTIWOOD
42	PARTICLE BOARD (VENEERED AND UNVENEERED LAMINATED/PLYWOOD	AEROLITE, GREENPLY, CENTURY, KITPLY, ARCHID, GREEN PANEL
43	ROAD MARKING PAINT WATER BASED / THERMOPLASHTIC	3M, VERTEX, ASIAN, BERGER
44	WPC DOOR FRAMES/PANELS/HYBRID SHEET/ FLOORING /JALLI	RAJSHRI PLASTIWOOD, BLACK KOBRA, SINTEX
45	ROAD FURNITURE & REFLECTIVE SIGN BOARDS (ROAD STUDS, SOLAR STUD, SPEED BREAKER, TRAFFIC CONE, SPRING, POST, MEDIAN MARKER, DELINERATOR, SAFETY CONES, SAFETY CONVEX MIRROR, Q-MANAGER, REFLECTIVE JACKETS, LIGHTS BETON, BOLLARD, ROAD BARRICADED CAUTION TAPE ETC)	3M, DARKEYE, CATEYE
46	ASBESTOS CEMENT PIPES	KIRTI, HYDERABAD ASBESTOS, EVEREST,GANGA ASBESTOS
47	HDHMR BOARDS	ACTION TESA, GREENPLY, ARCHID, GREEN PANEL, CENTURY PLY

(APPROVED LIST OF MANUFACTURERS/ MAKES (Contd…)

48	CPVC PIPES & FITTING	PLASTO, PRAYAG, VECTUS, SUPREME, FINOLEX, DUTTON, ASTRAL
49	UPVC PIPES & FITTINGS	PLASTO, PRAYAG, SUPREME, FINOLEX, DUTTON
50	SANITARYWARE	MARC, JAQUAR, KOHLER, CERA, PARRYWARE, HINDWARE, PRAYAG
51	GALVALUME SHEETS	JSW, TATA, JINDAL, DYNA ROOFING SYSTEM KAMDHENU
52	DI FLANGED PIPES & FITTINGS	CHANDRANCHAL, TATA METALIK, JINDAL SAW, ELECTRO STEEL CASTING LTD
53	CI/ DI VALVES	CHANDRANCHAL, TATA METALIK, JINDAL SAW, ELECTRO STEEL CASTING LTD, ISI MARKED
54	CHAIN LINK FENCING/BARBED WIRE/ PUNCHED TAPE CONCERTINA COIL/ RAZOR BLADE TAPE FENCING	A-1 FENCE, ARMSTRONG WIRE, MAIMOM ROGER
55	GALVANISED STEEL WELD MESH FENCING	A-1 FENCE, MULTIWELD WIRE CO LTD, JOHNSON SALES CORP
56	PLAIN & PRE LAMINATED MDF & HDF BOARDS	PDE COR (PIONEER PANEL PRODUCT), GREEN PANEL
57	GLASS REINFORCED GYPSUM (GRG) CEILING TILES	DIAMOND CEILING, GYPCORE, MARUTI GYPSUM, SHERA
58	GYPSUM BOARDS/ JOINTING & FINISHING MATERIAL/ PARTITION & CEILING FRAMING WORK / PLASTERING MATERIAL / GYPSUM CEILING TILES	GYPROC, GYPCORE, INDIA GYPSUM, LAFARGE
59	MS BUTT HINGES	SURYA, MOWJEE, ARGENT, RS INDUSTRIES
60	LAMINATES DECORATIVE	VIRGO, ARCHIDLAM, SUNMICA, GREENLAM, GREEN PANEL
61	ACP (ALUMINIUM COMPOSITE PANEL)	ALUCOBOND, EUROBOND, ALSTRONG, ARMSTRONG, ALEX PANEL
62	MARINE PLYWOOD & PLYWOOD FOR CONCRETE SHUTTERING WORKS	ALLIANCE, ALUTECH, CITYBOND, GL PLY
63	PVC FLUSHING CISTERN	CERA, PRAYAG, HINDWARE, PARRYWARE, SUPREME, POLYPLAST
64	TOILET SEAT COVER	PRAYAG, PARRYWARE, CERA, COMMANDER, POLYPLAST
65	ALUMINIUM TOWERBOLT, HINGES, HANDLES & SLIDING DOOR BOLT	CROWN, ARGENT, CLASSIC, GLOBAL
66	SS TOWERBOLT & SLIDING DOOR BOLT	CROWN, MAWJEE, ARGENT, RS INDUSTRIES, GLOBAL
67	MS TOWER & SLIDING DOOR BOLT	CROWN, MAWJEE, ARGENT, RS INDUSTRIES, GLOBAL
68	BRASS TOWERBOLT	CROWN, MAWJEE, ARGENT, RS INDUSTRIES, GLOBAL
69	LOCKING SYSTEM (MOTRICE LOCK, NITHT LATCHES, CYLINDRICAL LOCK, EURO PROFILE CYLINDERS)	DORSET, GODREJ, HARRISON, LINK

(APPROVED LIST OF MANUFACTURERS/ MAKES (Contd…)

70	DOOR HARDWARE (DOOR CLOSURE, FLOOR SPRING, TOWERBOLT & HINGES)	DORSET, UNIVERSAL, GODREJ
71	WALL CARE PUTTY/ WHITE CEMENT	JK, BIRLA, TRIMURTI
72	FLAT GLASS (FLOAT GLASS, MIRROR, FROSTED GLASS, LACQUERED GLASS, EXTRA CLEAR GLASS, SOLAR CONTROL REFLECTIVE GLASS, HIGH PERFORMANCE LOW E GLASS, TINTED GLASS)	SISECAM, SAINT GOBAIN, MODI FLOAT
73	BOX TYPE STEEL WINDOWS	FRIENDS MANUFACTURING CO, LUXMI WELDING WORKS, JEW
74	WOODEN PANEL DOORS, WINDOWS & FRAMES, FLUSH DOORS	SATISH KUMAR & CO, GL PLY, JAIN DOORS PVT LTD
75	PCC MANHOLE & DRAIN COVER, EARTH PIT, CABLE ROUTE INDICATOR	MEGHA CONTRACTORS PVT LTD, HARIOM TILES & PAVERS , NITCO
76	PVC KITCHEN CABINETS & CUPBOARDS	GIZA, PRAYAG, CERA, PARRYWARE
77	CHEMICAL FOR ANTITERMITE TREATMENT	DE- NOCIL BOMBAY, PEST CONTROL OF INDIA, TRISHUL, CLOREX
78	READY MIX CONCRETE	M/S GRASIM READY MIX, M/S ACC LTD, M/S L&T, M/S RDC CONCRETE (INDIA) LTD, UNITECH
79	FLUSH DOORS IS-2191, 2202	M/S GREEN PLY INDUSTRIES, M/S CENTURY PLY, M/S DURA PLY, MYSORE CHIP BOARD, ARCHID, SWASTIK PLY WOOD, M/S JAIN WOOD INDUSTRIES
80	SOLID CORE PVC PANEL DOOR/WALL PANELING/FALSE CEILING/INTERNAL DOOR/WINDOW/KITCHEN CABINET	SINTEX, RAJSHRIPLASTIWOOD LTD,M/S DURA PLAST EXTRACTION PVT LTD, M/S FENESTA,GIZA
81	INTEGRAL WATER PROOFING COMPOUND	WATER SEAL, IMPERMO, STP NO 1, PEDILITE, ACCO PROOF, FOSROC, CICO. BENGAL BITUMEN, IWL, DR FIXIT
82	WALL CARE PUTTY FOR BASE PREPARATION	BIRLA WALL CARE PUTTY, GOLD MOHAR, M/S SHALIMAR HARDWARE, BURGER, J&N, BIRLA LAVAL PLAST,CORAL,ACRO (WALLGUARD), TRIMURTI
83	DRY DISTEMPER / OIL BOUND DISTEMPER	GOODLASS NEROLAC PAINTS, JENSON &NICHOLSON, BERGER PAINTS, ICI DULUX, ASIAN PAINTS
84	WATER PROOFING FOR CEMENTIOUS SURFACE IS-2645	ACROCRETE & ACROCOTE, CKO, FOSROC, KAMPER SYSTEM, HI TECH, CORAL
85	BITUMINOUS PRODUCT INCLUDING APP MEMBRANE	M/S FARIDABAD SPINNING & WOOLEN MILLS PVT LTD, 837, SP MUKHERJEE MARG DELHI, M/S STP LTD (FORMERLY SHALIMAR TAR PRODUCTS) M/S BITUFELT PVT LTD 123/377 FAZALM GANJ KANPUR 208012, TEXAS, TEXAS INDIA LTD, MULTIPLAS, IWL CHANNAI.TORCH TAR APP MEMBRANE, DELHI
86	SGSW PIPES (IS-651) ISI MARKED	DEVRAJ INDUSTRIAL, BURAN, PERFECT SANITARY PIPE, SUPERME PIPE AND FITTINGS
87	PVC RAIN WATER/SEWAGE PIPES (IS-4985)	SUPREME, FINOLEX, RELIANCE, PRINCE
88	HYDROLIC DOOR CLOSER/FLOOR SPRING	EVEREST, UNIVERSAL, GODREJ
89	CERAMIC TILE ADHESIVE	FERROUS-CRETE,CORAL,UNISTONE
90	CONSTRUCTION CHEMICAL COMPOSITE	SIKA, XYPEX, MYK SCHOMBURG
91	STAINLESS STEEL RAILING	KICH/JINDAL/TATA
92	FIBRE BOARD(MEDIUM DENSITY)	NOVAPAN, BHUTAN BOARD, KIT PLY, MYSORE CHIP BOARD LTD, GREEN PANEL MAX (PLAIN EXTERIOR GRADE MDF/ PRELAM/ VENEERED MDF/ GREEN FLOORMAX LAMINATED)

(APPROVED LIST OF MANUFACTURERS/ MAKES (Contd…)

93	PPR PIPE	SFMC, SAFE, WETFLOW, KPT
94	DRAPERY RODS	VISTA, LEVALOR, JINDAL, MAC
95	MS PIPES	APL APPOLO, BHUSHAN, JINDAL
96	GUN METAL GATE VALVE	SANT, L & T, LEADER, KIRLOSKAR
97	EXTERNAL ACRYLIC EMULSION PAINT	BERGER (WEATHER COAT LONG LIFE 10), ASIAN (APEX ULTIMA), NEROLAC (EXCEL MICA MARBLE), DULEX (WEATHER SHIELD MAX)
98	WELDED WIRE MESH/FABRIC	MULTIWELD WIRE CO LTD, MUMBAI, JAINSONS SALES CORP, DELHI, JAISONS INDUSTRIES
99	PVC TILES/SHEETS	PRIMER POLYFILM LTD, WONDER FLOOR, ARMSTRONG
100	GLASS MOSAIC TILES	JOHNSON, ITALIA, CORAL MOSAIC, NITCO
101	HDPE STORAGE WATER TANKS	SINTEX, SPL, POLYCON, ROTEX, SHEETAL, VECTUS
102	GI PIPE	VPL, JINDAL, TATA, OSWAL, BST
E/M PORTION		
103	MCCB	INDO ASIAN, L&T, LEGRAND, ABB, SIEMENS, HAVELLS
104	MCB/ ELCB / RCCB/ DB/ ISOLATORS	INDO ASIAN, L&T, LEGRAND, ABB, SIEMENS, HAVELLS, POLYCAB, KINJAL
105	LT CABLES, 1100 VOLTS , XLPE	KEI, GLOSTER, POLYCAB, HAVELLS, RR KABLE, GRANDLAY
106	CHANGE OVER SWITCHES	INDO ASIAN, HAVELLS, L&T, LEGRAND, ABB, SIEMENS
107	VOLTAGE STABILIZERS UPTO 25 KVA	APLAB,AE, BRENTFORD, V GUARD, ANDREW YULE, VOLINA
108	VOLTAGE STABILIZERS ABOVE 25 KVA (SERVO CONTROLLED)	AE, BRENTFORD, V GUARD, APLAB, ANDREW YULE
109	UPS (OFF LINE & ON LINE) ABOVE 10 KVA UPS (OFF LINE & ON LINE) UPTO10 KVA	EMERSON, TATA LIBERT, APLAB, APC ,AE, ITE GURGAON, VINITEC, SINETRAC
110	LT CAPACITOR BANKS	HAVELLS, L&T, LEGRAND, ABB, SIEMENS
111	PVC COPPER WIRES – 650/1100 VOLTS	FINOLEX, GRANDLAY,HAVELLS, POLYCAB, GLOSTER
112	GI/MS CONDUIT/ERW CONDUIT PIPE	BHARAT STEEL TUBES, VPL, AKG, BEC
113	CASING CAPPING	KALINGA, BAJAJ, PLAZA, FINOLEX, TIRUPATI, POLYPACK, AKG, HARSH, VISHAL PIPES, SUPREME, DAUPHIN, DIPLAST, PALICAN, SWASTIC.
114	PVC RIGID PIPE	FINOLEX, ASTRAL, AKG, PRINCE, NORPACK
115	PT SWITCH / SOCKETS/ CEILING ROSES/ FAN RAGULATORS	ANCHOR, LEGRAND, CRABTREE, SCHNEIDER, SSK, PHILIPS, BAJAJ, KINJAL
116	MODULAR SWITCHES / SOCKETS/FAN RAGULATORS	LEGRAND, CRABTREE, SCHNEIDER, HAVELLS, POLYCAB. INDO ASIAN, ANCHOR, KINJAL
117	TUBE LIGHT/STREET LIGHT FITTINGS & ACCESSORIES – HPSV (70/150/250/400)	HAVELLS, HPL, HALONIX, POLYCAB, SURYA

(APPROVED LIST OF MANUFACTURERS/ MAKES (Contd…)

118	BOX TYPE DECORATIVE TUBE LIGHT FITTING	HAVELLS, HALONIX, POLYCAB, SURYA
119	ANY TYPE OF LED FITTINGS /LUMINAIRE (INTERIOR & EXTERIOR)	HAVELLS, WIPRO, BAJAJ, CG, POLYCAB, EVEREADY
120	LED FLOOD / STREET LIGHT / SECURITY/ BOLARD/ GATE LIGHT FITTINGS	HAVELLS, WIPRO, BAJAJ, CG, POLYCAB, EVEREADY
121	SK FITTING	VIKAS, PRAKASH, JEMCO, SPARKS, HOMEDEC
122	ELECTRIC CALL BELLS	ANCHOR, LEGRAND, CONA, HAVELLS, INDO ASIAN, KINJAL
123	GEYSERS	CROMPTON, BAJAJ, VENUS, USHA, AO SMITH, RECOLD, HAVELLS
124	SMC POLE BOXES	SINTEX, SHAHSWAT, FINOLEX
125	FLAME PROOF ELECTRICAL FITTINGS/ ACCESSORIES/ PANELS/ JUNCTION BOX/ SWITCHES & SOCKETS	BALIGA LIGHTING EQUIPMENTS, CROMPTON GREAVES, BAJAJ, SUDHIR, SHYAAM, GE LIGHTING, KINJAL
126	ACSR CONDUCTOR	ALIND, DYNAMIC CABLES, COYOTE, STERLITE POWER.
127	LAMINATED SHEET COVER	HYLUM, FORMICA, GREEN LAMINATED BOARDS
128	LED AVIATION OBSTRUCTION LIGHT	BAJAJ, CROMPTON GREAVES, WIPRO, PHILIPS, SURYA, ADHUNIK SWITCH GEAR
129	BUS BAR TRUNKING	L&T, INDO ASIAN, SCHNEIDER, SIEMENS, GE, LEGRAND.
130	HIGH VOLTAGE INSULATING MAT	PREMIER POLYFILM, SUNTEX INSULATICA , SAFE VOLT, SINTEX
131	CONTROL CABLES	L&T, FINOLEX, UNIVERSAL, ASIAN, CABCOM, GRANDLAY, NICCO , KINJAL
132	FIBRE OPTIC CABLES	LAPP, STERLITE, KEC, D-LINK, FINOLEX
133	LEAD ACID BATTERIES	AMARON, STANDARD FURKKAVA, EXIDE
134	POWER CONTACTORS	HAVELLS, INDO ASIAN, HPL
135	BATTERY CHARGERS, ACDB, DCDB, ELDB	SINETRAC, BPS, POWER ONE
136	LED LIGHT FITTING & LUMINARIES FOR ROAD AND STREET LIGHTING (OUT DOOR) AND LED LIGHTS, PANELS & LUMINARIES INDOOR	HAVELLS, WIPRO, BAJAJ, CG, POLYCAB, EVEREADY
137	INDUSTRIAL PLUG & SOCKETS	INDOASIAN, BCH, LEGRAND, HAVELLS, SIEMENS, KINJAL
138	SMC CROSS ARMS /LADDERS MAT/TRENCH COVER FOR HT/LT	SINTEX, MPI, SIMPEX, ASIAN, DEVI POLYMERS.
139	GI PIPES/ MS PIPES	VPL, JINDAL, TATA, OSWAL, BST
140	HDPE PIPES/ UPVC PIPE/ PPR PIPE	DUTRON, FINOLEX, SFMC, VPL, PRINCE, SUPREME
141	CI PIPES	ELECTRO-STEEL / LANCO, JINDAL, RASHMI, KESORAM, LECTROTHERM, TATA METALIKS, TATA KUBOTA
142	DI PIPES	ELECTRO-STEEL / LANCO, SRIKALAHASTHI, JINDAL, KESORAM, ELECTROTHERM, TATA
143	CI PIPE FITTING	SKF, SRIF, SINGHAL IRON FOUNDRY,HINDUSTAN ENGINEERING PRODUCTS, JINDAL, HEPCO(BINAYUDYOD)
144	DI PIPE FITTING	KARTAR, ELECTROSTEEL/ LANCO, KEJRIWAL, KESORAM, JINDAL, TATA METALIKS, TATA KUBOTA

(APPROVED LIST OF MANUFACTURERS/ MAKES (Contd…)

145	PCC CABLE COVER	MEHTAB TILES, PEARLESS INDUSTRIES, HINDMOZAIC GUJRAT, JJ CONCRETE LUCKNOW
146	SMART / TAMPER PROOF ENERGY METERS	ABB, L & T, SECURE, HPL, NEPTUNE
147	CEILING/EXHAUST/VENTILATING/WALL MOUNT FAN/ BLDC FAN (5 STAR RATED)	ATOMBERG/ HAVELLS/ CROMPTON/ GOLD MEDAL/ POLYCAB
148	LED LAMPS	HAVELLS, WIPRO, BAJAJ, CG, POLYCAB, EVEREADY
149	GI FITTINGS	JINDAL, TATA, BST, UNIK
150	HT XLPE CABLES 132/66/33 KV & 11 KV	UNIVERSAL CABLES, KEI,GLOSTER, POLYCAB, DYNAMIC CABLES, CMI, HAVELLS, CRYSTAL, GEMSCAB
151	SLUICE / NON RETURN / AIR RELEASE / PRESSURE RELEASE/ SPRING RELIEF/ BALL/STEAM/ FOOT VALVE	KIRLOSKAR, LEADER, ZOLOTO, L & T (AUDCO), SANT
152	WATER COOLER	VOLTAS, BLUESTAR,SHRIRAM (USHA) AND SIDHWAL
153	BATTERY	EXIDE/AMRON/MICROTEK/AMCO/ARISE
154	PRECISION AC	UNIFLAIR/BLUESTAR/EMERSON
155	BATTERY MODULAR	EXIDE, AMCO, AMARAJA, ARISE, STANDARD
156	SPLIT AC/ WINDOW AC	VOLTAS, HITACHI, DAIKIN, O- GENERAL CARRIER, FEDDERS, LLYOD
157	CONTACTOR & STARTERS	GENERAL CARRIER, FEDDERS, L&T, HAVELLS
158	CABLE GLAND	LLYOD, GRANDLAY, HAVELLS
159	CAPACITORS	SIEMENS, ABB, USHA, HAVELLS
160	BMS (BUILDING MANAGEMENT SYSTEM)	SIEMENS, HONEY WELL, SCHNIDER
161	WATER LEAKAGE DETECTION SYSTEM	SIEMENS, SCHINIDER, ABB
162	REFRIGERATOR	SAMSUNG/LG/ VOLTAS/ VIDEOCON/ KELVINATOR/ WHIRLPOOL,
163	CENTRIFUGAL PUMPS	KIRLOSKAR, KSB, GRANDFOS, ITT
164	LT PANELS/ APFC PANELS/PUMP HOUSE PANELS/AC CONTROL PANELS	SIEMENS, L & T, ABB, HENSEL, STANDARD,INDO ASIAN, HAVELLS, UNIVERSAL SWITCH GEAR, PSC, GM&G, ITE GURGAON, NEPTUNE, PRECISION SYSTEM CONTROL
165	VOLTMETER/ AMMETER/ FREQUENCY METER/ PF METER	L&T, AE, SCHENEIDER, LEGRAND, HAGER, ,HAVELLS
166	MOTAR STARTER	L&T, BCH, SCHNEIDER, SIEMENS, ABB
167	SINGLE PHASE PREVENTORS/ PHASE SEQUENCE CORRECTOR	L&T, SIEMEN, SINETRAC, MINILEC, LEGRAND, SCHNEIDER, ABB
168	SUBMERSIBLE CABLE	FINOLEX/L&T/ANCHOR/HAVELLES/POLYCAB
169	FIRE HYDRANT VALVE	MINIMAX, FIREX, SAFEX, FLAME GUARD
170	WATER LEVEL INDICATOR	RECKTRONIC DEVICES SYSTEM, UNIVERSAL SWITCH GEAR CONTROL

(APPROVED LIST OF MANUFACTURERS/ MAKES (Contd…)

171	SUBMERSIBLE PUMP	CALAMA, KIRLOSKER, BEACON, KSB, MATHER & PLATT, GRANDBASS, WASP, ITT, CRI
172	NON CLOG SEWAGE PUMP	KIRLOSKAR, BEACON, KSB, CROMPTON
173	COPPER WINDING WIRE	ATLAS, HTP, BIC, KINJAL
174	RUBBER COUPLING	UNIQUE FORM –FLESE, HIC, MAHALAXI/METRO
175	DESERT COOLERS	SYMPHONY, KHAITAN, KENSTAR
176	DEEP FREEZER	BLUE STAR, VOLTAS, SHRI RAM (USHA)
177	HT/LT XLPE AERIAL BUNCHED CABLES	CRYSTAL, PARAFLEX, NATCAB
178	HT/LT CABLE JOINTS HEAT SHRINKABLE/ COLD SHRINKABLE/ PUSH ON TYPE	COMPAQ, MS-SEAL, MSS, M-SEAL, DENSON
179	LT AIR CIRCUIT BREAKER	YAMUNA, DENSONS, INDO ASIAN, STANDARD, HAVELLS,
180	COMPOSITE POLYMERIC INSULATOR	YAMUNA, DENSONS, JAIPURIA, POWER GRID, MG MATERIALS
181	FIBRE GLASS REINFORCED PRODUCTS 11 KB & 33KV CROSS ARMS/ TOP HAMPER	YAMUNA, DENSONS, JAIPURIA
182	POWER CONTACTOR	INDO ASIAN, HAVELLS, L&T, SIEMENS, LEGRAND
183	STREET LIGHT TIMERS	INDO ASIAN, HAVELLS, L&T, SIEMENS, LEGRAND
184	HT/LT STEEL TUBULAR SWAGED POLES	SHIVAM METAL SHAPERS, SURYA, HPL
185	ELECTRIC MOTOR	HPL, BHARAT BIJLEE, KIRLOSKAR ELECTRIC, CROMPTON GREAVES, ABB, SIEMENS, GEC.
186	WINDOW TYPE/ SPLIT AIR CONDITIONERS (5 STAR RATED AND ANNUAL USAGE IS > 1000HRS, 3 STAR FOR OTHER PLACES LIKE CONFERENCE ROOMS)	VOLTAS, LG, BLUE STAR, HITACHI, CARRIER, O-GENRAL
187	CLASSETTE TYPE AC	BLUE STAR, LLOYD, VOLTAS, PANASONIC, TOSHIBA, LG.
188	PILLAR TYPE AC	LLYOD, LG, BLUE STAR, WHIRLPOOL, PANASONIC
189	DUCTABLE AC	LLYOD, VOLTAS, HITACHI, LG, BLUE STAR
190	AIR CRUTAIN	CROMPTON GREAVES, ALMONARD, ROBUSTIX
191	FIRE DAMPER	DYNA CRAFT, RAVISTAR, MMAYO, CARRY AIR, AIRFLOW, RUSKIN
192	ALUMINIUM EXTRUDED, ALUMINIUM GRILL/ DIFFUSER	DYNA CRAFT, RAVISTAR, MMAYO, SERVED GREEN HECK, BRIGHTFLOW
193	CHILLERS/ CHILLING UNITS	CARRIER, VOLTAS, BLUESTAR, KIRLOSKAR
194	COOLING TOWERS	MIHIR, PAHARPUR, ADVANCE, DELTA
195	VARIABLE FREQUENCY DRIVES (VFD)	DANFOSS, HONEYWELL, ABB, AUTOMATION, SIMPHOENIX
196	POT/ Y-STRAINER	EMERALD, RAPID COOL, LEADER
197	PRESSURE GAUGE/ DIAL TYPE THERMOMETERS	FIEBIG, H. GURU, EMERALD, AUTOMAT LUB SYSTEM.
198	INSULATION (FIBRE GLASS)	LLYOD, SHRI FIRE PACK, IMPEX INSULATION, TRANS INSULATION

(APPROVED LIST OF MANUFACTURERS/ MAKES (Contd…)

199	INSULATION (EXPENDED POLYETHYLENE)	TOSHIBA, STYRENE PACKAGING, CMAYICORN, ECOTHEM INSULATION
200	INSULATION (CROSS POLYTHENE FOAM)	SUPREME, PARAMOUNT, AEROFLEX, TROCELLONE
201	TF THERMOCOL FOR PIPE INSULATION	PK PACKING, BIRDSELL, PROPACK, STYRENE PACKAGING, RYCON SERVICES, KIWI THERMO PACK, SAHARA THERMO PACK.
202	CONTROLS (3 WAY VALVES)/ THERMOSTATS	ANERGY, SIEMENS, HONEYWELL
203	EXTRUDED ALUMINIUM GIRLLS/ DIFFUSERS/ FIRE DAMPERS	DYNA CRAFT, MMAYO, SERVEX
204	GI SHEETS FOR DUCTING	SAIL, TATA, JINDAL
205	ACTUATOR	SIEMENS, ADVANCE, CASTLE, NIBCO, AIP
206	STRIP HEATERS	RAYCOLD, DASSPASS, AUE, HONEYBEE
207	HUMIDSTAT/ THERMOSTAT	DANFOSS, HONEYWELL, SIEMENS, JOHNSON, RAPIDCOOL
208	AIR FILTER/ HEAP FILTER	LLOYD, MIHIR, DANFOSS, INDFOSS
209	FACTORY BUILT DUCTS	FLEX, ROLASTAR, TECHNO AIR, CAMDUCT ECODUCT
210	FLEXIBLE CONNECTOR	CORI ENGINEER, RESISTO FLEX, DUNLOP
211	HOT WATER GENERATOR	THERMAX, INDCON 3M, RAPID CONTROL, ZENITH.
212	DUCT FLANGE	ZECO, ROLSTAR, ECODUCT
213	EXPANSION VALVE	DANFOSS, INDFOSS, SPORLON, ALCO
214	FLOW SWITCH	RAPID COOL, SIEMENS, BCH, L&T
215	3-WAY MODULATING VALVE	HONEYWELL, SIEMENS, ANERGY, INDFOSS, DANFOSS, JOHNSON
216	ROOM THERMOSTAT/ AHU THERMOSTAT	HONEYWELL, SIEMENS, DASPASS, ESCORTS, INDFOSS
217	HP/LP CUTOUTS/ CONTROLS	HONEYWELL, PENN, DANFOSS, INDFOSS
218	AIR WASHERS	BREEZE AIR, NICOTRA, PECMA AIREF, AIRFLOW
219	BUTTERFLY VALVE	AUDECO, ADVANCE, DEEPAK
220	FIRE EXTENGUISHER ALL TYPE	MINIMAX, FIREX, NEWAGE, NITIN, CEASEFIRE
221	FIRE HOSE PIPE & ACCESSORIES	NEWAGE, DUNLOP, COSMOS
222	FIRE PANEL	MATHER & PLANT, MINIMAX, SAFEX, AGNI, HONEYWELL, SYSTEM SENSON, FIREX, NOTEFIRE.
223	HOOTER/ SIREN	MINIMAX, MELTRON, AHUJA, PHILIPS, AGNI
224	MANUAL CALL BELL POINT	AGNI, SURKSHA, MINIMAX, FIREX, KINJAL
225	SMOKE/ HEAT/ IONISATION DETECTOR ADDRESSABLE & NON ADDRESSABLE	APPOLO, EDWARDS, HONEYWELL, SYSTEM SENSER.
226	WATER SPRINKLER HEAD	APPOLO, TYCO, CEASEFIRE, SAFEX, MINIMAX, STARTECH FIRE SYSTEM, ELECTRONIC CONTROLS

Contractor

AGE (Contracts)
for Accepting Officer